



Flexiv Elements

User Manual

V2.10.5

CONTENTS

Preface	1
1 Introduction	2
2 Elements Basics	4
2.1 Robot System Status	4
2.2 Coordinate Systems	8
2.3 Freedrive	11
2.4 Go To	13
2.5 Variables	14
3 Flexiv Tab	17
3.1 Start Up the Robot	17
3.2 Data Sync Portal	18
3.3 Data Sync Process	19
4 Projects Tab	20
4.1 File Lists	20
4.2 Edit Panel	21
4.3 Execute Panel	29

CONTENTS

5 Monitor Tab	34
5.1 General	34
5.2 I/O	35
5.3 Active Project	36
6 Apps Tab	37
6.1 Introduction	37
6.2 Trajectory Editor	38
6.3 Plan Log Viewer	40
6.4 Robot Log	42
7 Settings Tab	45
7.1 Introduction	45
7.2 General	46
7.3 License	47
7.4 Version & Update	48
7.5 Robot Mounting	49
7.6 Safety	50

CONTENTS

7.7 Tool	53
7.8 Workpiece Coordinates	55
7.9 Plan ID Editor	56
7.10 Fieldbus	57
7.11 Remote Mode	58
7.12 Internal Plans	59
7.13 Self-Examination	60
7.14 Joint Torque Sensor Calibration	61
7.15 F/T Sensor Calibration	62
7.16 Kinematics Model Validation	63
7.17 Backup & Restore	66
8 Primitive Commands	67
8.1 Introduction	67
8.2 I/O Command Instruction	69
<i>8.2.1 Introduction</i>	69
<i>8.2.2 Configuration Overview</i>	70

CONTENTS

8.2.3 Configure in the Main Plan	71
8.2.4 Configure in the Subplan	78
8.2.5 Configuration Result	85
8.3 Tool Command Instruction	86
8.3.1 Select the tool	86
8.3.2 Move to the "home position" with the tool	87
8.3.3 Move to a desired gripping pose	88
8.3.4 Pick up the workpiece by configuring Tool Command	89
8.4 AI Command Instruction	92
8.4.1 Introduction	92
8.4.2 Add the object	93
8.4.3 Clear the original information of AI Computer	94
8.4.4 Move to the original pose	96
8.4.5 Take the picture for the target point with AI Command	97
8.4.6 Move to target point of the workpiece	99

CONTENTS

9 Communications	101
9.1 Remote Mode Configuration Instruction	101
9.1.1 Introduction	101
9.1.2 List of I/O Signals	102
9.1.3 Remote Mode - I/O	104
9.1.4 Remote Mode - Fieldbus	111
9.1.5 Remote Mode - RDK.	118
10 Appendix	119
10.1 Troubleshooting	119
10.1.1 Summary of Common Issues	119
10.1.2 Control Box Power On Issue	120
10.1.3 Light Strip Issue	121
10.1.4 Network Issue	122
10.1.5 Servo On Issue	124
10.1.6 Fault State Issue	125

Thank you for choosing Flexiv. This manual provides the essential information you need to use **Flexiv Elements**, an intuitive graphical interface on the Teach Pendant. You can operate the adaptive robot to perform complex tasks by creating programs via a drag-and-drop function module in Flexiv Elements.

This manual is intended for the user who is expected (though not required) to be familiar with elementary concepts of programming. No special knowledge about robots in general or Flexiv robots in particular is required. Information in this manual is subject to change due to improvement or redesign of the product.

You can also refer to the following related manuals for information that can help guide you through using Flexiv robots, **Flexiv Elements**, and other peripheral equipment:

- ***Flexiv Rizon Quick Start Guide*** with brief instructions for installing and programming the robot.
- ***Flexiv Rizon User Manual*** with detailed instructions for robot installation, operating, programming, and maintenance and repair. It also contains important safety instructions that must be strictly observed.
- ***Trajectory Editor User Manual*** with detailed instructions for using **Trajectory Editor**, an application in the **Apps** tab in Flexiv Elements designed specifically for viewing, creating, and managing robot trajectories.
- ***Anybus Gateway User Manual*** with detailed instructions for setting up **Anybus Gateway**, a communication type in the **Settings** tab in Flexiv Elements.

For more information, please visit our website at **www.flexiv.com** or call us at **+1 650 440 7808** or **+86 4008 888 105**.

This manual consists of the following chapters:

Chapter 1 Introduction

Helps you quickly understand how to read this manual by providing a general introduction of the chapters and legends.

Chapter 2 Elements Basics

Describes some of the basic concepts and functions used across different tabs in Flexiv Elements, including robot system status, coordinate systems, **Freedrive** function, **Go To** function, and variables.

Chapter 3 - 7 Elements Tabs

Provide detailed instructions on the functions and usage of the five main tabs in Flexiv Elements, namely, the **Flexiv** tab, the **Projects** tab, the **Monitor** tab, the **Apps** tab, and the **Settings** tab. These chapters help you quickly get familiar with the user interface and the programming workflow.

Chapter 8 Primitive Commands

Contains the instructions for the usage of **I/O Command**, **Tool Command**, and **AI Command**, the commands used by primitives, through examples.

Chapter 9 Communications

Introduces the communication types used by the robot system. Currently this chapter only covers **Remote Mode**.

Chapter 10 Appendix

Contains **troubleshooting** information to help you identify, isolate, and resolve software and hardware issues.

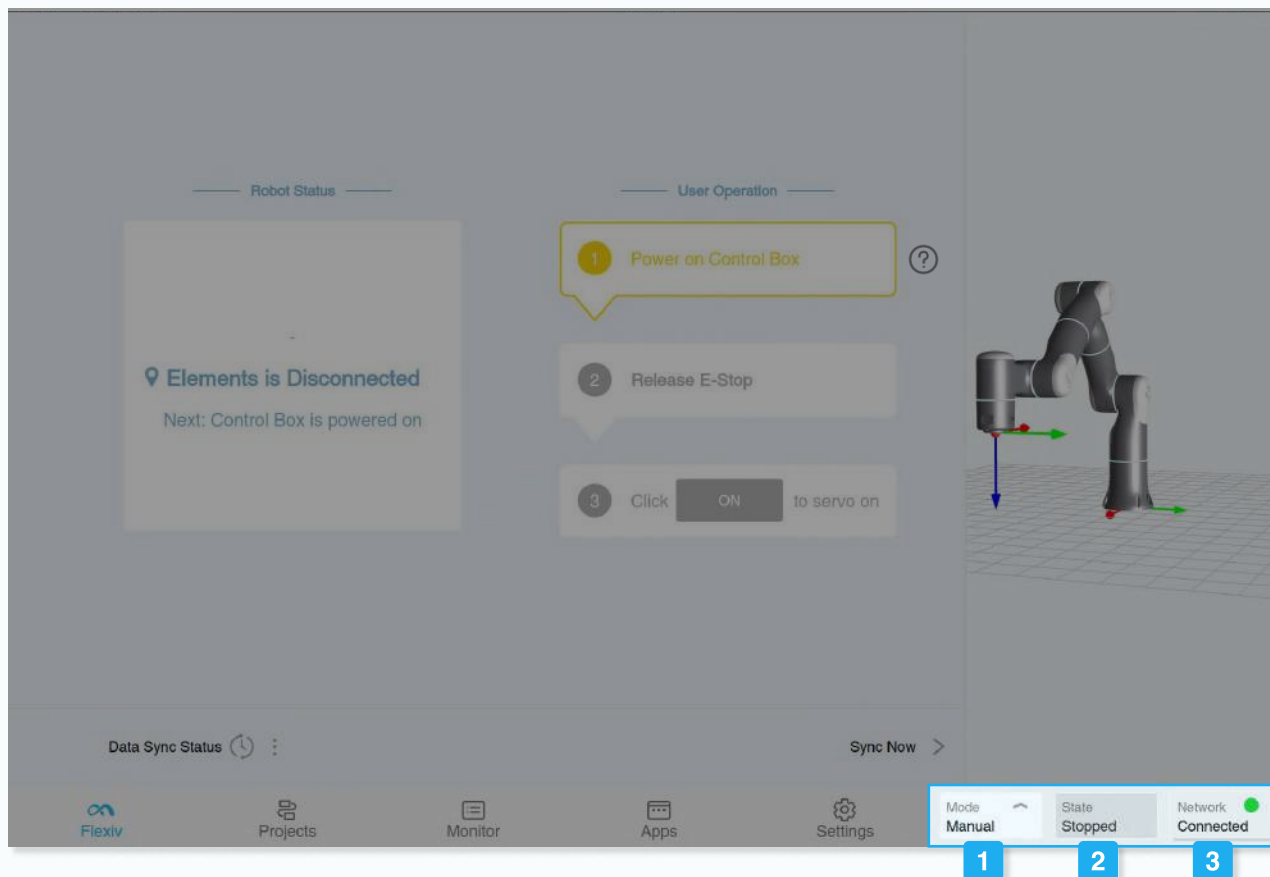
All of the procedures in this manual are organized in a sequential, step-by-step format. Legends/symbols are used to explain the functions of Flexiv Elements in this manual. They can help you quickly navigate certain chapters, sections or steps of your interest. See the table below for their respective usage.

Legend		Usage	
1 2 3 4		Panel No.	Explain the interface by dividing the panels
		Panel Area	
1 2 3 4		Step No.	Explain the interface by breaking down the steps
		Step Area	
Common term: The explanation of common terms		Explanatory style of common terms	
Flexiv term: The explanation of Flexiv terms		Explanatory style of Flexiv terms	

2.1 Robot System Status

Part 1/3 - Location of Robot System Status

The bottom-right corner of the Flexiv Elements main interface displays the status of the robot system, including operation mode, operation state, and the connection status between Flexiv Elements and the robot.



- 1 Operation mode
- 2 Operation state
- 3 Connection status between Flexiv Elements and the robot

2.1 Robot System Status

Part 2/3 - Lists of Robot System Status

1 List of Operation Modes

Mode Manual	In Manual mode, the robot motion is controlled by the operator and robot TCP speed is limited to 250 mm/s or less. The robot will only run when the operator holds down the enabling button on the Motion Bar. If the enabling button is released, the robot will stop immediately.	 Lower position on Motion Bar
Mode Auto	In Auto mode, the robot is capable of running at full speed and can automatically execute the user-customized plans that consist of Primitives (i.e. motion commands) on Flexiv Elements.	 Upper position on Motion Bar
Mode Auto (Remote)	In Auto(Remote) Mode, the robot is controlled remotely by the host computer through signal input and can automatically execute commands from the host computer.	 Upper position on Motion Bar
Mode Unknown	Flexiv Elements and the robot are disconnected	
Mode Unmatched	The mode on the Motion Bar is unmatched with the mode on Flexiv Elements	

2.1 Robot System Status

Part 2/3 - Lists of Robot System Status

2 List of Operation States

State Unknown	Flexiv Elements and the robot are disconnected
State Init	Robot is powered on for the first time and initialization is not completed yet
State Stopped	Robot is stopped
State Freedrive	Freedrive is enabled
State Recovery	Robot is in recovery state and can be moved back within safety limits in Manual mode
State Working	Robot is executing (Manual or Auto)
State Fault	An error has occurred <u>*You can refer to P125 to see how to clear fault</u>

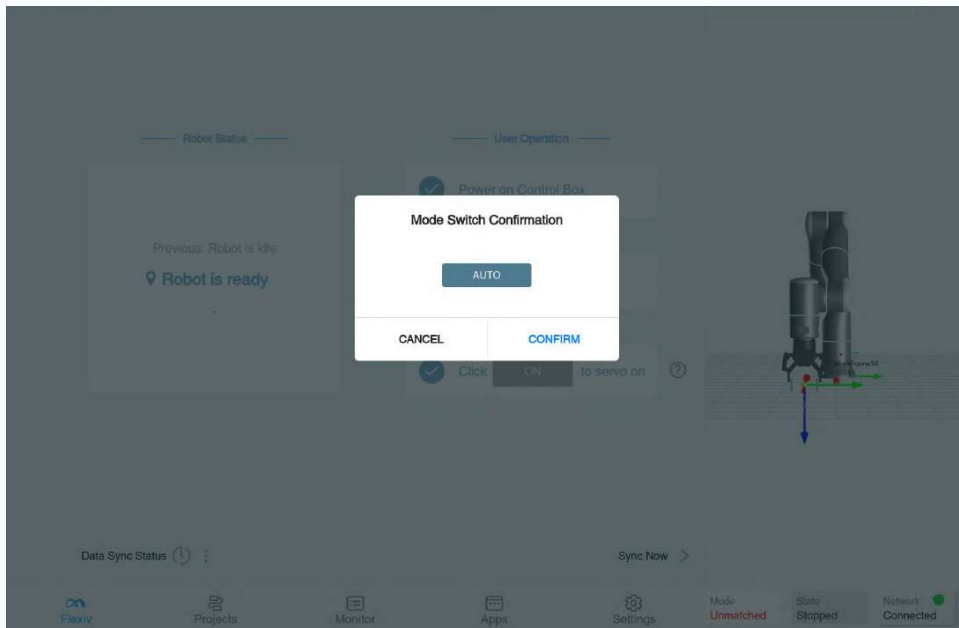
3 List of Connection Statuses Between Flexiv Elements and the Robot

Network Connected	Flexiv Elements and the robot are connected <i>*You can tap it to disconnect Flexiv Elements and the robot</i>
Network Disconnected	Flexiv Elements and the robot are disconnected <i>*You can tap it to connect Flexiv Elements and the robot</i>

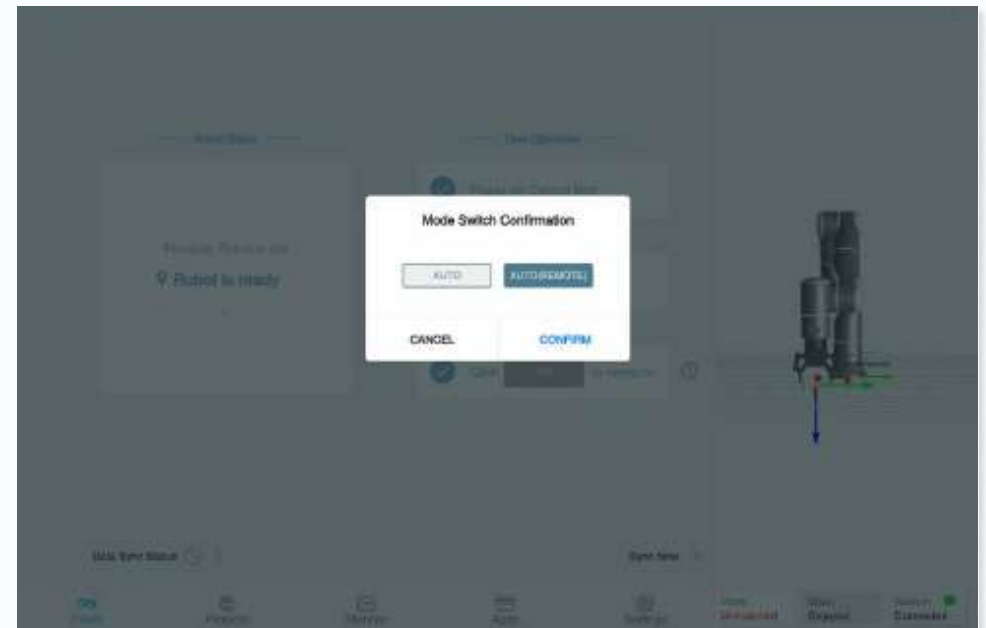
2.1 Robot System Status

Part 3/3 - Mode switch from Manual to Auto / Auto (Remote)

If you need to switch from Manual mode to Auto mode, slide the switch to Auto mode (upper position ) on the Motion Bar. Then Flexiv Elements will prompt either one of the dialogs below based on the state of Remote Mode.



When Remote Mode is **disabled**, Elements will prompt whether to switch to Auto mode. Tap **CONFIRM** to switch.

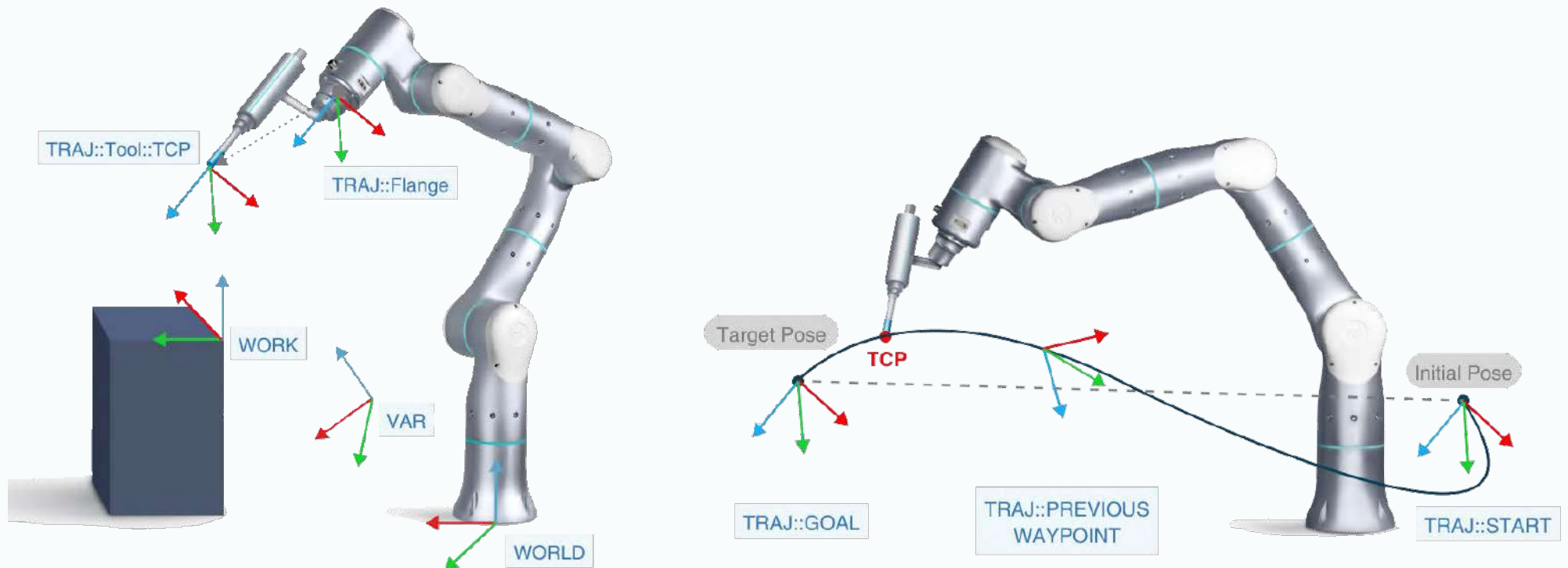


When Remote Mode is **enabled**, Elements will prompt you to choose whether you want to switch to Auto mode or Auto (Remote) mode. Select the mode and tap **CONFIRM** to switch.

2.2 Coordinate Systems

Part 1/2 - Illustrations

The Cartesian coordinate systems used by Flexiv robots are shown in the illustrations below.



2.2 Coordinate Systems

Part 2/2 - Cartesian coordinate systems

The following Cartesian coordinate systems are defined for Flexiv robots:

Coordinate System	Description
WORLD	The WORLD coordinate system is a fixed Cartesian coordinate system that is located at the center of the robot base.
WORK	The WORK coordinate system is a Cartesian coordinate system that defines the position of the workpiece. It is relative to the WORLD coordinate system. It can be created and edited in Settings > WorkCoord .
VAR	The VAR coordinate system is a Cartesian coordinate system that is represented by a user-defined COORD variable. The configuration of the VAR coordinate system can be changed during the plan execution. The following VAR coordinate systems can be used: VAR::Name is defined by a COORD variable configured in Projects > VARIABLES > Plan Variables PVAR::Name is defined by a COORD variable configured in Projects > VARIABLES > Project Variables GVAR::Name is defined by a COORD variable configured in Projects > VARIABLES > Global Variables

2.2 Coordinate Systems

Part 2/2 - Cartesian coordinate systems

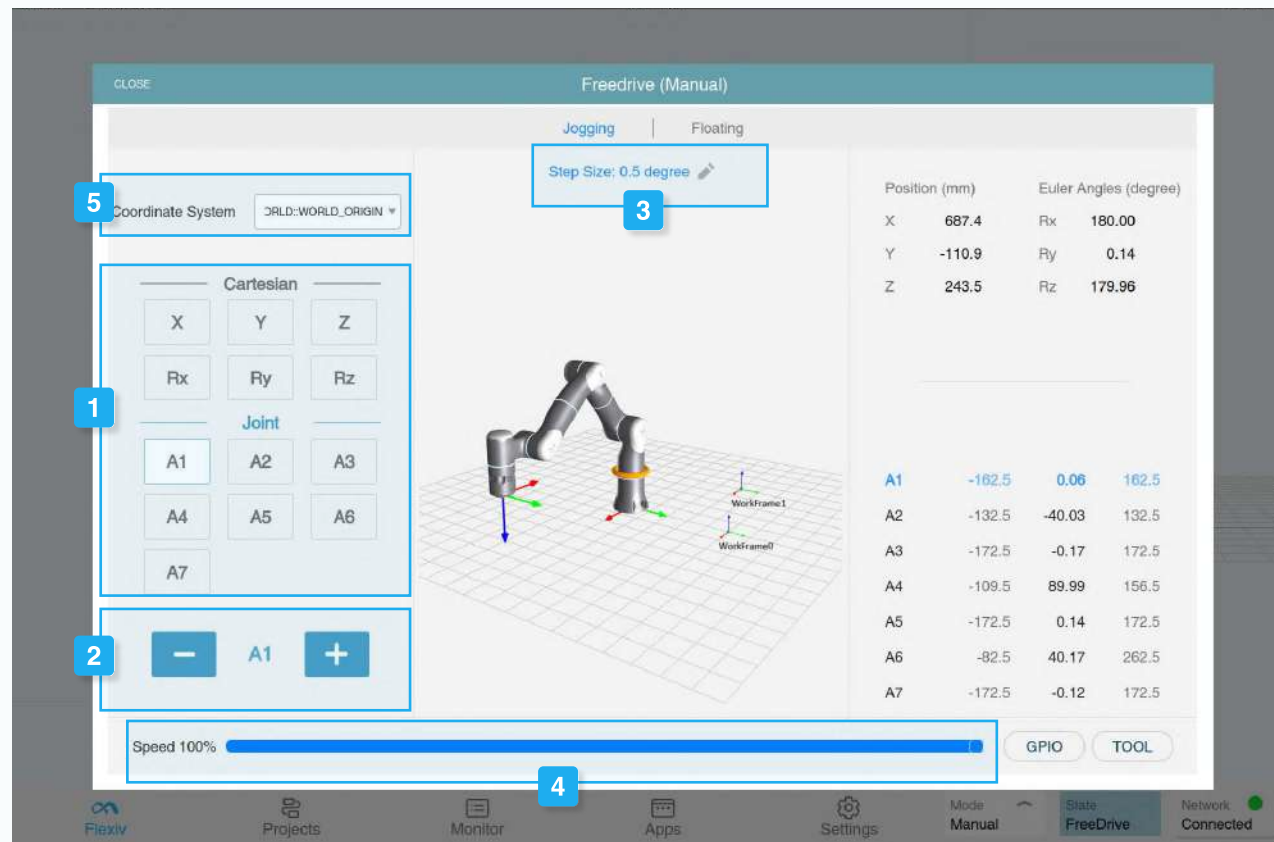
The following Cartesian coordinate systems are defined for Flexiv robots:

Coordinate System	Description
TRAJ	The TRAJ coordinate system is a dynamic Cartesian coordinate system that is located at a reference coordinate on the robot's moving trajectory. It can be changed along with the change of the trajectory. The following TRAJ coordinate systems can be used: TRAJ::Flange, also called the Flange coordinate system, is located at the center of the robot flange. It can be used in Freedrive mode. TRAJ::Tool::TCP, also called the TCP coordinate system, is located at the TCP (Tool Center Point). It is relative to the Flange coordinate system. It can be created and edited in Settings > Tool. TRAJ::START is located at the initial TCP pose upon the start of primitive execution. It can be used to define the offset of a waypoint relative to the initial TCP pose in the TCP coordinate system. TRAJ::GOAL is located at the target TCP pose of the primitive. It can be used to define the offset of a waypoint relative to the target TCP pose in the TCP coordinate system. TRAJ::PREVIOUSWAYPOINT is located at the previous waypoint on the trajectory. It can be used to define the offset of a waypoint relative to the previous waypoint in the TCP coordinate system.

2.3 Freedrive

Freedrive is a human-robot collaboration function that allows you to move the robot into desired positions and poses. It contains two functions: **Jogging** and **Floating**. To access the Freedrive interface, you can press the **Freedrive** button (⊙) on the Motion Bar or tap **TEACH** on point configuration-related interfaces.

Tab 1/2 - Jogging



Jogging allows you to jog the robot to the synchronization position of all joints and Cartesian poses.

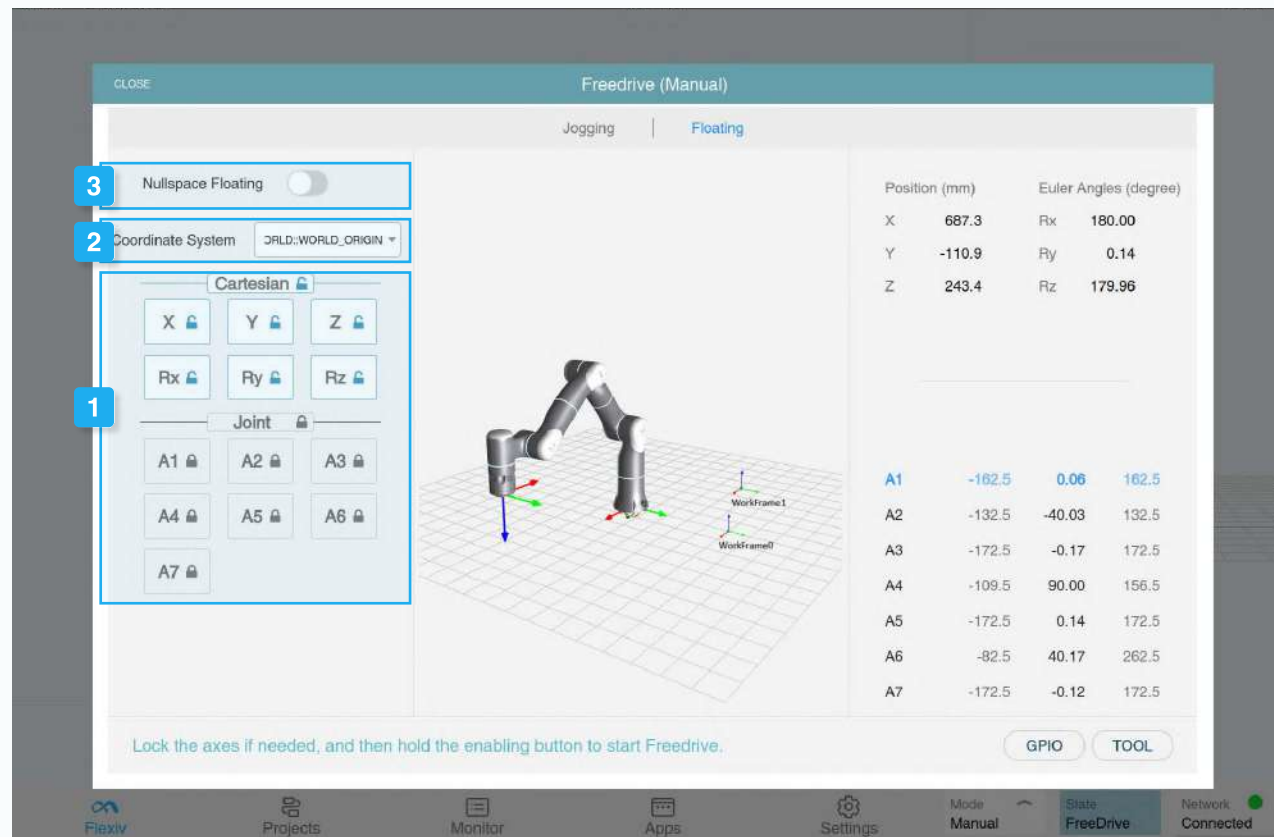
- 1 Lock/unlock the Cartesian axis or specific joint
- 2 Buttons can be long pressed or tapped
- 3 Edit step size in degrees
- 4 Adjust robot speed
- 5 Select the coordinate system you want to use

2.3 Freedrive

Tab 2/2 - Floating

Floating allows you to manually pull the robot into desired positions and poses.

Make sure to avoid any external contact and re-calibrate the tool prior to using this function for better Freedrive performance.

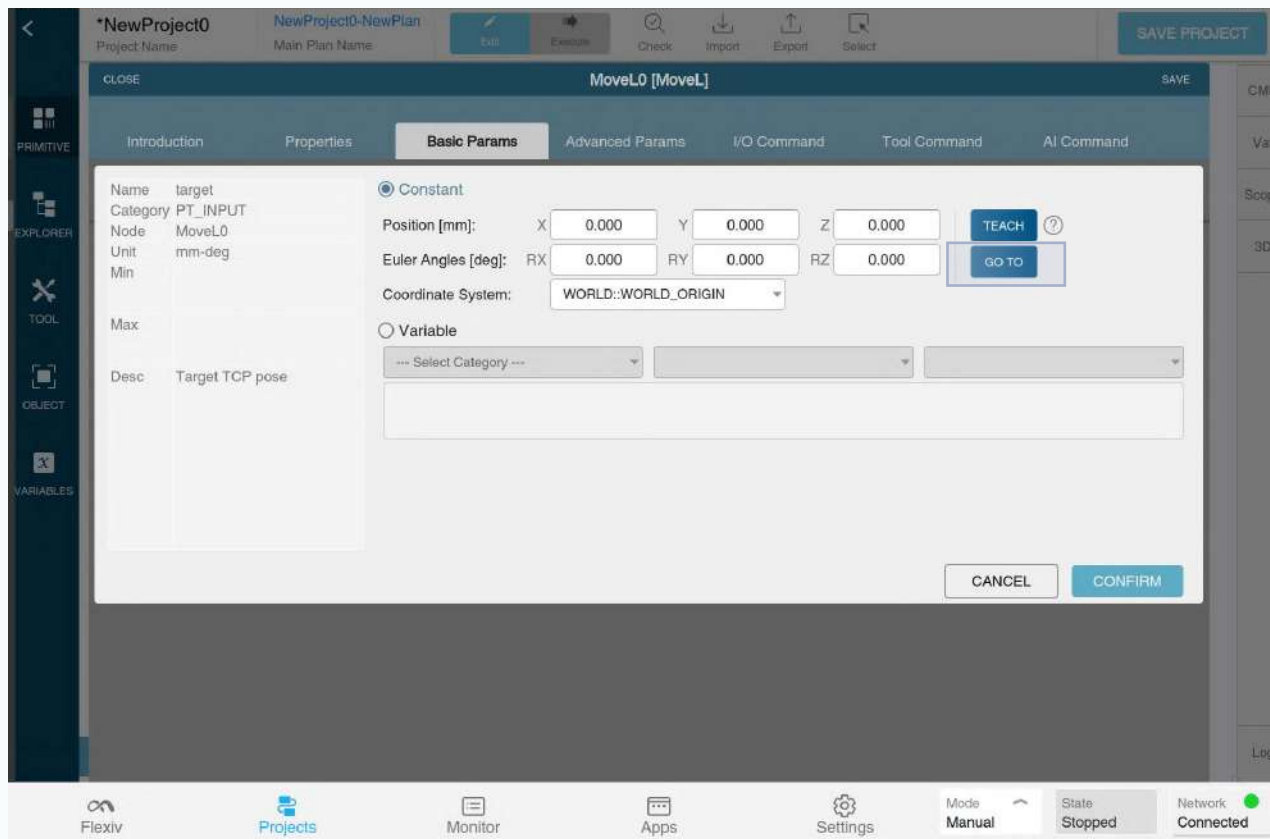


- 1 In this tab, all axes and joints are locked by default. To enter the Floating mode, unlock **Cartesian** or **Joint** first.
- 2 Only when **Cartesian** is unlocked can you switch between different coordinate systems.
- 3 Once you toggle on the **Nullspace Floating** switch, the robot configuration will be automatically adjusted (make sure that the robot can safely perform the movement without colliding with any obstacles).

***Warning:** Floating can also be used in Auto Mode for a smoother teaching experience. However, compared to Manual Mode, the robot runs at a higher speed in Auto Mode, which may cause collisions between the robot and the operator or between the robot and the obstacle.

2.4 Go To

GO TO is a function that enables you to move the robot from its current pose to the pose you defined. When using this function, avoid any potential collisions between the robot and nearby equipment.



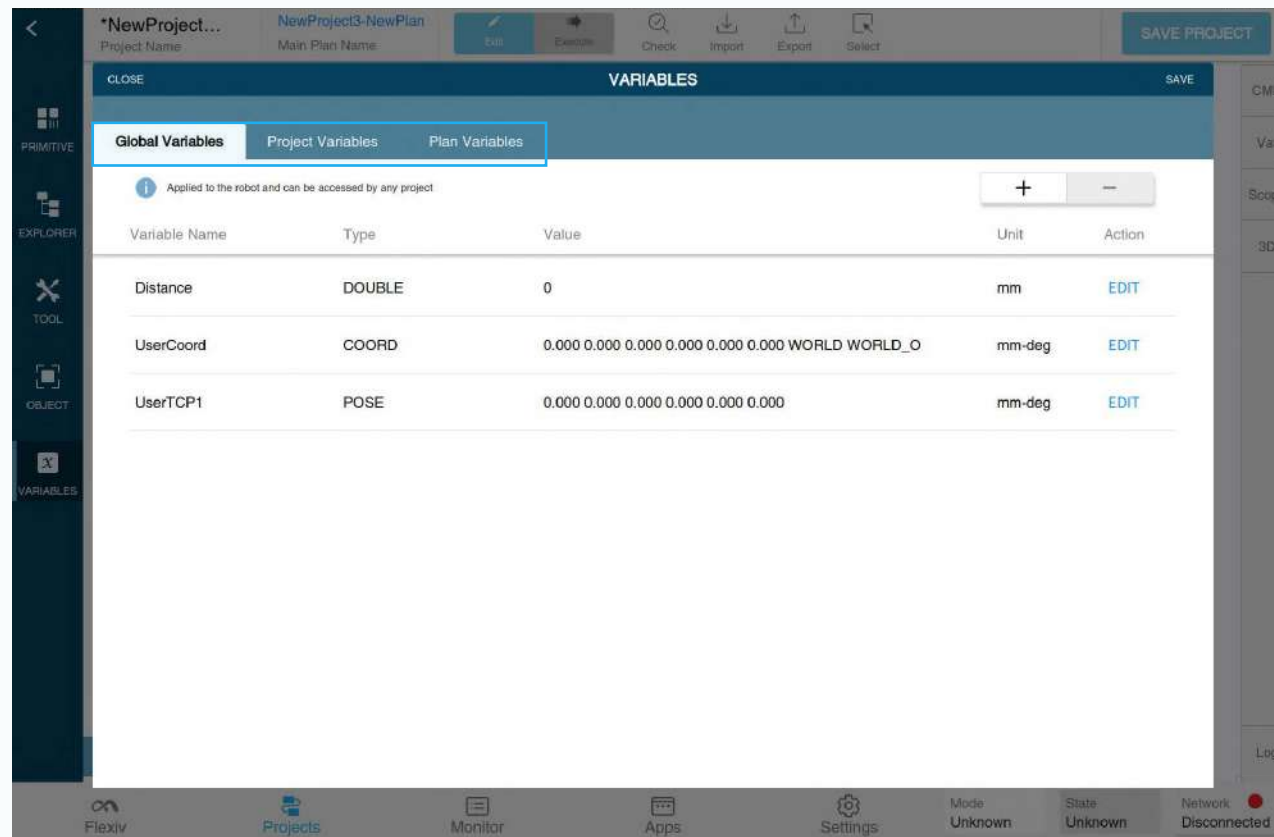
Tap **GO TO** to bring up the GO TO interface and move the robot to the pose you defined.

2.5 Variables

Variables can be defined in projects and applied for counting, operation, logic control, signal interaction, etc.

Part 1/2 - Variable Scopes

Depending on the usage scenarios of variables, you can define variables in the following three scopes:



Global Variable: It can be used in all projects. It is involved in the calculation when the plan is running. Its calculated value is updated when the plan is stopped or terminated, and the updated value is always saved even if the Control Box is powered off.

Project Variable: It can be used in all plans in the current project. It is involved in the calculation when the plan is running. Its value will be initialized after the project is assigned again or the Control Box is powered off.

Plan Variable: It can be used in the current plan and its parent plan. It is involved in the calculation when the plan is running. Its value will be initialized after the project is assigned again or the Control Box is powered off.

***Note:** Global variables should be unique across all the projects. On the other hand, project variables and plan variables have just local uniqueness at the project and plan level. Changes to global variables can only be saved in Stopped state.

2.5 Variables

Part 2/2 - Variable Types

Variable types include basic variable type, basic array type, and general array type. The basic array type is composed of multiple integer variables or multiple floating variables. It has a fixed array length and is denoted by "VEC". The general array type is composed of multiple basic variables or basic arrays. It has an unlimited array length and is denoted by "ARRAY".

Basic variable types and their corresponding general array types:

Basic Variable Type	Definition	Range	Corresponding General Array Type
BOOL	Boolean type variable Commonly used for logical judgment or to represent the on/off status of digital input and output signals	TRUE/FALSE	ARRAY_BOOL
INT	Integer type variable Commonly used for counting	-99999999~99999999	ARRAY_INT
DOUBLE	Floating type variable Commonly used for parameter calculation	-99999999.000~99999999.000	ARRAY_DOUBLE
STRING	String type variable Commonly used for user-defined message	Unlimited character length	ARRAY_STRING
POSE	Pose type variable Commonly used to represent TCP pose or define the TCP variable	X Y Z : -99999999.000~99999999.000 Rx Ry Rz : -180.000~180.000	ARRAY_POSE
COORD	Coordinate type variable Commonly used to represent the TCP pose in a defined coordinate system or define the variable type coordinate system	X Y Z : -99999999.000~99999999.000 Rx Ry Rz : -180.000~180.000	ARRAY_COORD

2.5 Variables

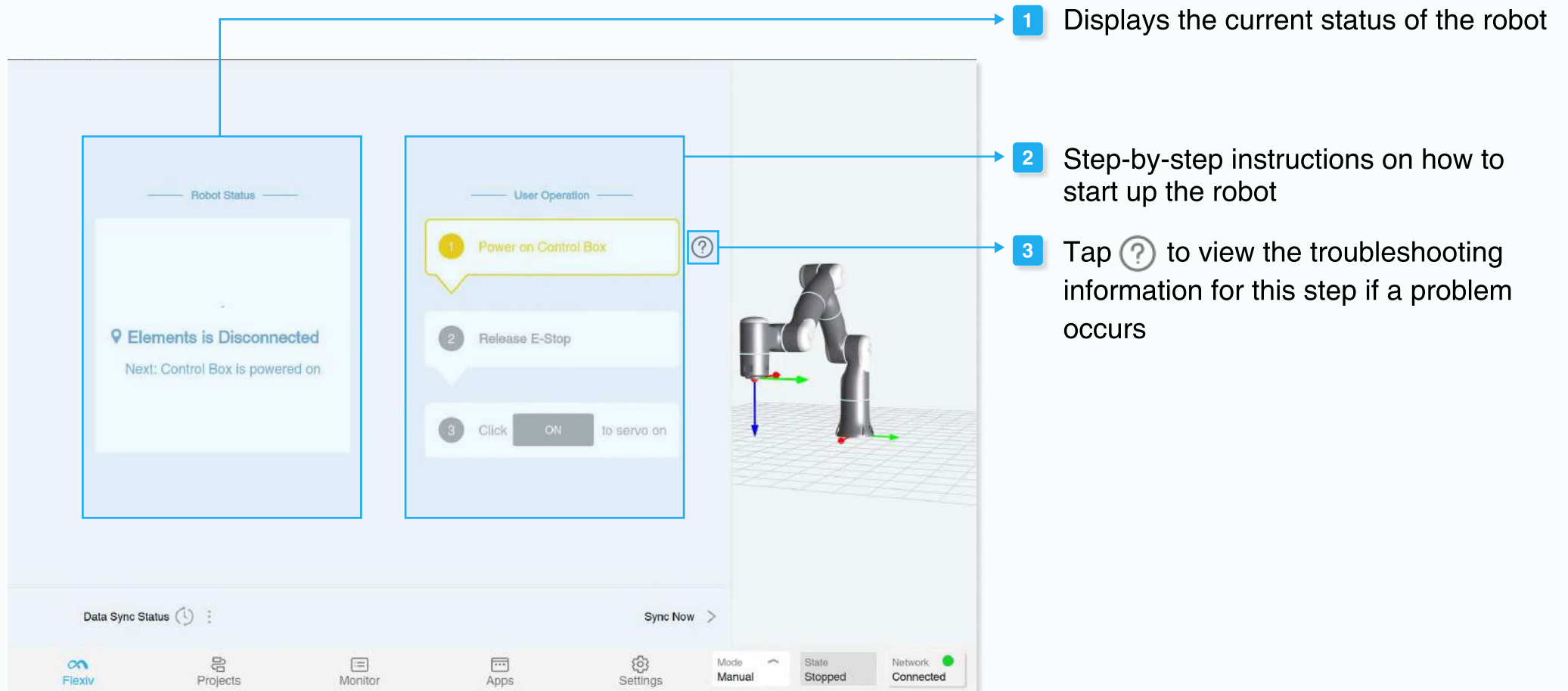
Part 2/2 - Variable Types

Basic array types and their corresponding general array types:

Basic Array Type	Definition	Corresponding General Array Type
VEC_2i	An array composed of 2 INT variables	ARRAY_VEC_2i
VEC_6i	An array composed of 6 INT variables	ARRAY_VEC_6i
VEC_2d	An array composed of 2 DOUBLE variables	ARRAY_VEC_2d
VEC_3d	An array composed of 3 DOUBLE variables	ARRAY_VEC_3d
VEC_4d	An array composed of 4 DOUBLE variables	ARRAY_VEC_4d
VEC_6d	An array composed of 6 DOUBLE variables	ARRAY_VEC_6d
VEC_7d	An array composed of 7 DOUBLE variables	ARRAY_VEC_7d

3.1 Start Up the Robot

This interface is mainly used for the initialization and startup of the robot. You can view the status of the robot in panel **1** and follow the instructions in panel **2** to servo on the robot system. You can also tap **?** to view the troubleshooting information for the corresponding step.

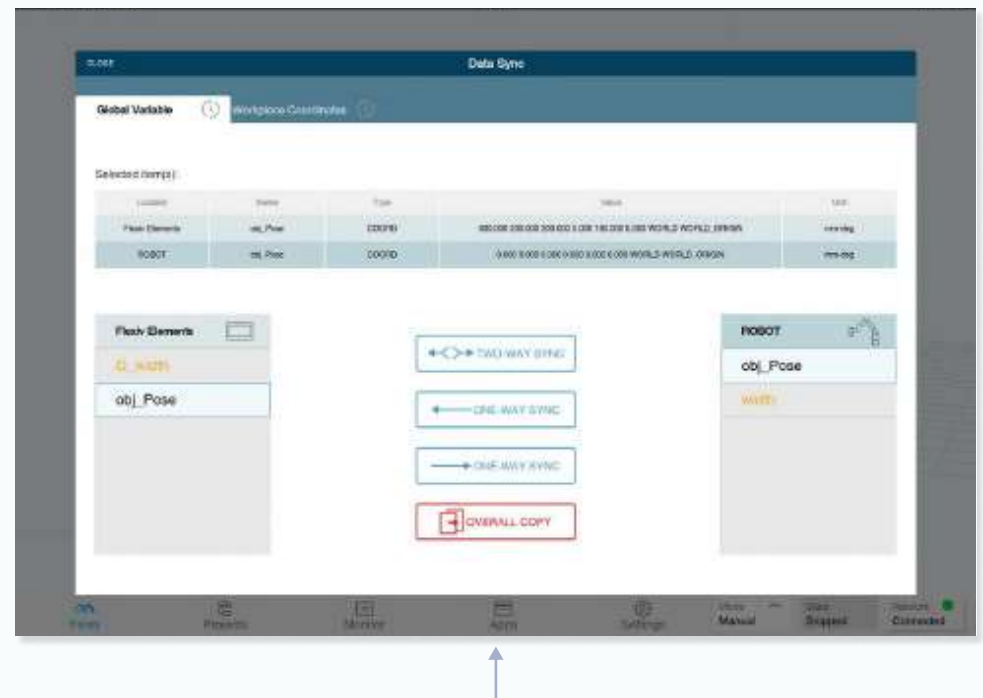
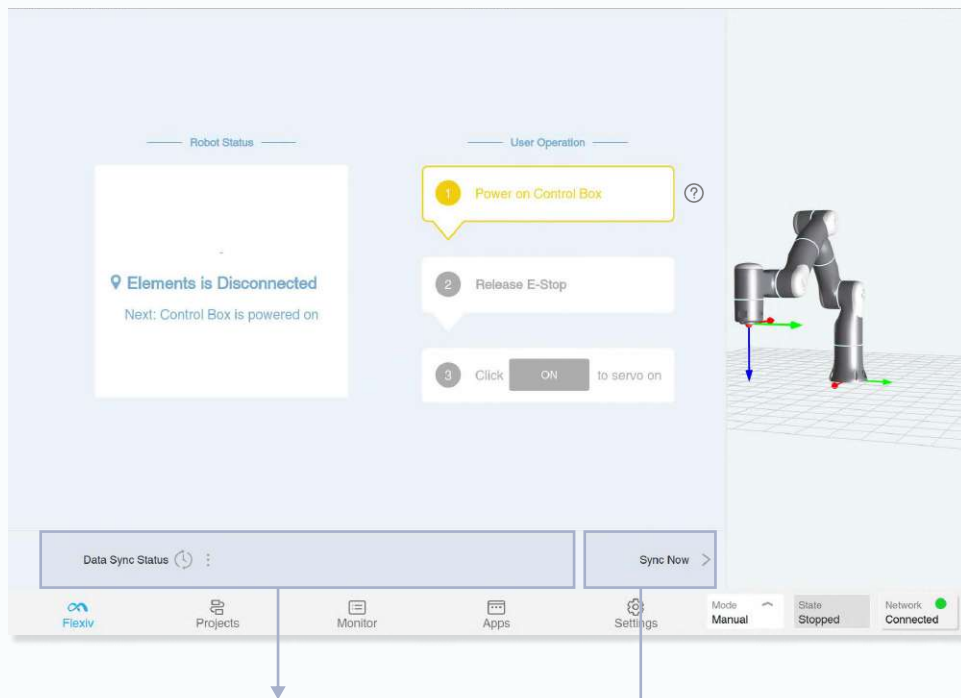


The screenshot shows the Flexiv robot startup interface. It is divided into two main panels: 'Robot Status' on the left and 'User Operation' on the right. The 'Robot Status' panel displays a message: 'Elements is Disconnected' with a location pin icon, and a sub-message: 'Next: Control Box is powered on'. The 'User Operation' panel contains a list of steps: 1. 'Power on Control Box' (highlighted in yellow), 2. 'Release E-Stop', and 3. 'Click ON to servo on'. A blue box highlights the first step, and a blue arrow points from it to a question mark icon. A 3D model of the robot arm is shown on the right side of the interface. At the bottom, there is a 'Data Sync Status' section with a clock icon and a 'Sync Now' button. The bottom navigation bar includes icons for Flexiv, Projects, Monitor, Apps, Settings, Mode (Manual), State (Stopped), and Network (Connected).

- 1** Displays the current status of the robot
- 2** Step-by-step instructions on how to start up the robot
- 3** Tap **?** to view the troubleshooting information for this step if a problem occurs

3.2 Data Sync Portal

This interface provides a portal for data synchronization between Flexiv Elements and the robot. It is typically used for transmitting data and resolving conflicts when pairing Flexiv Elements with the robot.



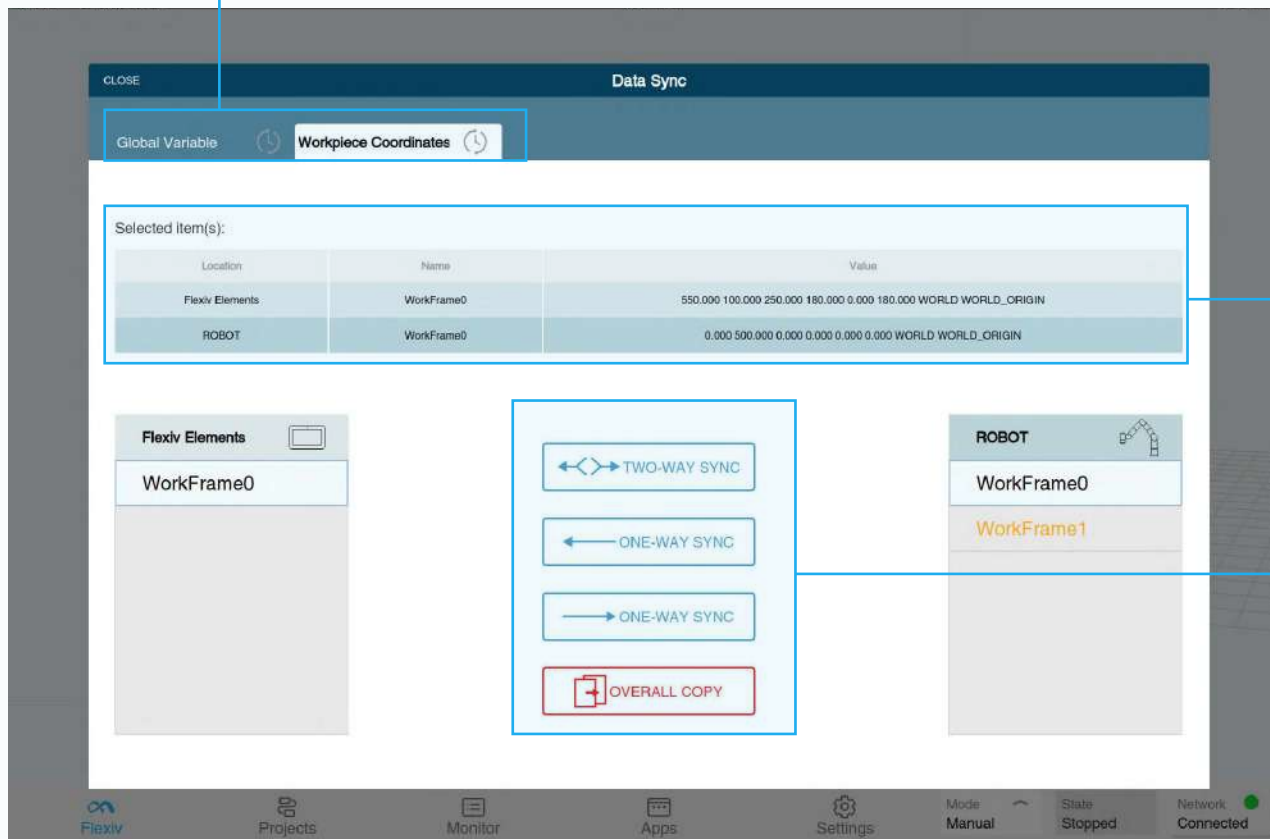
1 Current data synchronization status

2 Tap **Sync Now** to enter the data synchronization page

3.3 Data Sync Process

1 Data sync status:

⌚ Waiting for sync
✅ Sync completed



2 Detailed comparison table of data with the same name to help you resolve conflicts

3 Four options to synchronize data:

↔ TWO-WAY SYNC

Sync data in both directions if there are no conflicting names

← ONE-WAY SYNC

Overwrite the data in Flexiv Elements with the data in the robot with the same name

→ ONE-WAY SYNC

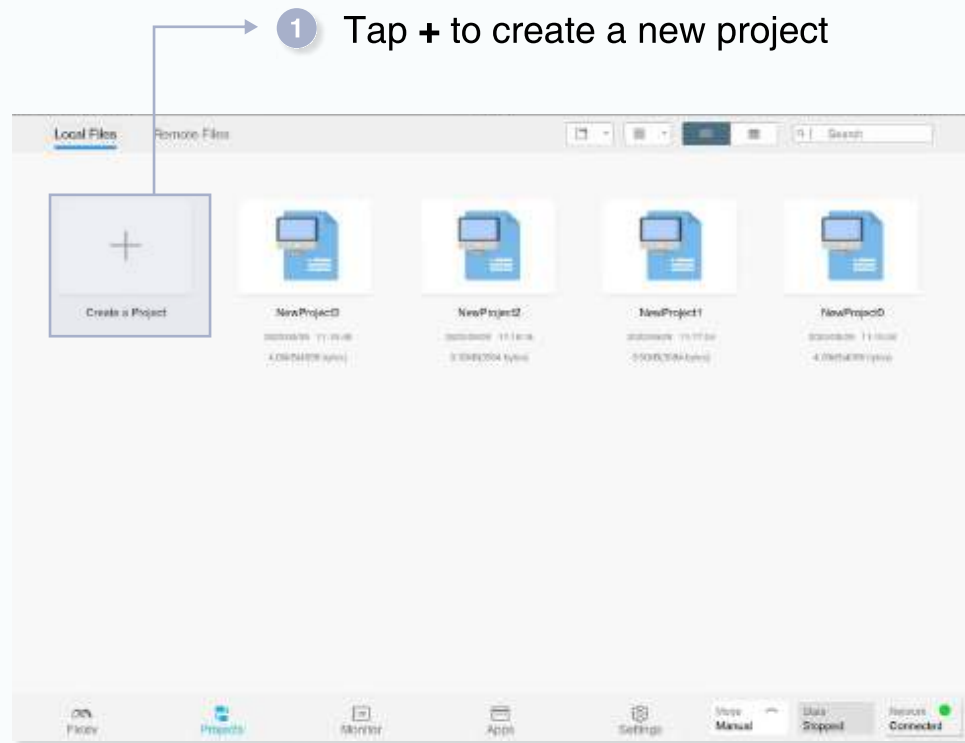
Overwrite the data in the robot with the data in Flexiv Elements with the same name

📄 OVERALL COPY

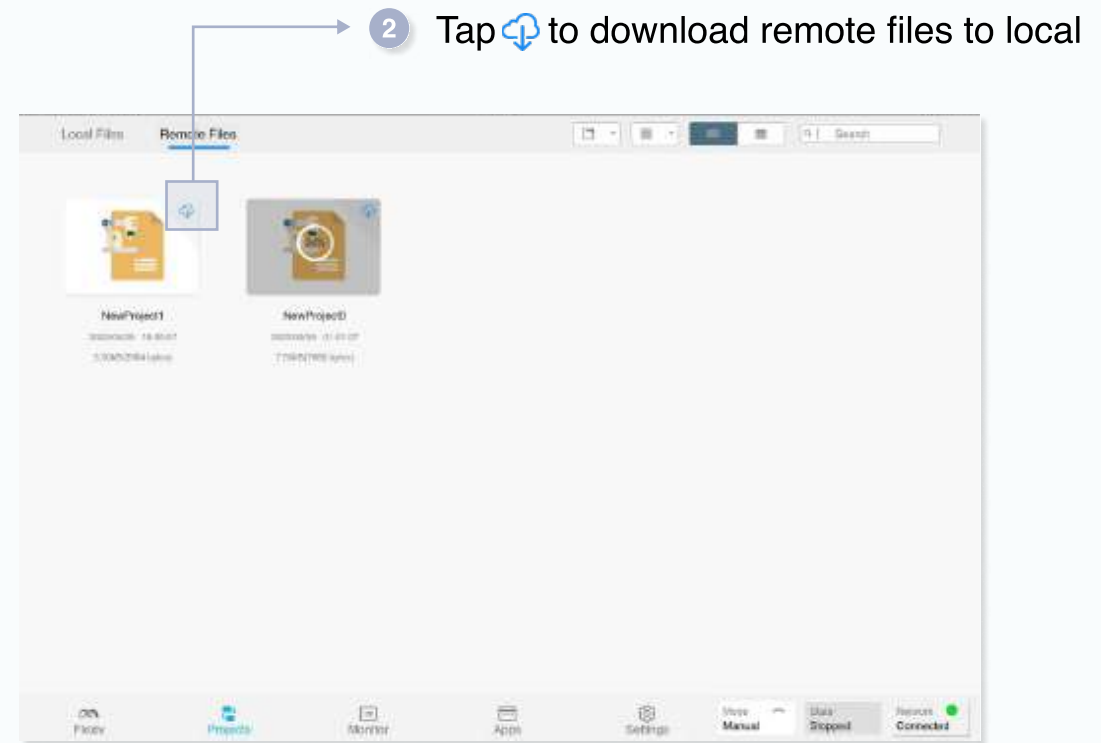
Completely overwrite the other side with all the data from one side

4.1 File Lists

This interface in the **Projects** tab displays a list of local files (Flexiv Elements side) and remote files (robot side) in the **Local Files** tab and the **Remote Files** tab, respectively.



Local Files (Flexiv Elements)

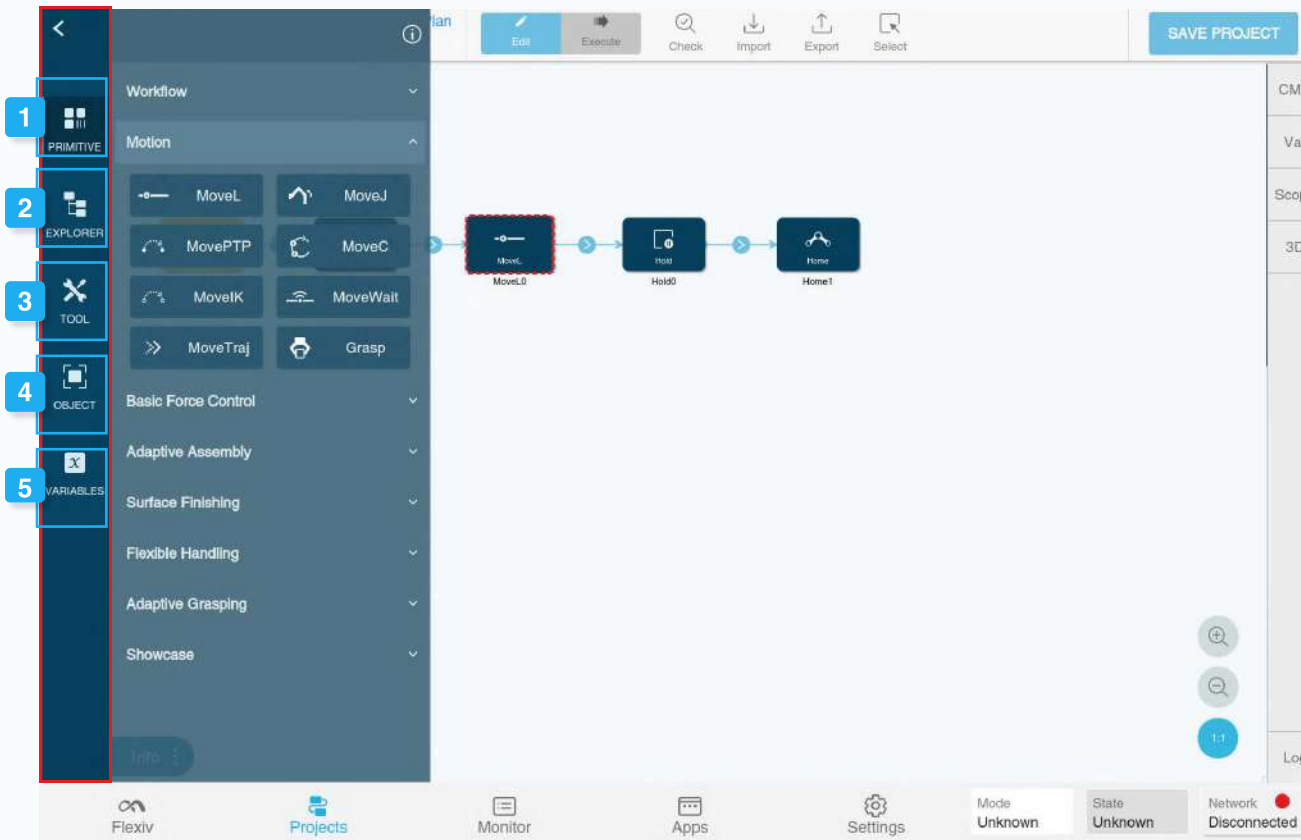


Remote Files (Robot)

4.2 Edit Panel

Part 1/3 - Left Menu Bar

The left menu bar provides you with the nodes, actions, tools, and variables to program the action workflow.



1 PRIMITIVE: *Primitives are the actions/nodes that can be used to operate the robot and tools.* You can use a sequence of primitives to control the robot to perform low-level and/or high-level tasks. Primitives are also the basic building blocks of plans, which are typically used to run complete applications.

2 EXPLORER: Shows the hierarchical structure of the project.

3 TOOL: Shows the available tools in the current project, such as grippers, suction cups, grinding heads, etc.

4 OBJECT: Shows a list of the AI photo objects used by the plan.

5 VARIABLES: Supports viewing and editing **Global Variables, Project Variables, and Plan Variables.**

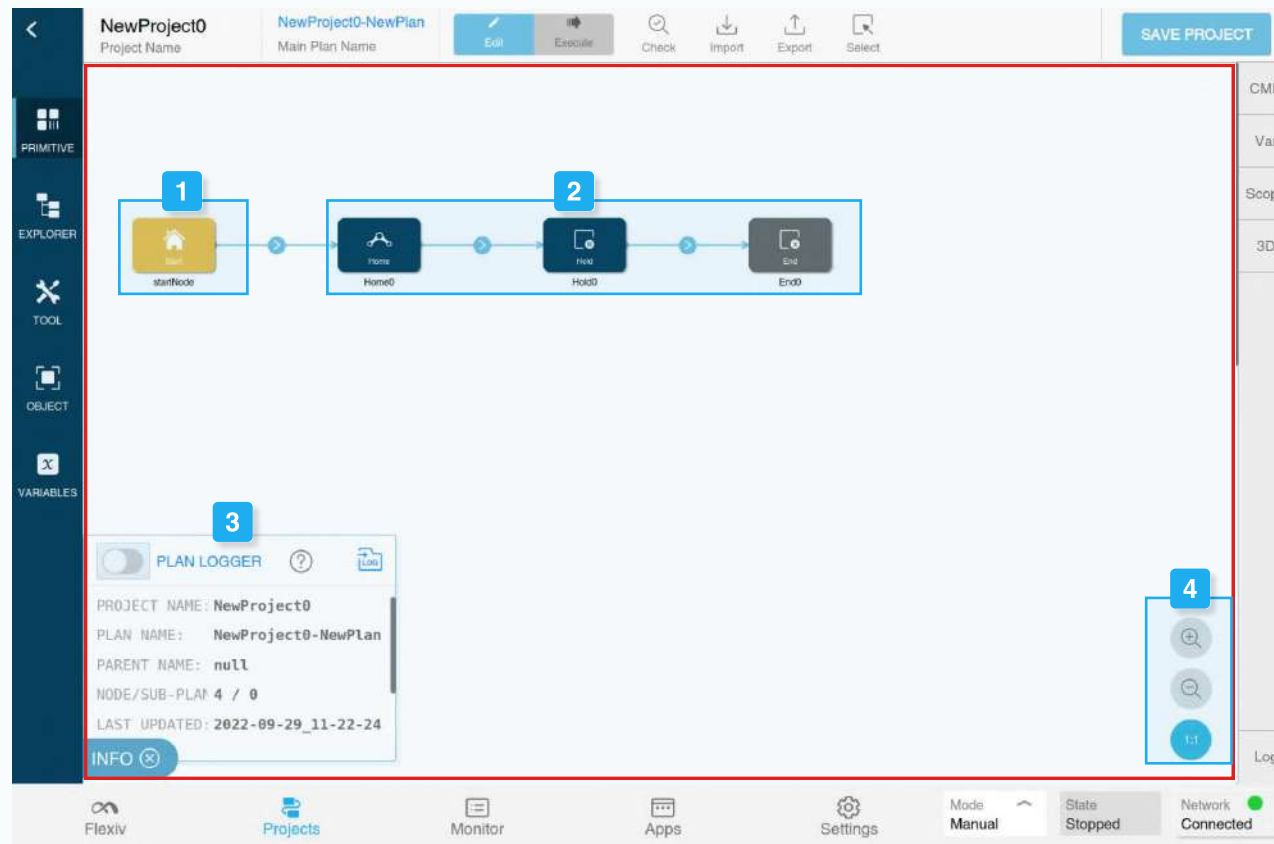
Please refer to P14 for details.

***Note:** The latest version of the **Primitives Manual** can be found at: <http://www.flexiv.cn/en/primitives/>.

4.2 Edit Panel

Part 2/3 - Primitive Canvas: Overview

The primitive canvas provides you with an area for creating and editing projects. It allows you to: drag and drop nodes from the **PRIMITIVE** list to add a primitive to the project; edit a node by tapping the node > **Edit** and edit the parameters in the pop-up window; make transitions between nodes and edit the transition conditions by tapping the blue arrow.



- 1 **startNode** is the entry point of the plan. It is set as the first node by default.
- 2 Nodes represent specific robot operations. You can drag nodes out from the **PRIMITIVE** list.
- 3 Information panel of the current plan
 - Toggle to  to enable PLAN LOGGER. Once enabled, a log file (.csv) will be generated when the robot runs the current plan.
 - Tap  to quickly redirect to **Apps > Plan Log Viewer** to view the most recent log file.
 - Info label status

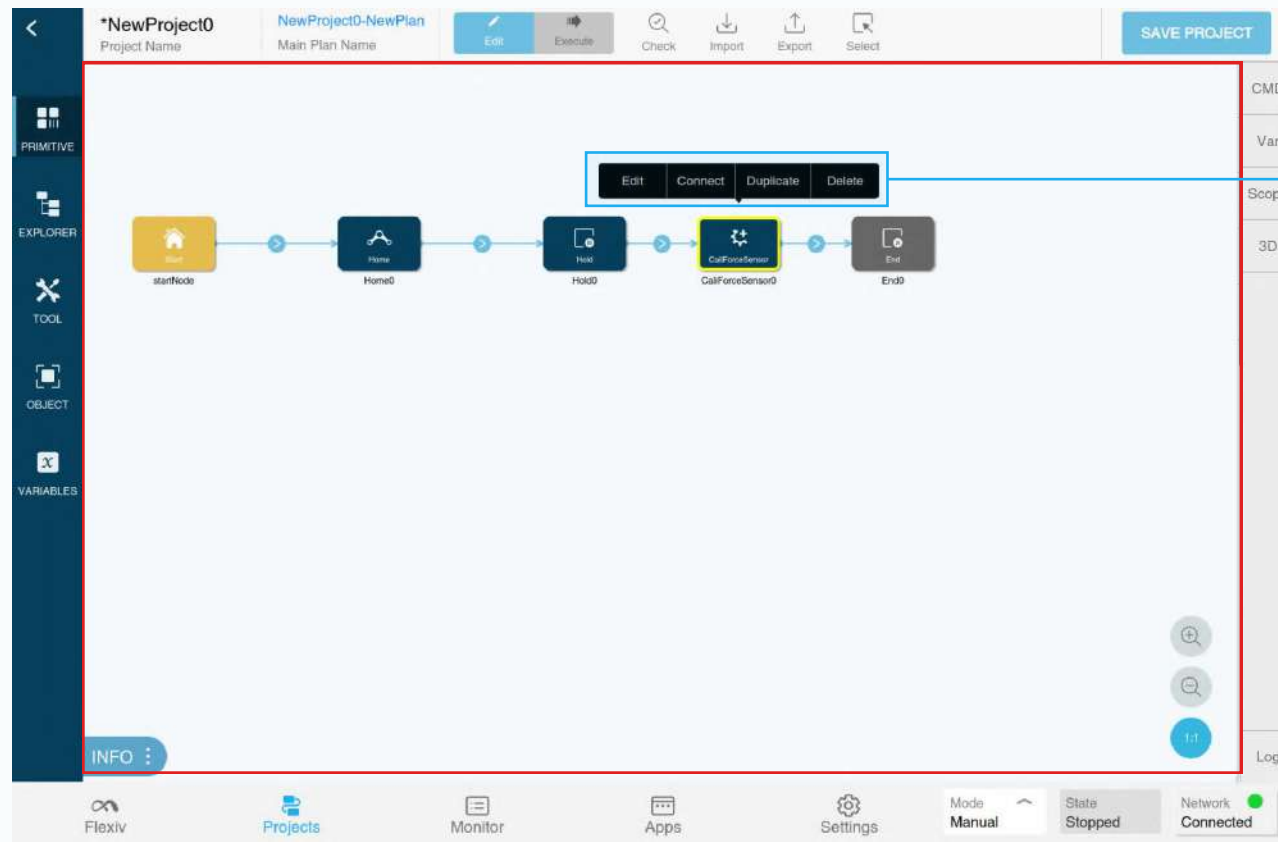
INFO (X) (unfolded) ← Tap → (folded) INFO :

- 4  Zoom in  Zoom out  Fit

4.2 Edit Panel

Part 2/3 - Primitive Canvas: Node Operation

Four options are available after you tap a node: **Edit**, **Connect**, **Duplicate**, and **Delete**.



1

Edit Duplicate Delete

Operations you can perform on the current node (see next page for more details on **Edit**)

2

Connection steps:

1. Tap a node

2. Tap **Connect**

3. Tap the next target node to connect

4.2 Edit Panel

Part 2/3 - Primitive Canvas: Node Operation

Tap the node > **Edit** to enter the edit interface. This interface contains the following tabs:

Introduction: A general introduction of the node, including description, GIF, example usage, input parameters, output parameters, and state parameters.

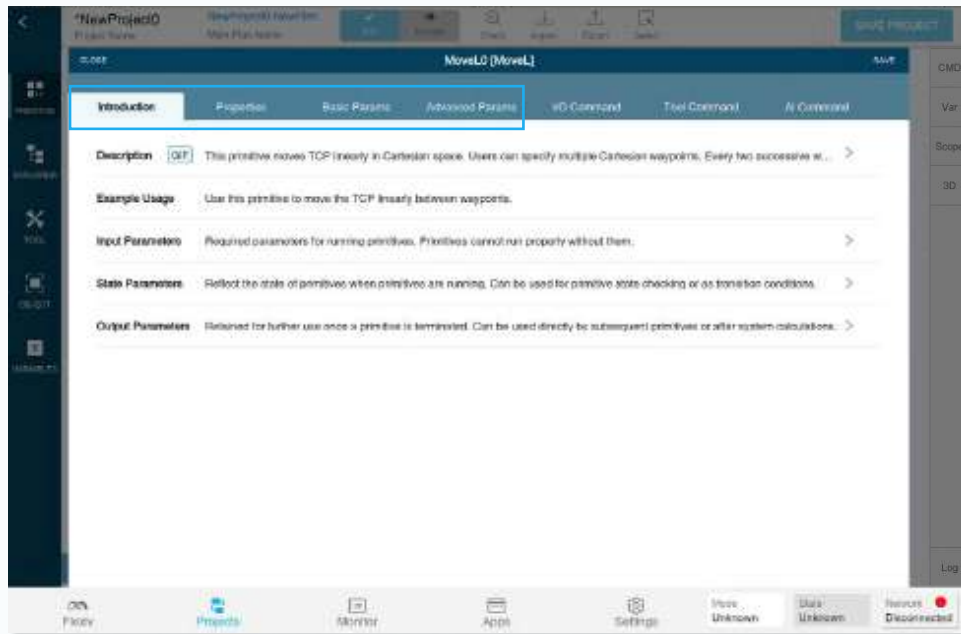
Properties: Basic information about the node, including the node name and TCP configuration.

Basic Params & Advanced Params: **Primitive** parameters that can be configured by users.

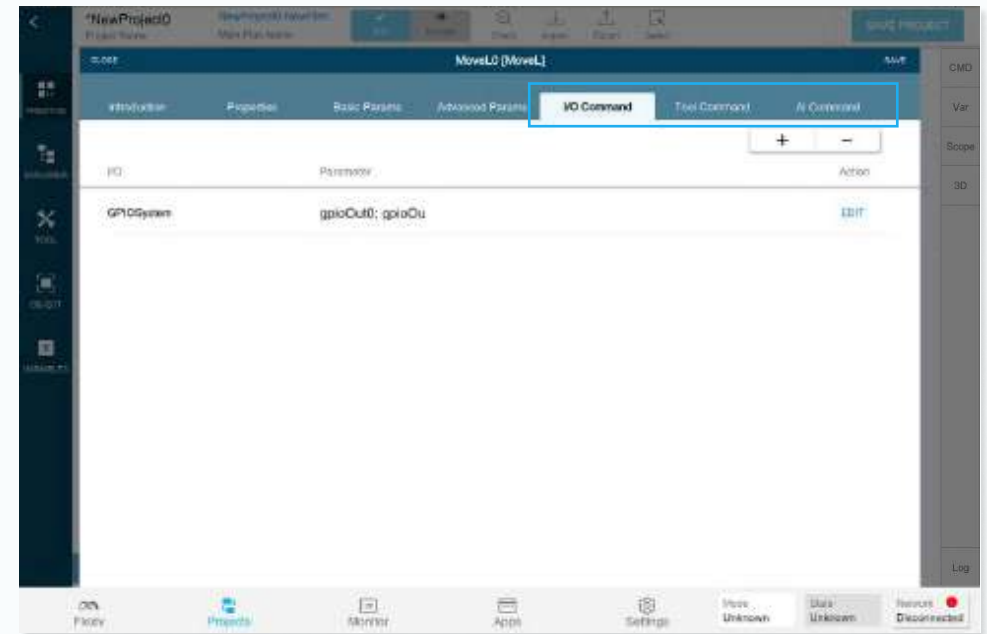
I/O Command: User-specified commands for interacting with I/O signals of external devices. The available options are GPIO and Fieldbus.

Tool Command: User-specified commands for interacting with tools, such as grippers or polishing tools.

AI Command: User-specified commands for interacting with the AI computer.



Tabs: Introduction / Properties / Basic Params / Advanced Params



Tabs: I/O Command / Tool Command / AI Command

4.2 Edit Panel

Part 2/3 - Primitive Canvas: Transitions between nodes

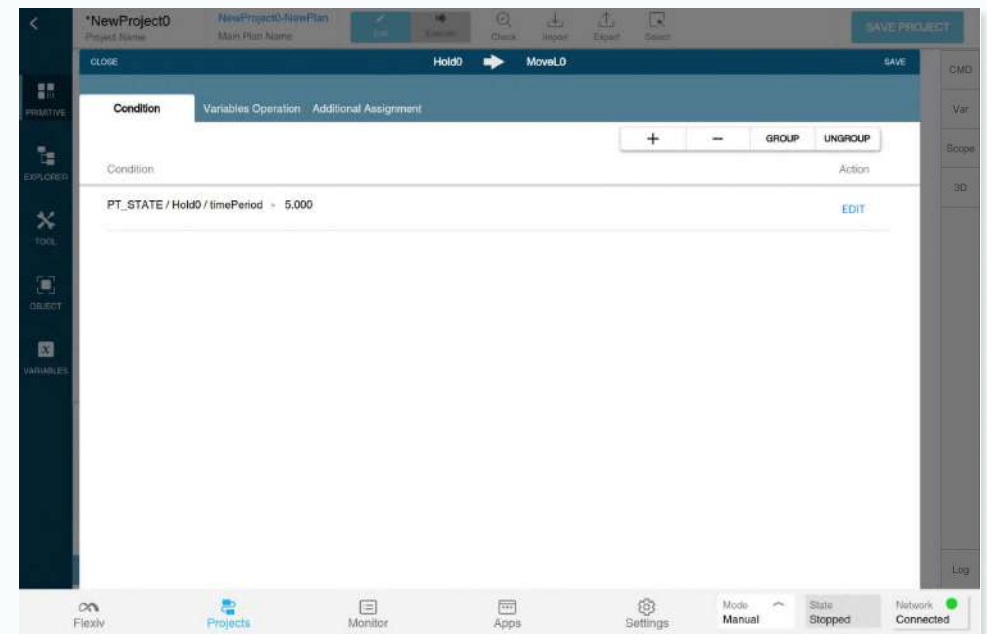
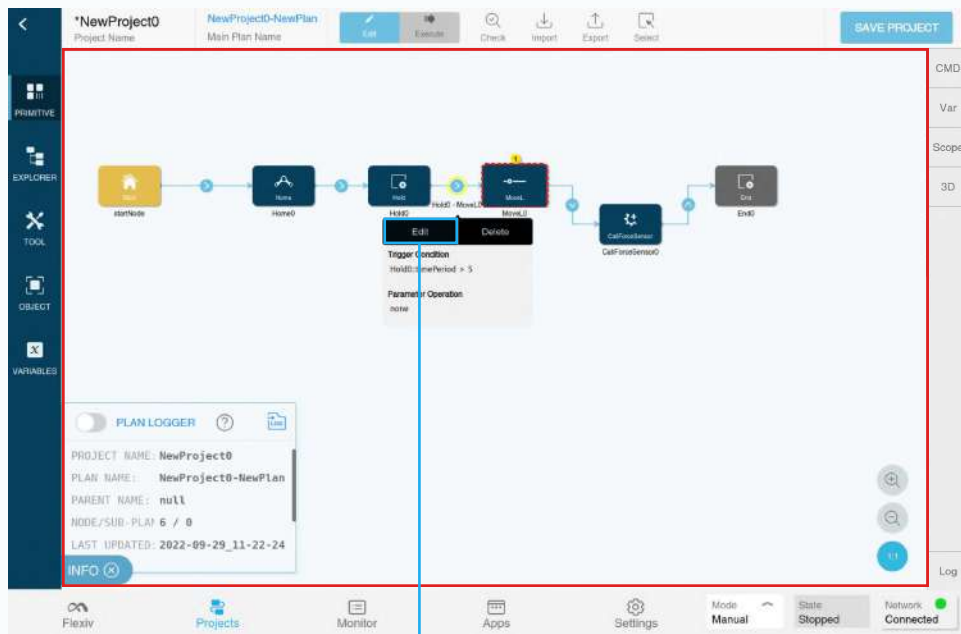
Tap  to **Edit** or **Delete** the transition between two nodes.

Tap  to enter the edit interface.

Condition: Used to edit the transition condition between two nodes

Variables Operation: Used to perform data operations before entering the next node

Additional Assignment: Used to perform additional data assignment before entering the next node

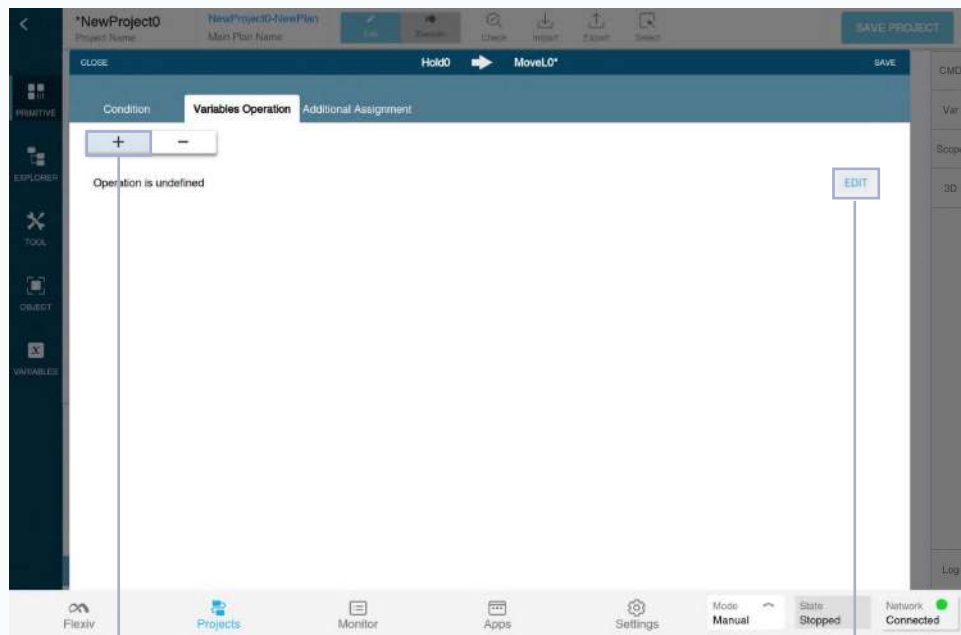


Tabs: Condition / Variables Operation / Additional Assignment

4.2 Edit Panel

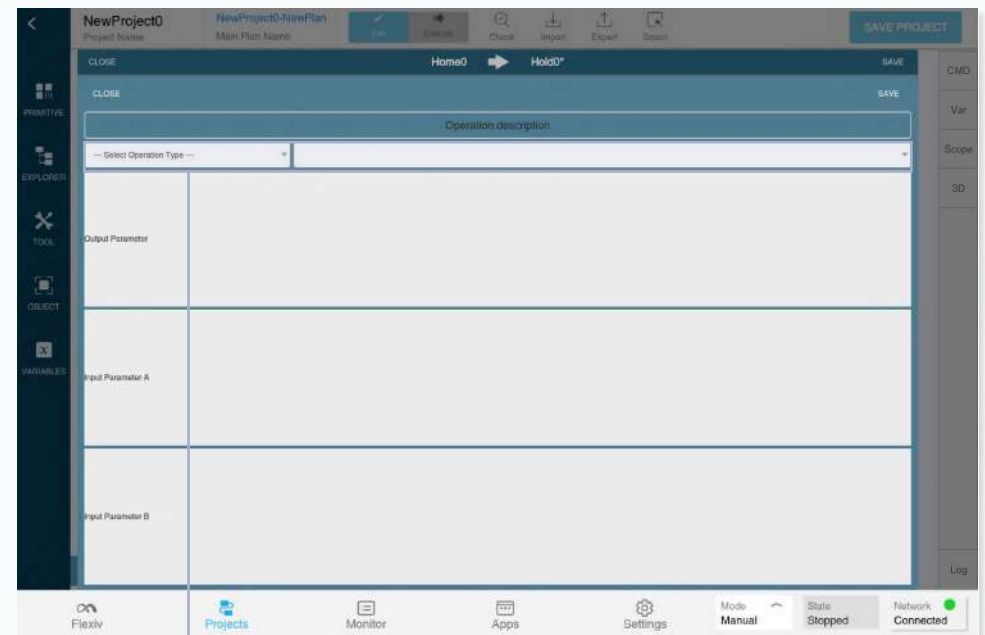
Part 2/3 - Primitive Canvas: Variables Operation

You can follow the steps below to edit variable operations.



1 Tap **+** to add an operation

2 Tap **EDIT** to enter the edit interface

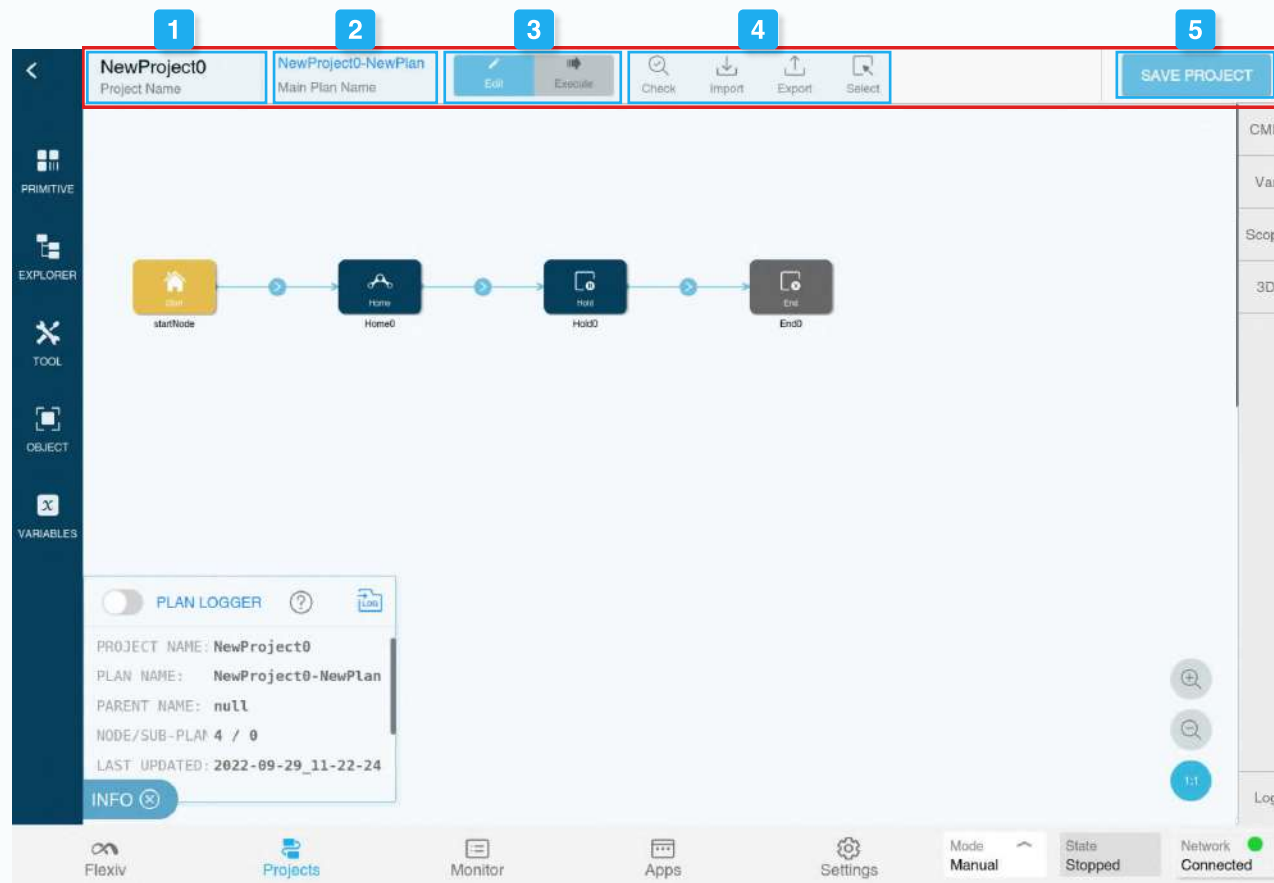


3 Select the operation type,
then select the parameter combination

4.2 Edit Panel

Part 3/3 - Header Menu Bar: Overview

The header menu bar contains fields/buttons that allow you to manage and execute your projects.



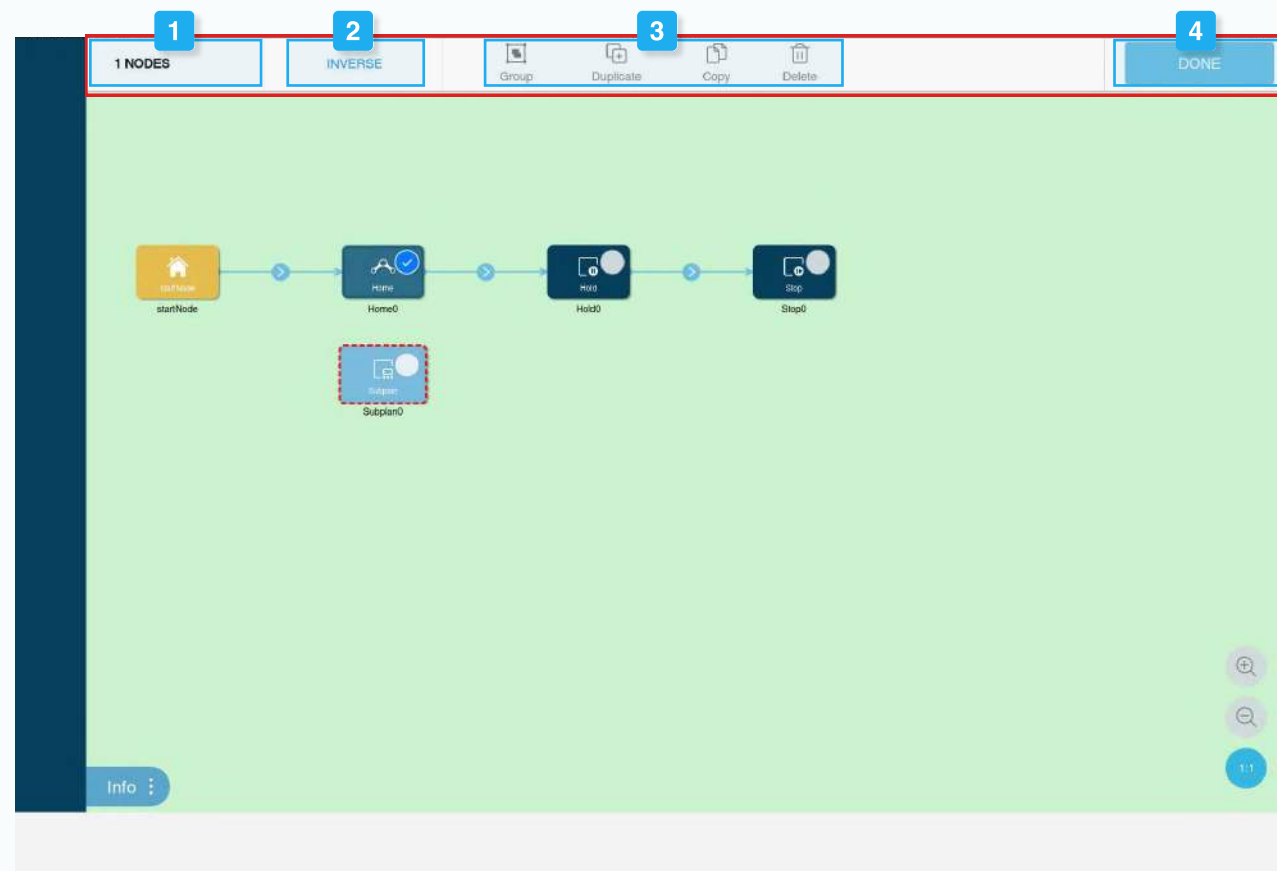
- 1** Project name
- 2** Main plan name (Tap to edit plan name and description)
- 3** Switch between edit mode and execute mode. You can edit the project in edit mode and execute the project in execute mode.
- 4**
 - Check the correctness of the project
 - Import project from local (Flexiv Elements)
 - Export project to local (Flexiv Elements)
 - Batch operations, including **Group**, **Duplicate**, **Copy**, and **Delete** (see next page for details)
- 5** Save project to local (Flexiv Elements)

4.2 Edit Panel

Part 3/3 - Header Menu Bar: Select mode

In edit mode, tap **Select** to enter select mode.

Select mode is mainly used to group multiple nodes together to complete batch operations.



Selected node



Unselected node



Subplan (grouped nodes)

1 Number of currently selected nodes

2 Invert selection

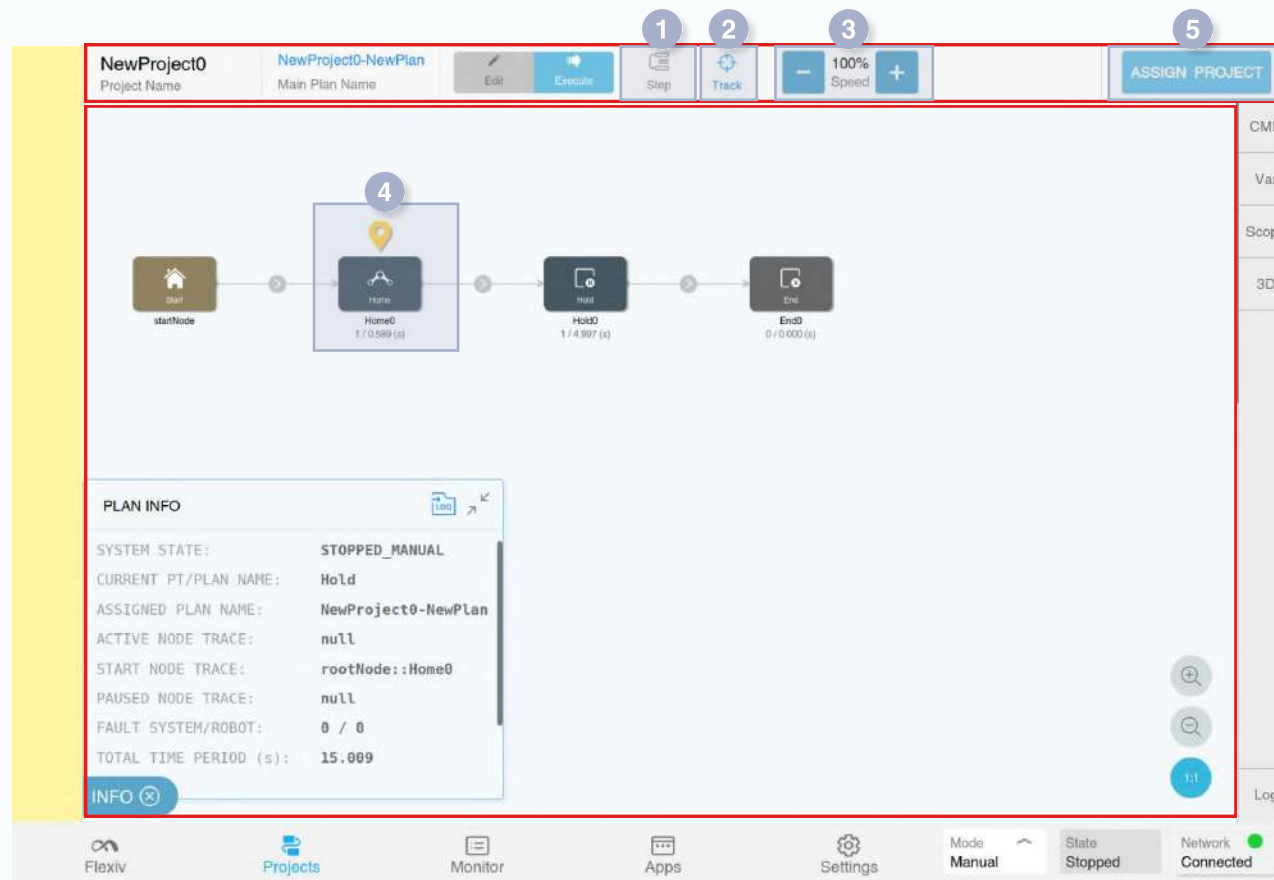
3 Batch operations

4 Exit select mode

4.3 Execute Panel

Part 1/3 - Header Menu Bar + Primitive Canvas

Before assigning a project:



- 1 Tap  to enter **single-step mode**:

Robot runs one node at a time.

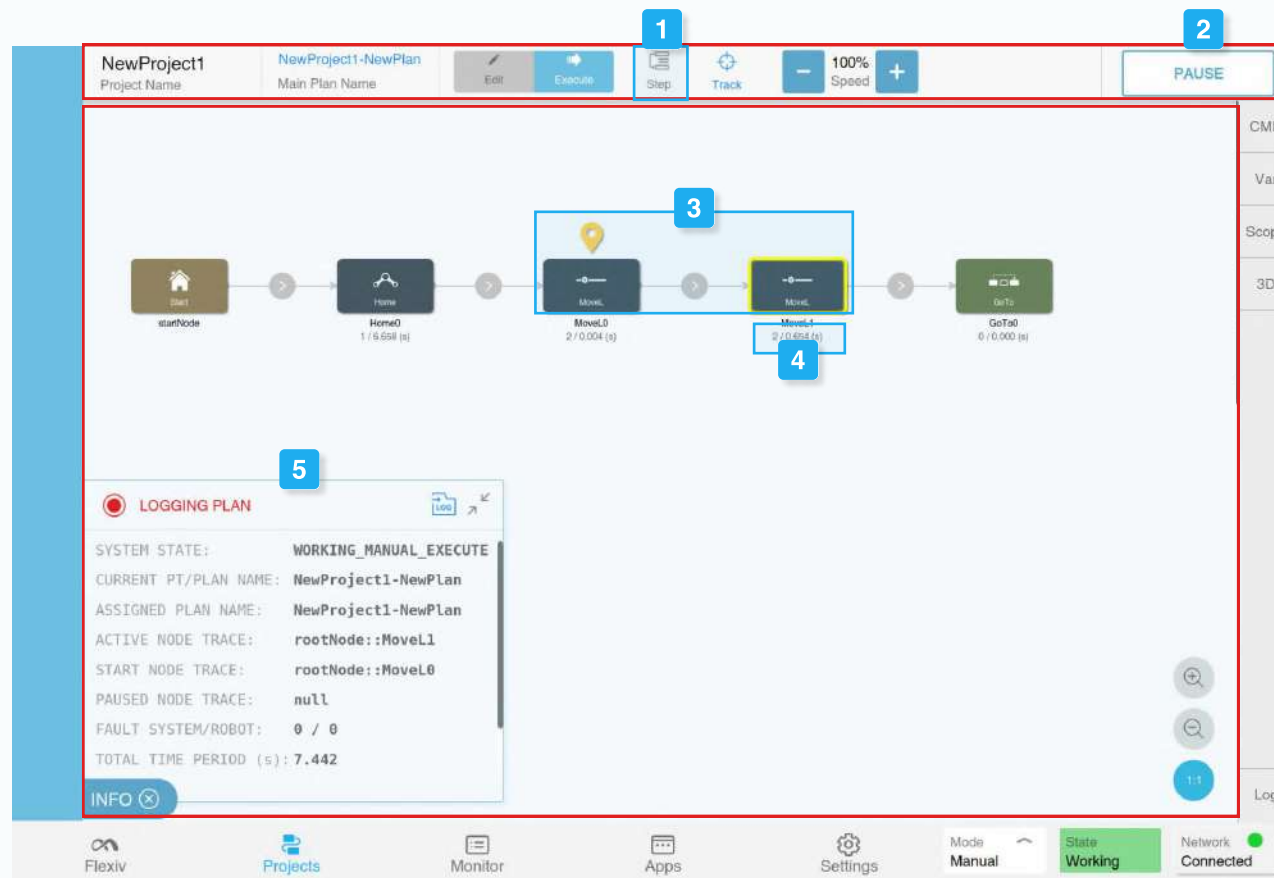
You need to press the **Start/Stop** button () on the Motion Bar to proceed to the next node.

- 2  track the nodes in the plan
Track
- 3 Adjust the motion speed while executing the project
 -  10% speed reduction per tap
 -  10% speed increase per tap
- 4 Define a new start node:
 1. Tap a node to define a new start point.
 2. **Start** will pop above the node, tap it.
 3.  will appear above the new start node.
- 5 Tap  to assign and execute the project.

4.3 Execute Panel

Part 1/3 - Header Menu Bar + Primitive Canvas

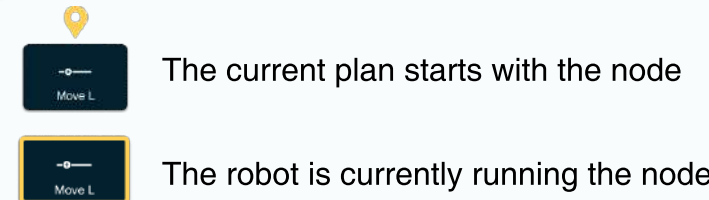
After assigning a project (**Working state** in the regular mode):



1 **Single-step mode** is off

2 Tap **PAUSE** to pause the plan

3 Color-coded states of nodes:

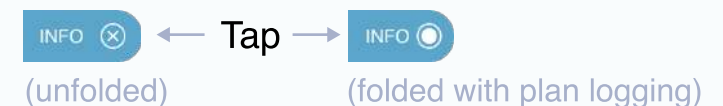


4 2 / 0.654(s): loop times / running time

loop times: The number of times the corresponding node is looped in the current plan.
running time: The total amount of time spent on running the node.

5 Information panel of the current plan

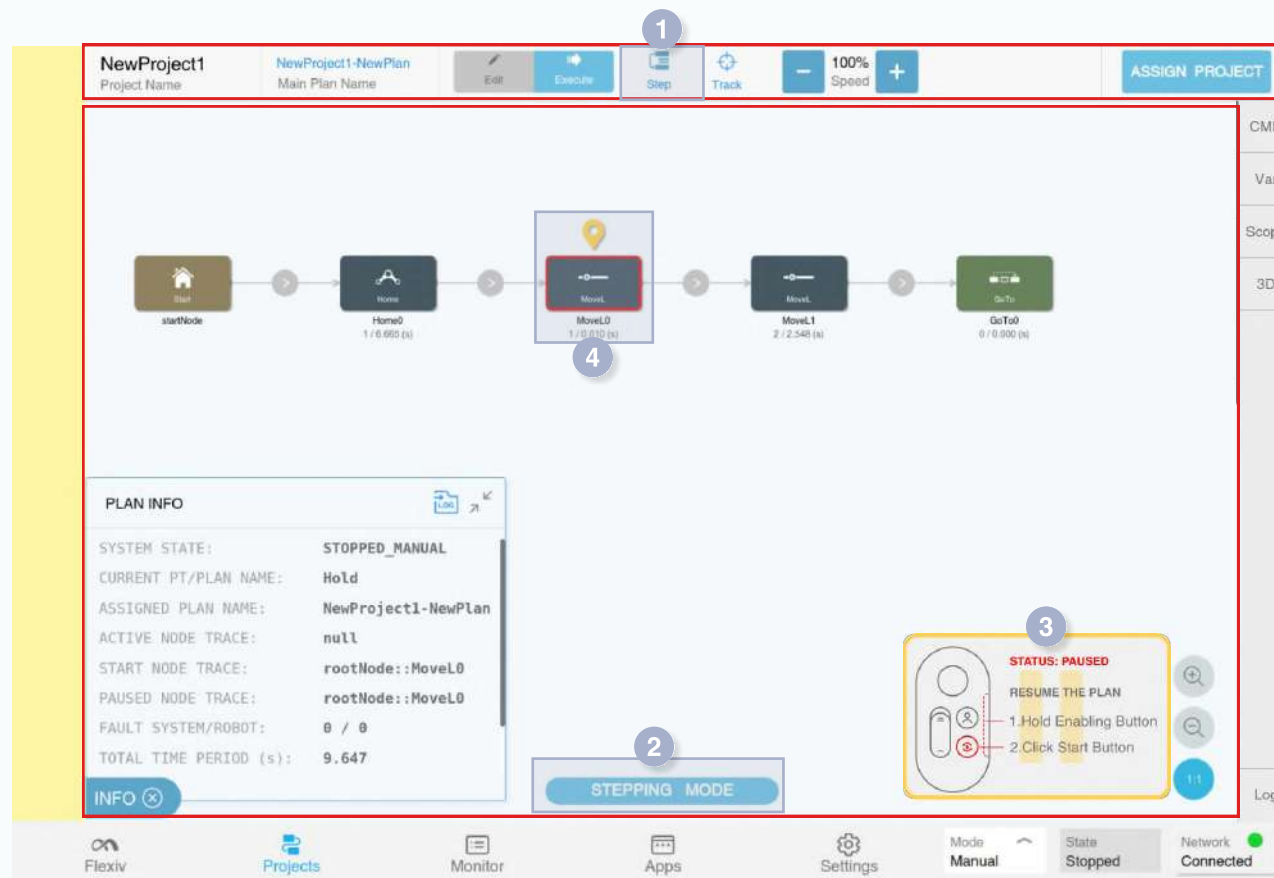
-  Plan logging is on
- Tap  to zoom out the panel
- Tap  to quickly redirect to **Apps > Plan Log Viewer** to view the most recently generated log file.
- Info label status



4.3 Execute Panel

Part 1/3 - Header Menu Bar + Primitive Canvas

After assigning a project (Stopped(Paused) state in the single-step mode):



1 Single-step mode is on

2 STEP MODE: Robot runs one node at a time. appears in the bottom center of the canvas

3 Follow the instructions to operate on the Motion Bar to resume the project

4 Color-coded states of nodes:

 Pause on the node

 Pause on the node and will re-run this node if the plan resumes

4.3 Execute Panel

Part 2/3 - Left Indicator Bar

The color of this bar indicates the current robot status and whether the enabling button on the Motion Bar is functioning.

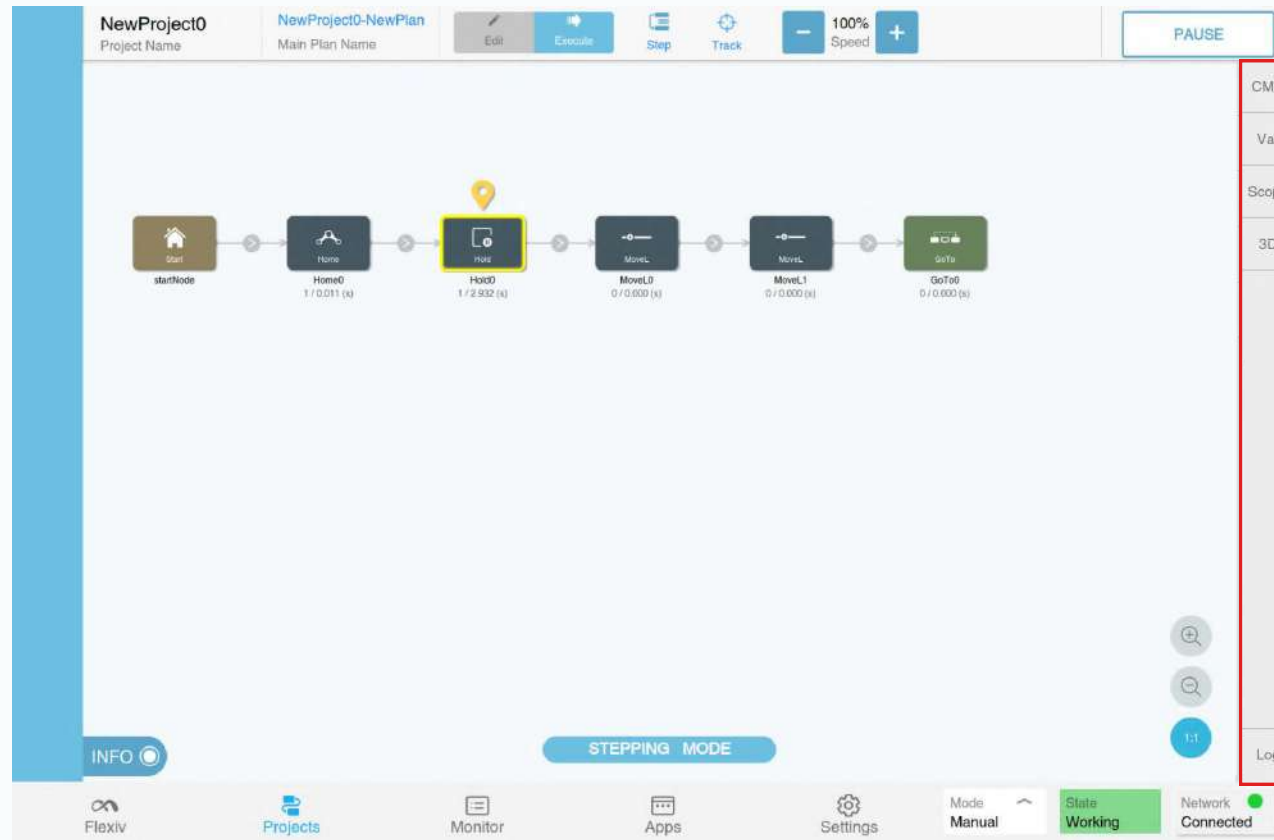


Color	Robot Status	Is Enabling Button Functioning?
	Init + E-Stop is pressed	No
	Manual Mode + Enabling Button is released	Yes
	Manual Mode + Enabling Button is pressed	Yes
	Auto Mode	
	Software Fault	No

4.3 Execute Panel

Part 3/3 - Right Menu Bar

The right menu bar is mainly used for monitoring the status of the plan and the robot through various real-time graphs and messages during execution.

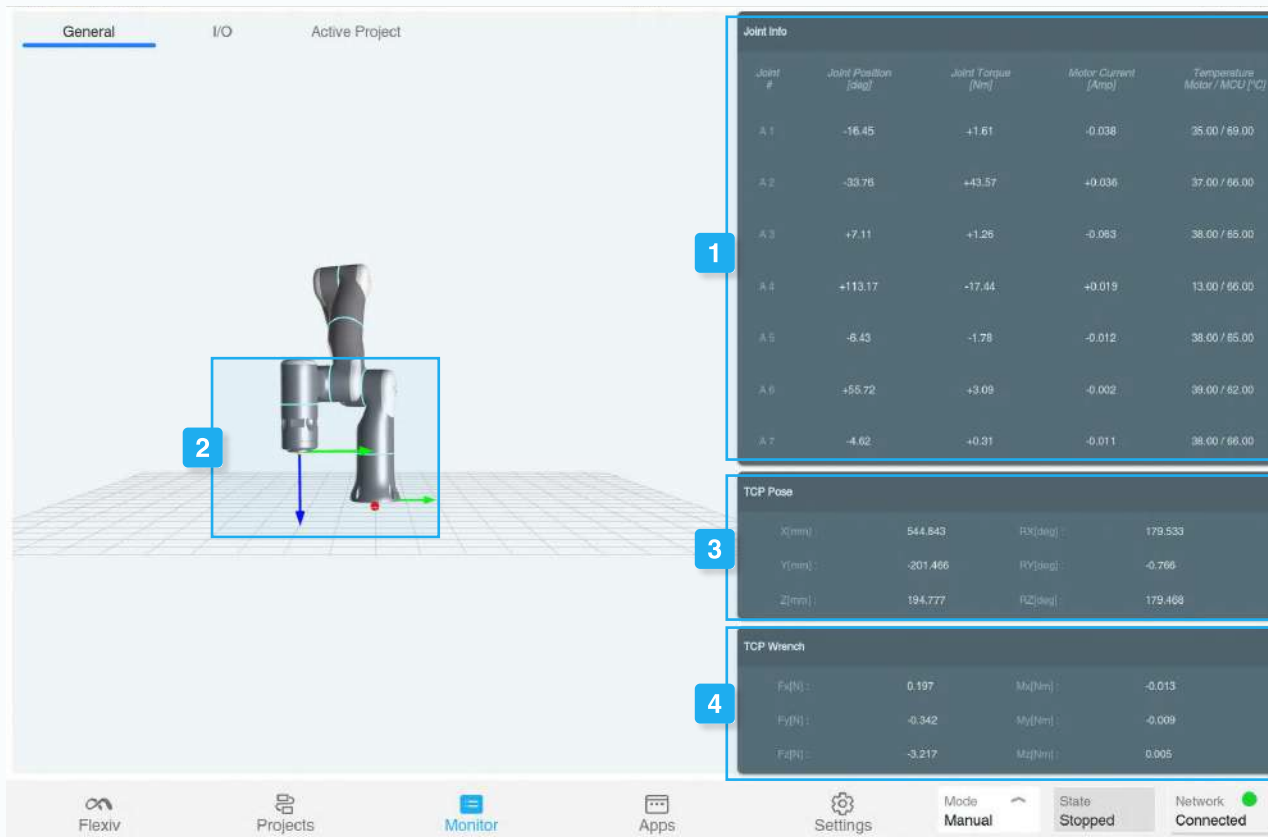


- 1 **CMD:** View the settings of commands (I/O command/tool command/AI command) of the current plan and quickly make edits.
- 2 **Var:** View the value of variables (global variable/project variable/plan variable) of the current project during execution.
- 3 **Scope:** View the real-time data charts of the robot's current operation (force, torque, velocity, acceleration, etc.) and analyze whether the robot is in good operating condition.
- 4 **3D:** View the real-time 3D visualization of the robot during execution.
- 5 **Log:** View the log data generated by the robot and AI computer programs during execution.

The **Monitor** tab allows you to monitor the status of the robot in real time. It contains three subtabs: **General**, **I/O**, and **Active Project**.

5.1 General

The **General** tab monitors the real-time conditions of the robot, including its **joint info**, **TCP pose**, and **TCP wrench**.

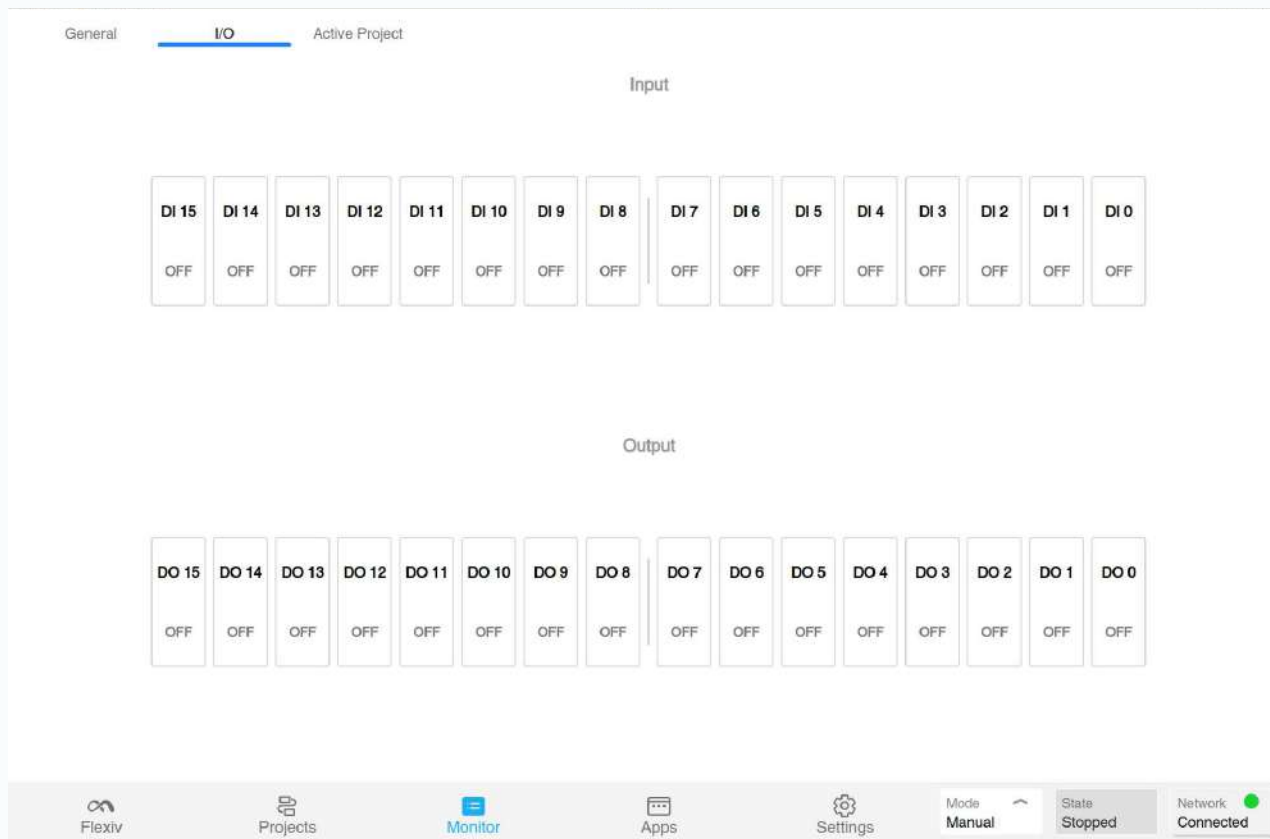


- 1 Joint Info:** A1-A7 joints' real-time position, torque, motor current, and temperature.
- 2 XYZ Axes:** X, Y, Z axes are represented as red, green, and blue, respectively.
- 3 TCP Pose:** TCP's real-time position relative to Base Coordinate.
- 4 TCP Wrench:** TCP's each direction's real-time force and moment in Cartesian Coordinate System.

The **Monitor** tab allows you to monitor the status of the robot in real time.

5.2 I/O

The **I/O** tab monitors the real-time status of the **GPIOs** on the Control Box.



The screenshot displays the 'I/O' tab in the Flexiv software interface. At the top, there are tabs for 'General', 'I/O' (which is selected), and 'Active Project'. Below the tabs, the interface is divided into two main sections: 'Input' and 'Output'.

Input Section: This section contains 16 digital input pins, labeled DI 15 down to DI 0. Each pin has a corresponding status indicator below it, all of which are currently set to 'OFF'.

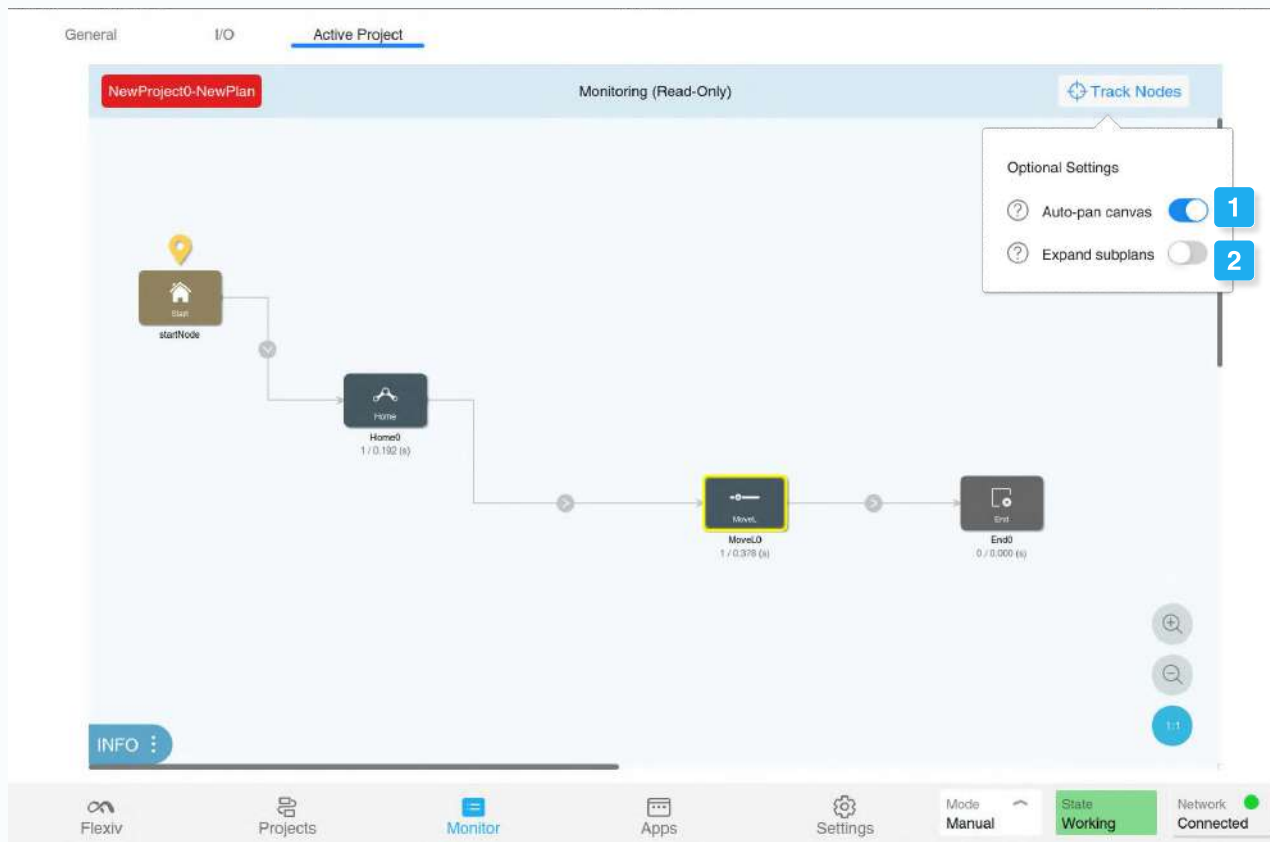
Output Section: This section contains 16 digital output pins, labeled DO 15 down to DO 0. Each pin has a corresponding status indicator below it, all of which are currently set to 'OFF'.

At the bottom of the interface is a navigation bar with several icons: Flexiv logo, Projects, Monitor (highlighted in blue), Apps, Settings, Mode (set to Manual), State (set to Stopped), and Network (set to Connected with a green indicator).

The **Monitor** tab allows you to monitor the status of the robot in real time.

5.3 Active Project

The **Active Project** tab monitors the real-time status of the active (running) projects.



1 Auto-pan canvas (enabled by default):

- Automatically pan the canvas when nodes are out of canvas bounds.

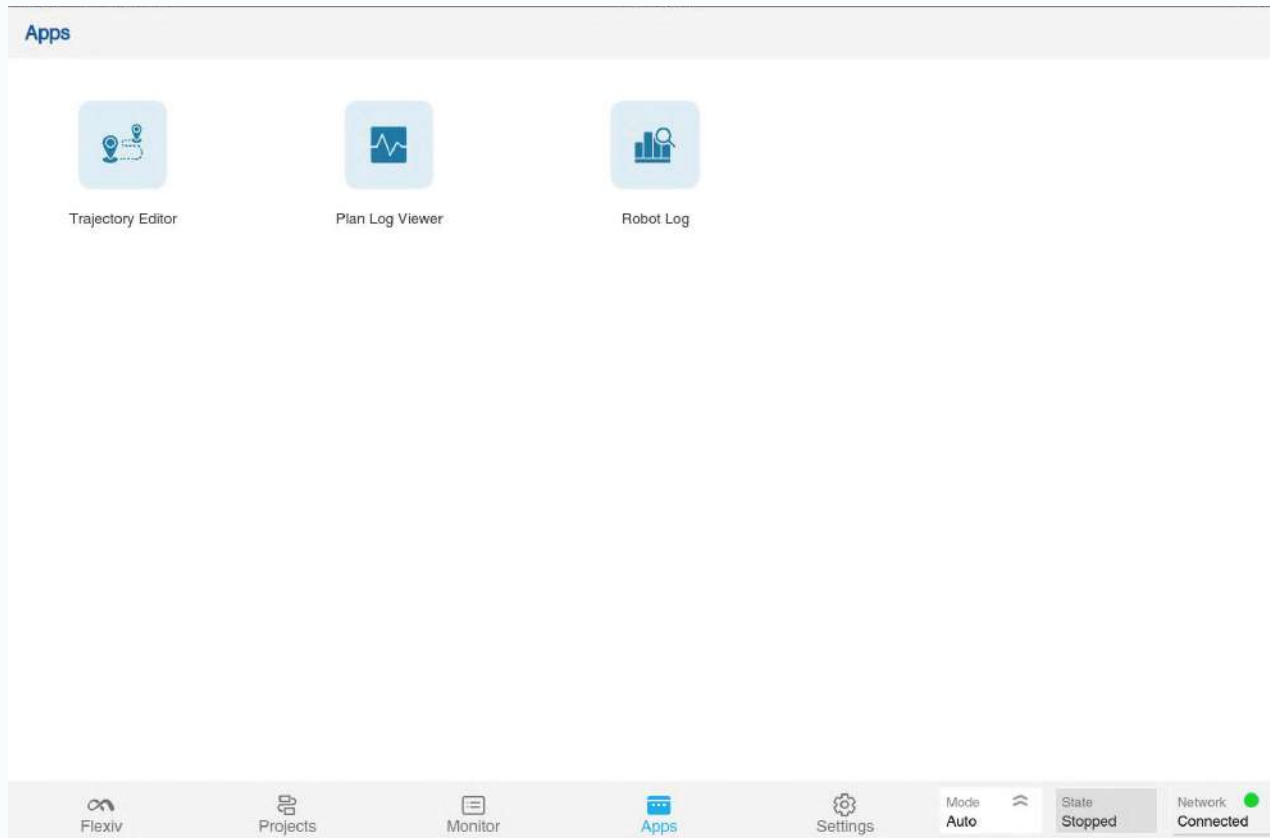
2 Expand subplans (disabled by default):

- Automatically expand the subplans when running Plan node(s). You can also manually return to the parent plan by tapping **BACK TO PARENT**.

***Note:** In **Remote Mode**, the active projects are assigned to the robot through RDK or PLC.

6.1 Introduction

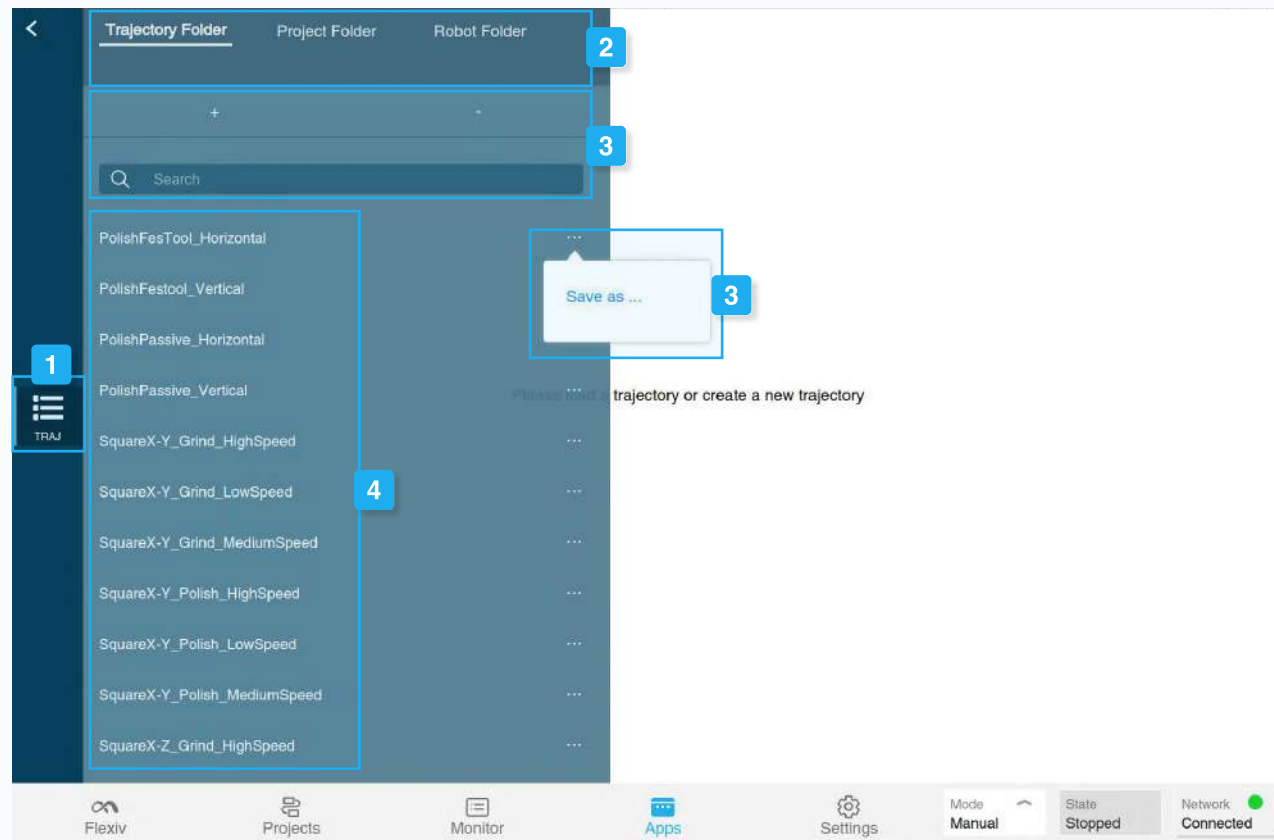
The **Apps** tab provides you with commonly used applications that contain a variety of functions such as trajectory editing and log viewing. Currently, three applications are available in this tab: **Trajectory Editor**, **Plan Log Viewer**, and **Robot Log**.



6.2 Trajectory Editor

The **Trajectory Editor** app is designed to create and edit robot trajectories that can be used in projects. You can refer to the ***Trajectory Editor User Manual*** for detailed instructions.

Part 1/2 - File Menu



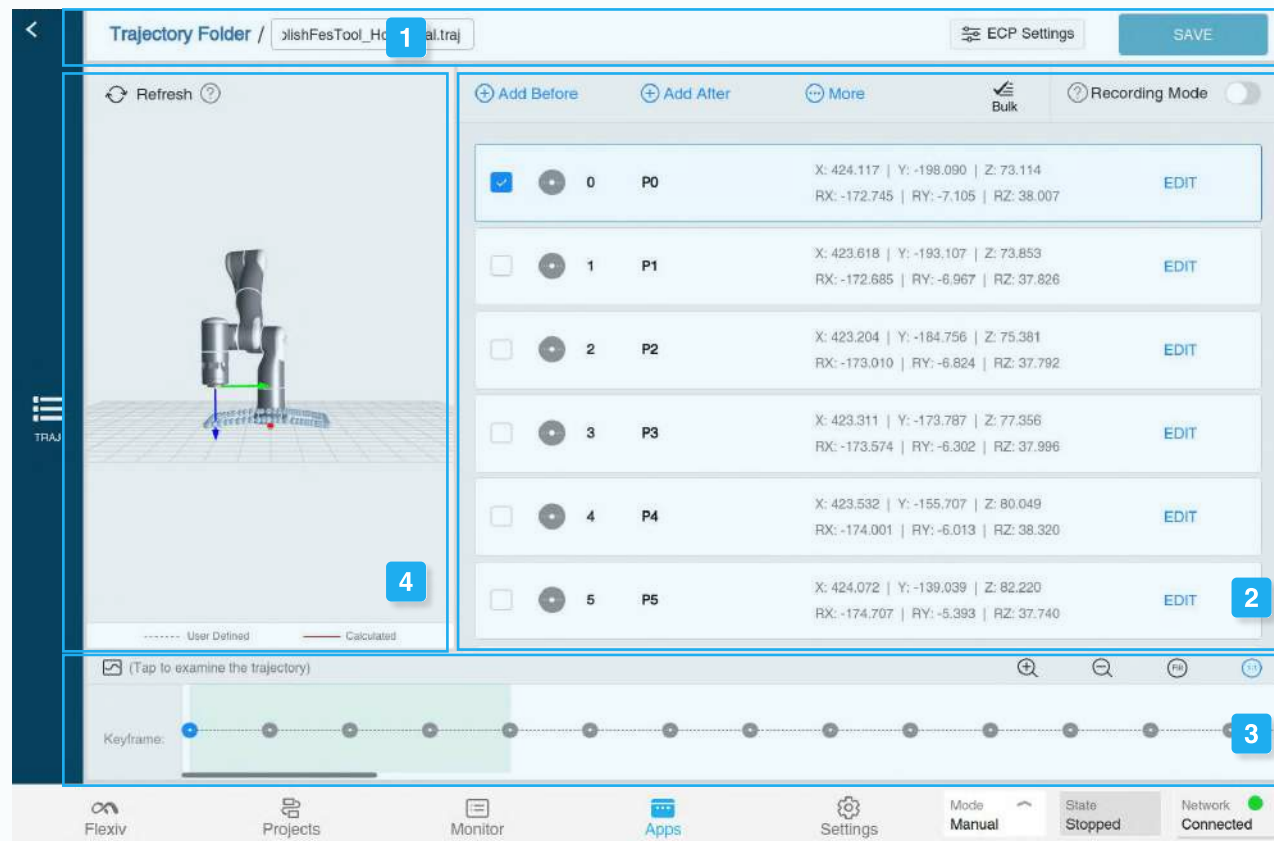
Flexiv Elements is connected to the robot

- 1 Tap  to open the file menu.
- 2 Folder tabs: **Trajectory Folder** tab, **Project Folder** tab, and **Robot Folder** tab. **Robot Folder** tab is only available when Flexiv Elements is connected to a robot.
- 3 Actions:
 - Create new:** Tap  to create a new trajectory file.
 - Delete:** Tap  to enter the delete mode.
 - Search:** Type in keywords in the search bar  to quickly search for files.
 - Save as:** Tap  to store a copy of a file to a desired project.
- 4 List of trajectory files: A list view of all the trajectory files in a project or a folder.

6.2 Trajectory Editor

The **Trajectory Editor** app is designed to create and edit robot trajectories that can be used in projects. You can refer to the ***Trajectory Editor User Manual*** for detailed instructions.

Part 2/2 - Main Interface



1 Title Bar: 1. Folder / file name (leftmost); 2. **ECP Settings**: Tap to add an ECP property to a trajectory file or remove an ECP property from a trajectory file; 3. **SAVE**: Tap to save the changes you make.

2 Action & List Panel: 1. **Add Before & Add After**: Tap to add all types of patterns; 2. **Recording Mode**, once enabled, can be used to add pattern type **Waypoint**; 3. **Bulk**: Tap to enable/disable the bulk select function; 4. **More**: Tap to show more functions; 5. The **list panel** displays a list of all items in the trajectory file.

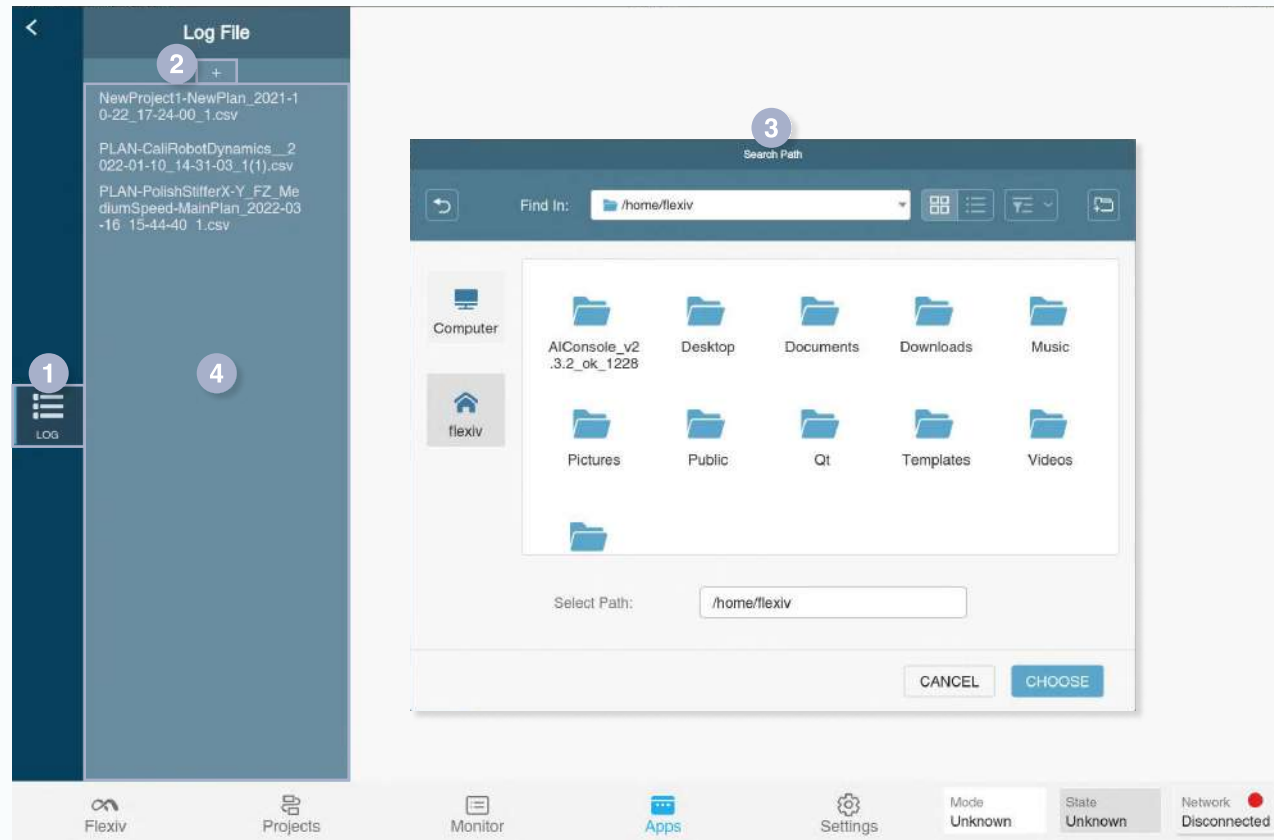
3 Track Panel: There is a one-to-one correspondence between keyframes and items in the list panel.

4 Preview Panel: Preview the sequence of waypoints or trajectory patterns in the current trajectory.

6.3 Plan Log Viewer

The **Plan Log Viewer** app is designed to display plan log files. It allows you to replay the robot's operations and visualize the recorded data. It provides an efficient way for robot performance diagnosis and troubleshooting.

Part 1/2 - Import Log File



Import a log file

- 1 Tap 
- 2 Tap +
- 3 Choose the log file from UI runtime directory
- 4 File name will appear in this panel, tap to view the historical data (see next page)

6.3 Plan Log Viewer

Key robot operation data (e.g. position, velocity, torque, acceleration, etc.) are visualized in time plot graphs, allowing you to analyze the data in detail and check any performance problems the system detects.

Part 2/2 - Visualize Recorded Data



1 Select the content you want to display in panel 2 from the drop-down menu

2 Switch between different views:

-  Default view (displays panel 1 & 2 in full)
-  Main panel view (displays panel 2 in full)
-  Robot 3D view (tap the icon to add panel 3 to default view or main panel view)

When you drag the cursor in the frames of the main panel, the 3D view will display the real-time robot configuration quality score accordingly.

3 Tap  and the icon will change to  to indicate that you can zoom in the corresponding frame to get a detailed view of the amplitude of the wave.

- **Measured:** The actual value measured by the system when the robot is running.
- **Forward:** The command value computed by the system using forward dynamics before the robot runs. This value is mainly used for debugging purposes.
- **Desired:** The expected value calculated by the system before the robot runs.

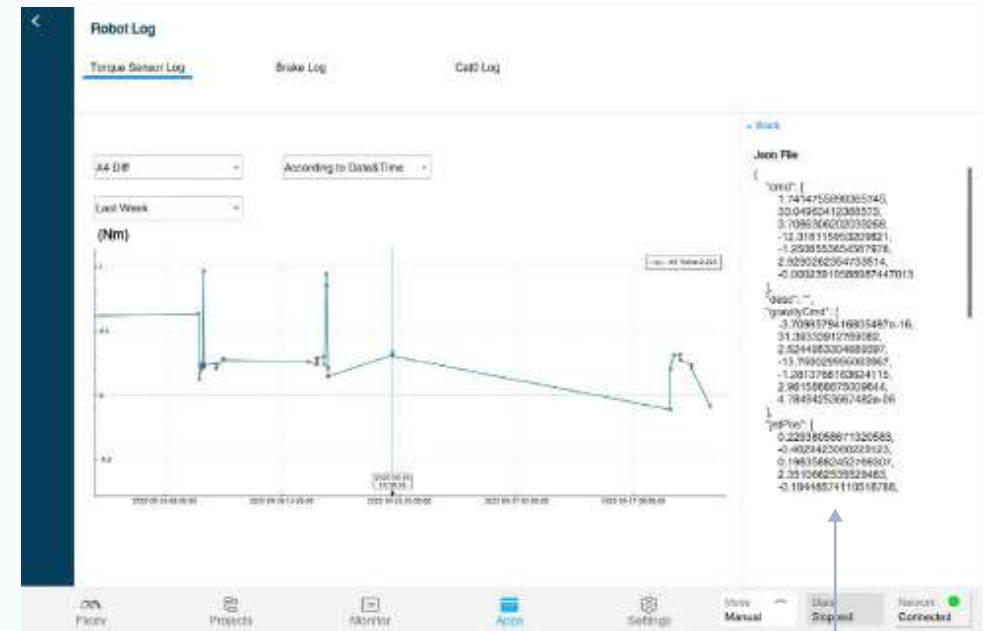
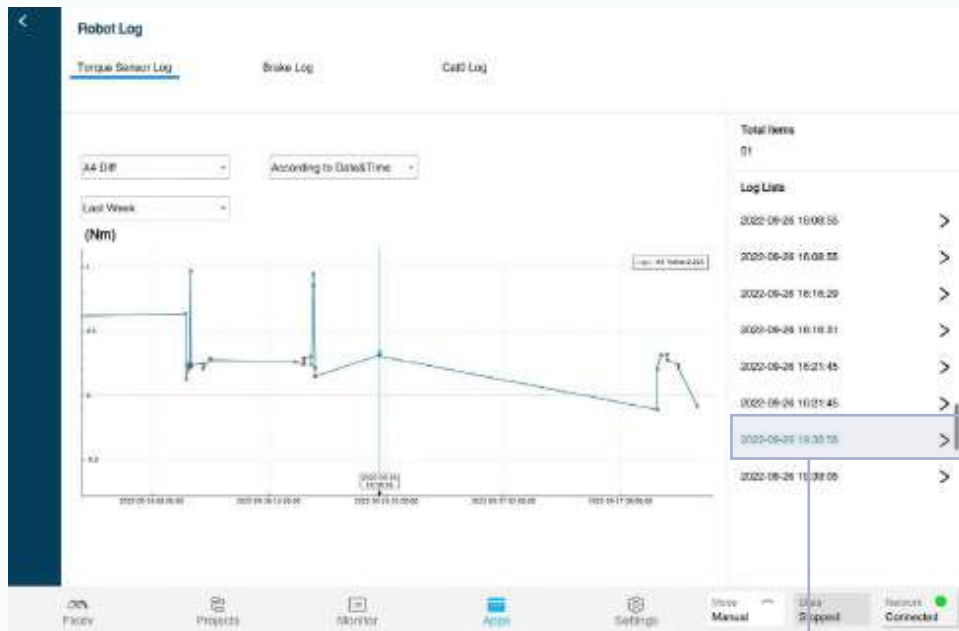
***Note:** Report and Plot Data are not available for now.

6.4 Robot Log

The **Robot Log** app displays three types of logs, namely, **Torque Sensor Log**, **Brake Log**, and **CAT0 Log**.

Tab 1/3 - Torque Sensor Log

Torque Sensor Log records sensor data in given circumstances and displays torque sensor deviation for the purpose of evaluating the accuracy of the sensor. You can query data by using the filters 1. Date & Time, 2. Brake times, and 3. CAT0 times.



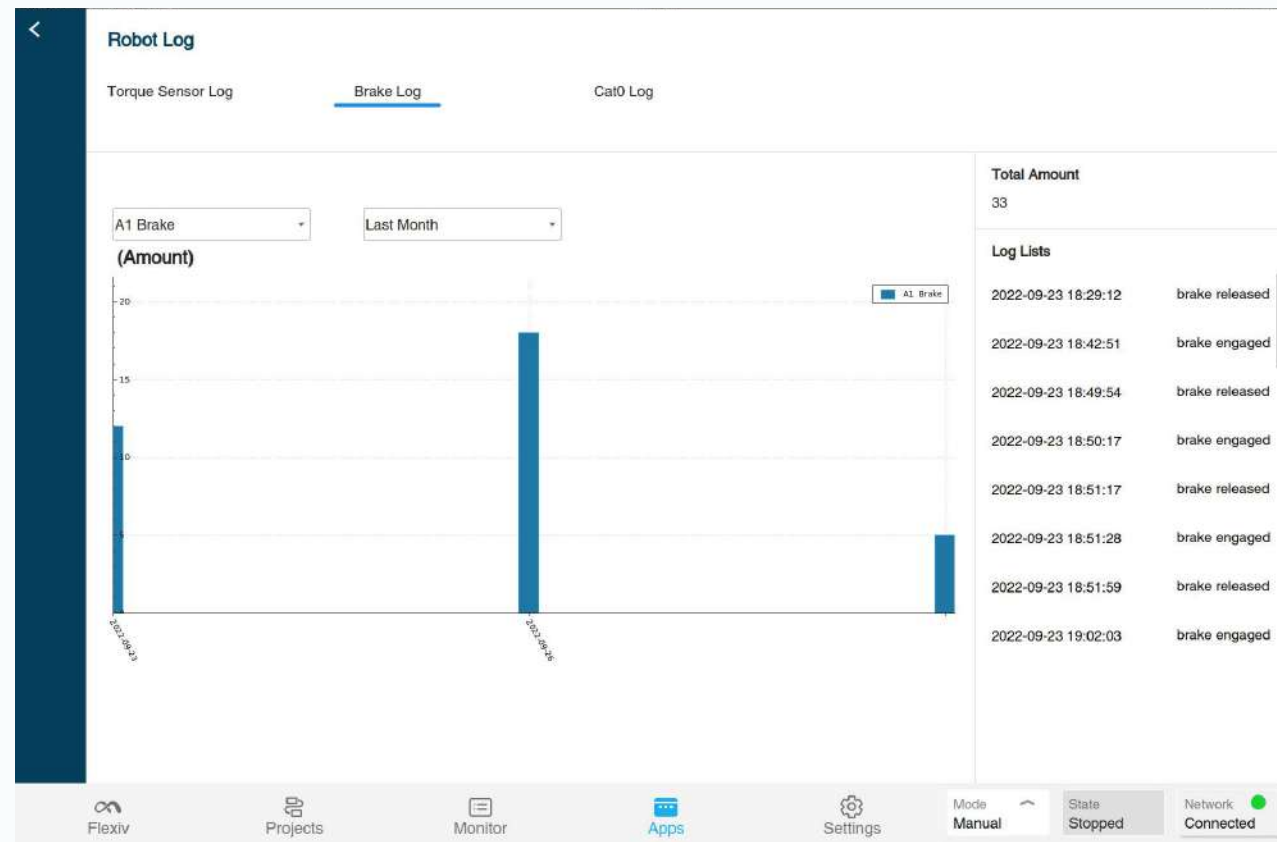
Tap to check the raw data

6.4 Robot Log

The **Robot Log** app displays three types of logs, namely, **Torque Sensor Log**, **Brake Log**, and **CAT0 Log**.

Tab 2/3 - Brake Log

Brake Log records the time each time the brake is released or engaged, and calculates the total amount. You can query data according to Date & Time.



6.4 Robot Log

The **Robot Log** app displays three types of logs, namely, **Torque Sensor Log**, **Brake Log**, and **CAT0 Log**.

Tab 3/3 - CAT0 Log

CAT0 error message is logged when CAT0 is triggered. You can query data according to Date & Time.

Robot Log

Torque Sensor Log

Brake Log

Cat0 Log

Last Month

Date & Time	Log	
2022-09-26 15:47:30	[ROBOT]Reached tcp force/torque limit. TCP force was violated.	More
2022-09-26 16:16:29	[ROBOT]Reached tcp force/torque limit. TCP force was violated.	More
2022-09-27 09:51:48	[SYSTEM]Joint SOL_A7 violated joint check on sequence number mismatch in communication.	More
2022-09-27 10:02:49	[SYSTEM]Joint SOL_A3 violated joint check on sequence number mismatch in communication.	More

Flexiv

Projects

Monitor

Apps

Settings

Mode

Manual

State

Stopped

Network

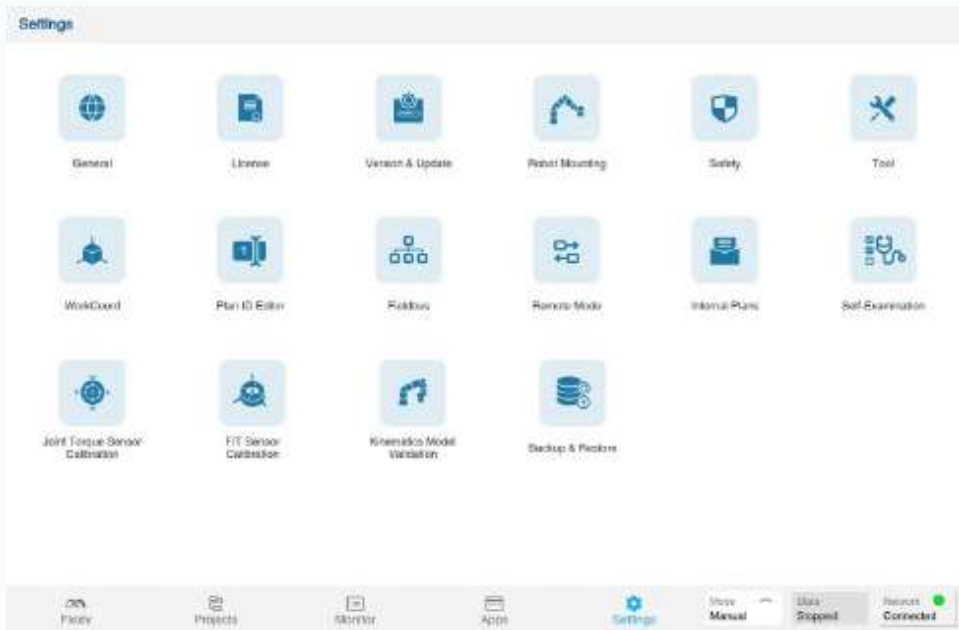
Connected

7.1 Introduction

In the **Settings** tab, you can configure and fine-tune the robot system's basic, safety, communication and other settings.

Setting Options

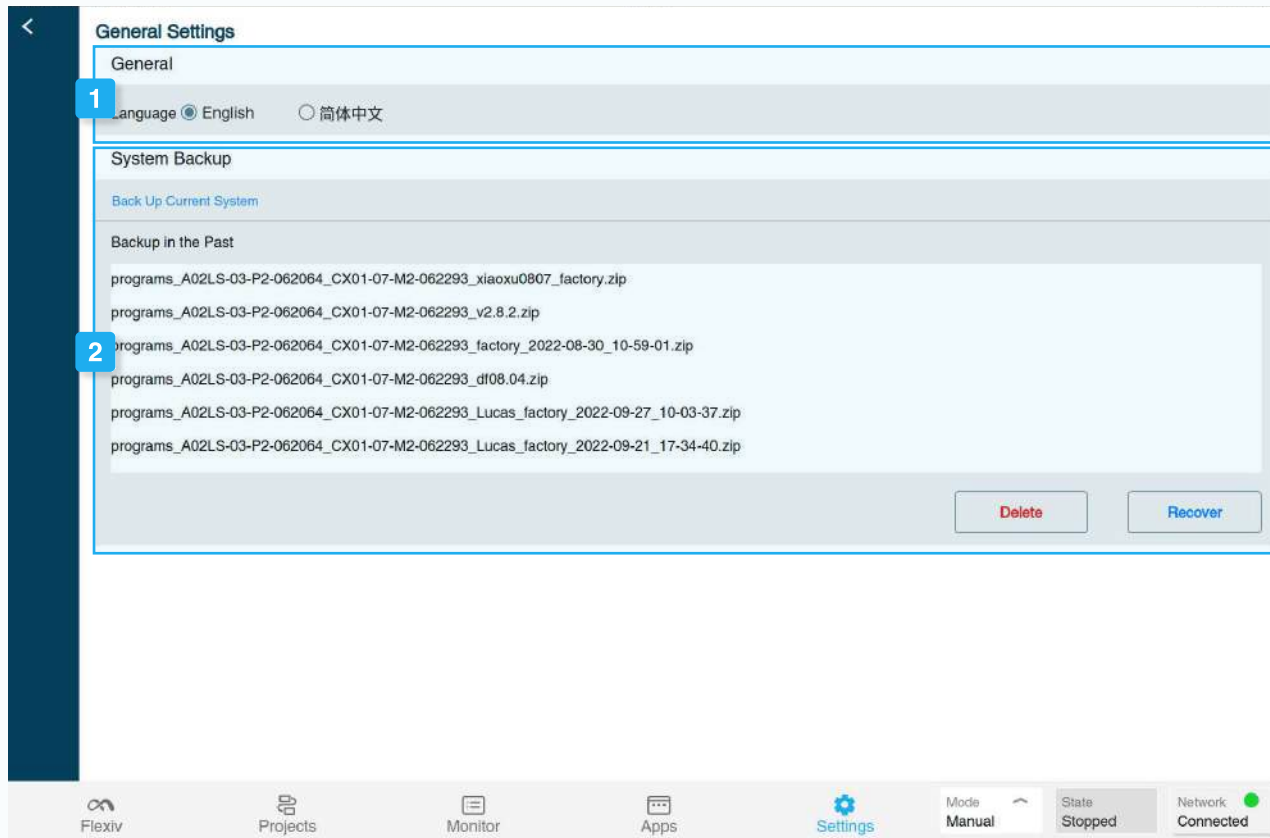
General / License / Version & Update / Robot Mounting / Safety / Tool / WorkCoord / Plan ID Editor / Fieldbus / Remote Mode / Internal Plans / Self-Examination / Joint Torque Sensor Calibration / F/T Sensor Calibration* / Kinematics Model Validation / Backup & Restore



***Note:** F/T Sensor Calibration is only available when Flexiv Elements is connected to a Rizon 4S robot.

7.2 General

In this tab, you can configure general system settings, including **Language** and **System Backup**.



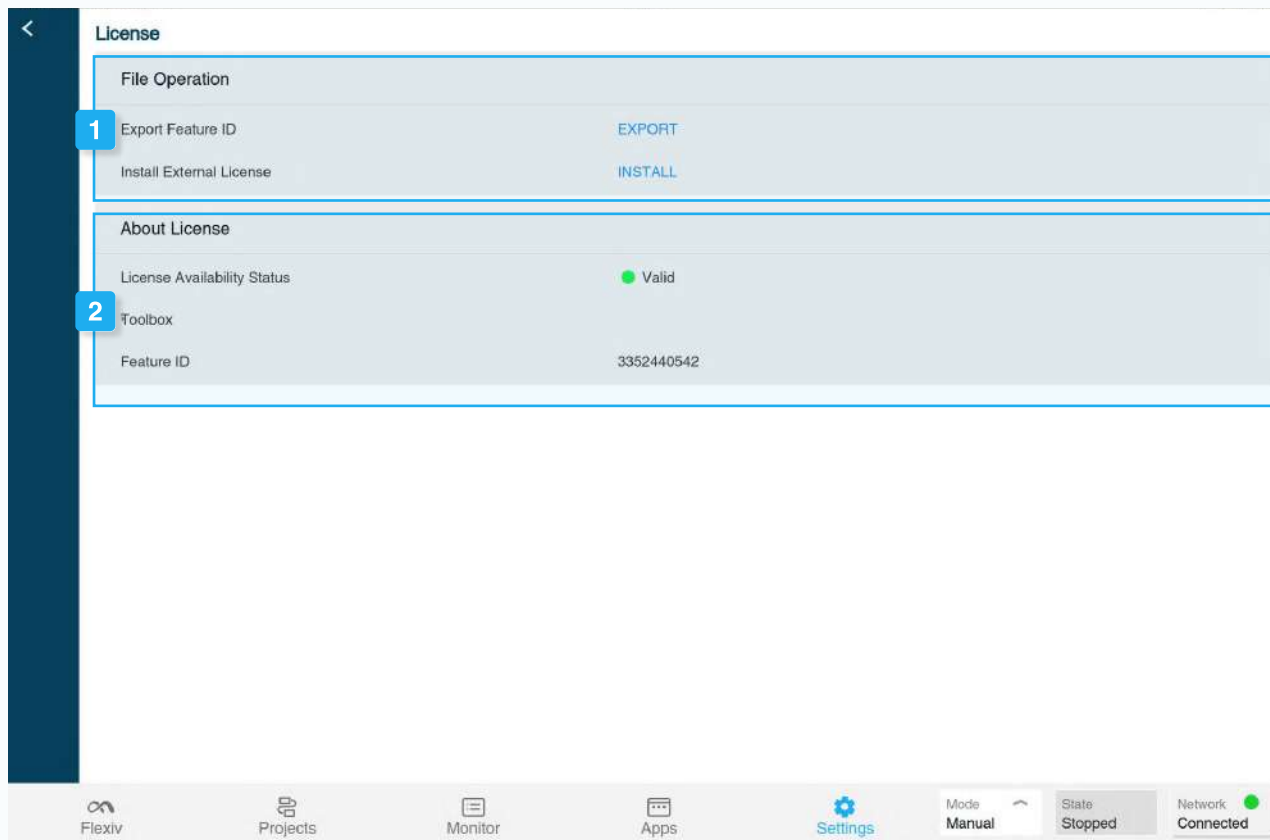
1 Language: Flexiv Elements currently supports two languages: **English** and **Simplified Chinese**.

2 System Backup:
Tap **Back Up Current System** to back up the whole files of the robot.

Choose a file from the **Backup in the Past** list, then tap **Delete** to delete the file or tap **Recover** to recover the file.

7.3 License

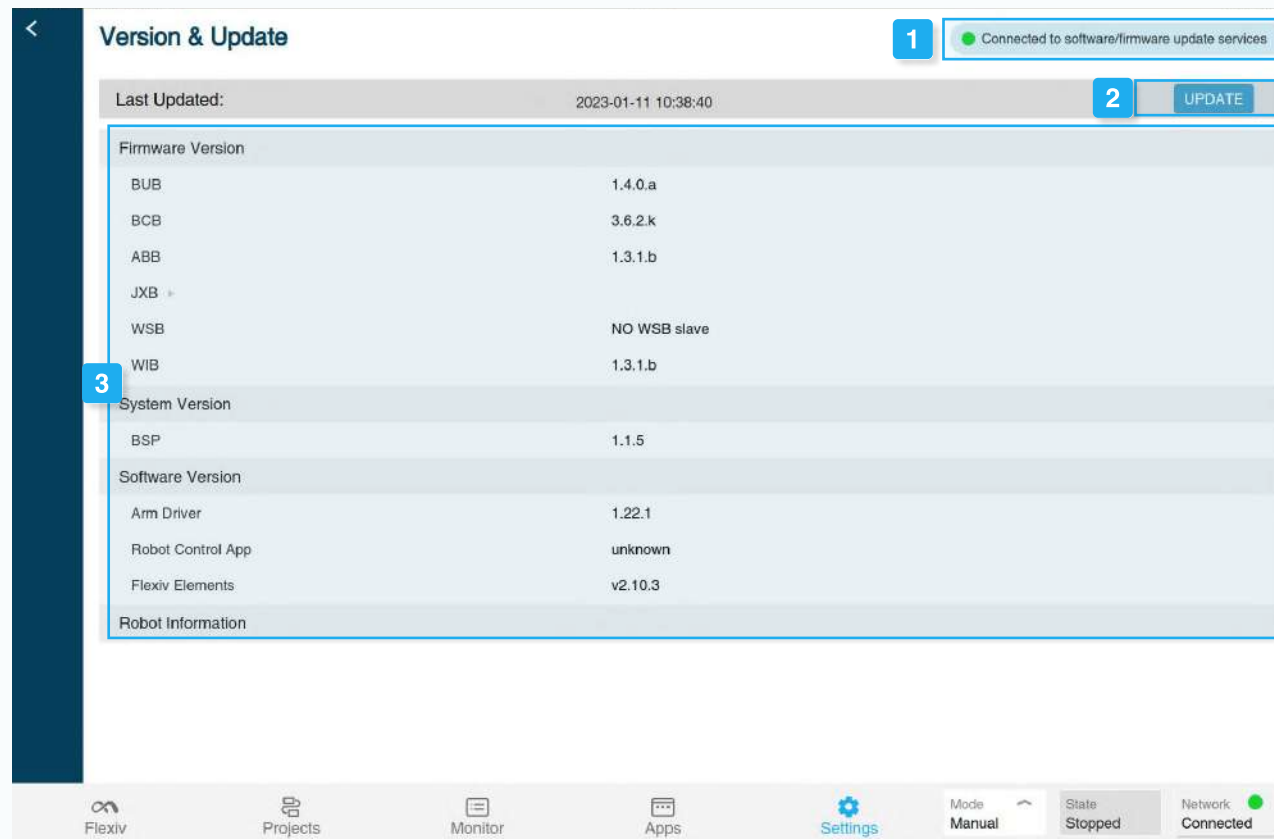
In this tab, you can configure robot license and view the license validity status.



- 1 **File Operation:** You can install purchased licenses on Flexiv Elements, and export the feature ID.
- 2 **About License:** Information of license status, purchased primitives, and feature ID (used to identify different hardware platforms).

7.4 Version & Update

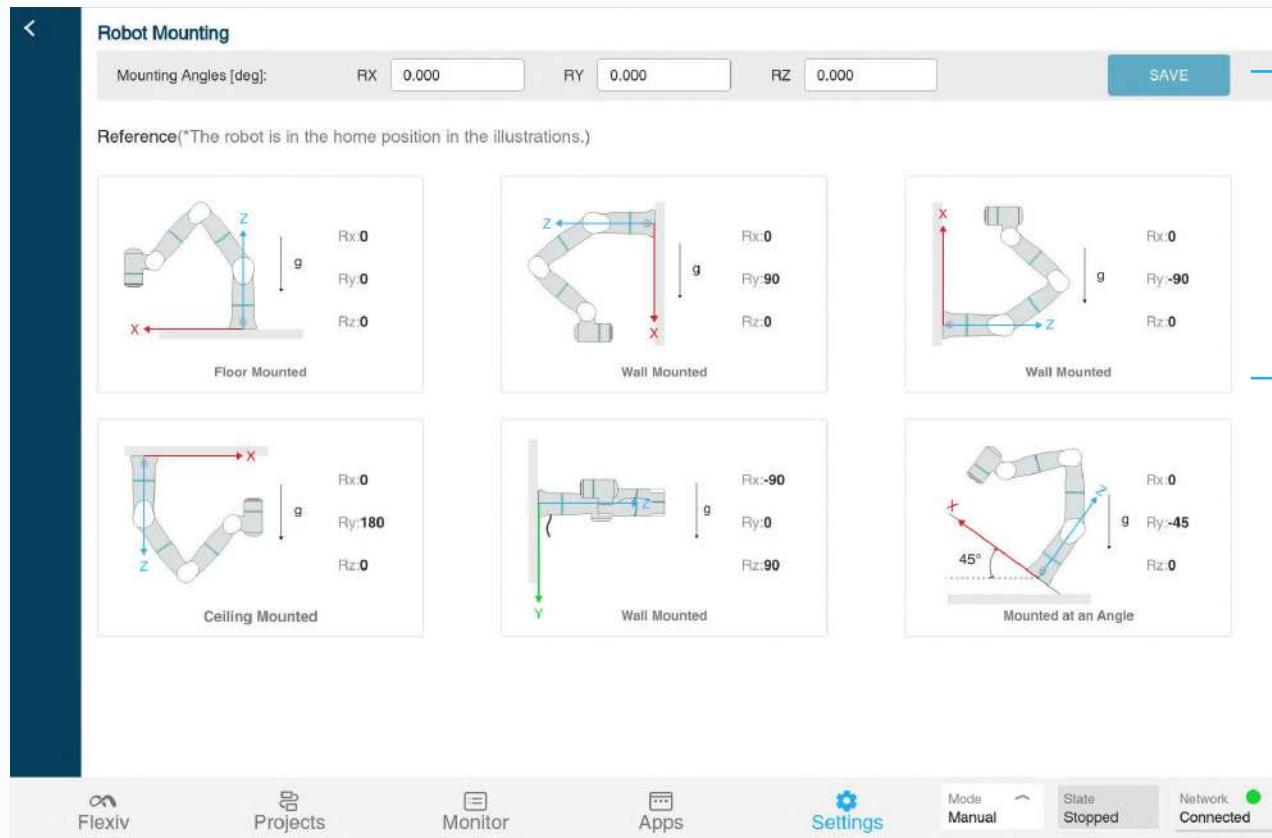
In this tab, you can view the robot info, software/firmware version, and update software and firmware.



- 1 Status of update service:
 1.  : Update service is ready;
 2.  : Update service is not ready.
- ***Note:** Keep Flexiv Elements and the Control Box connected to enable the update service.
- 2 Tap **UPDATE** and choose a file package to update the software/firmware.
- 3 Firmware version, system version, software version, and robot information of the currently connected robot.

7.5 Robot Mounting

In this tab, you can view and configure the mounting angles of the robot.



1 View/configure the mounting angles

2 Illustrations of mounting examples

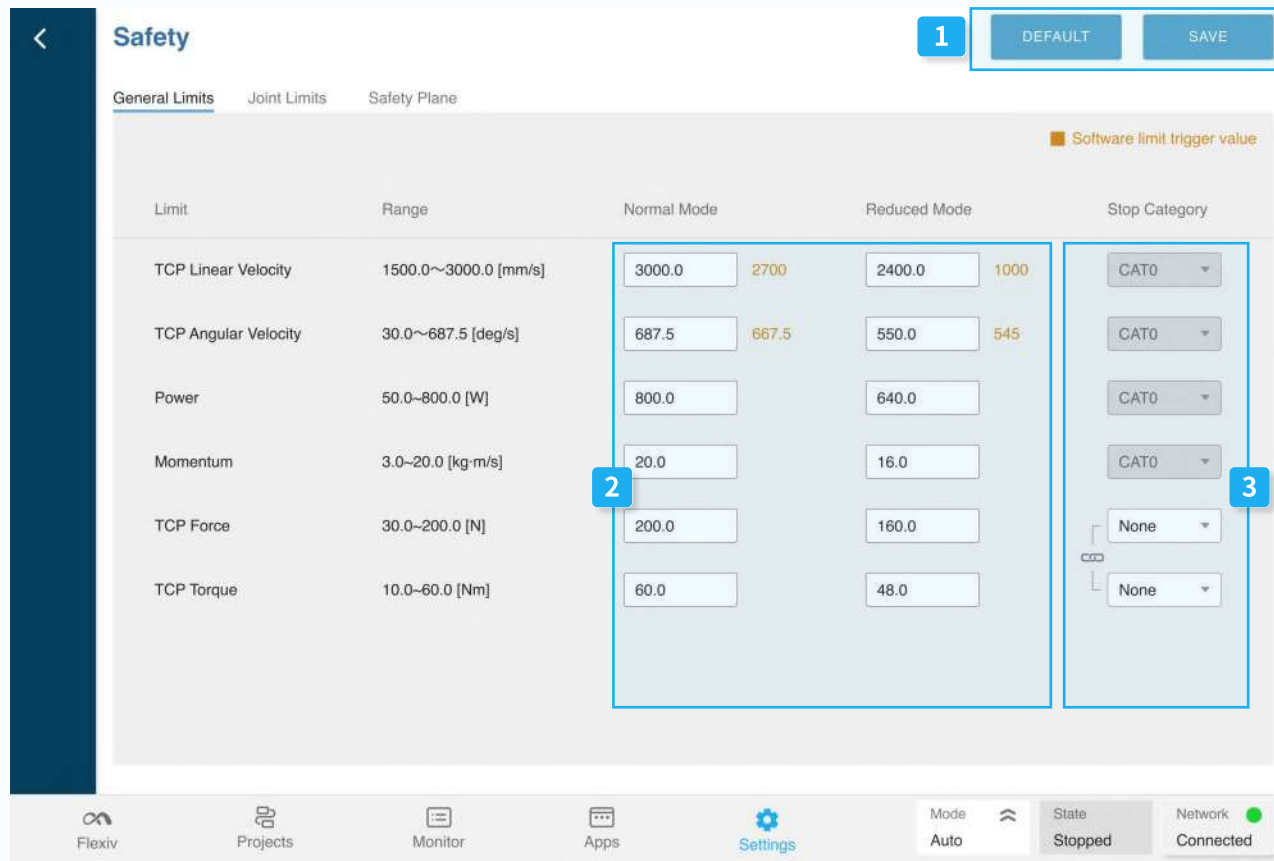
***Note:** If the robot is mounted on a flat table or floor, no change is needed in this interface. However, if the robot is ceiling mounted, wall mounted, or mounted at an angle, you need to configure the mounting angles.

7.6 Safety

In this tab, you can configure safety-related parameters (password is required to save changes).

Tab 1/3 - General Limits

In the **General Limits** tab, you can configure TCP-related limits and power limits for Normal Mode and Reduced Mode.



Limit	Range	Normal Mode	Reduced Mode	Stop Category
TCP Linear Velocity	1500.0~3000.0 [mm/s]	3000.0 2700	2400.0 1000	CAT0
TCP Angular Velocity	30.0~687.5 [deg/s]	687.5 667.5	550.0 545	CAT0
Power	50.0~800.0 [W]	800.0	640.0	CAT0
Momentum	3.0~20.0 [kg-m/s]	20.0	16.0	CAT0
TCP Force	30.0~200.0 [N]	200.0	160.0	None
TCP Torque	10.0~60.0 [Nm]	60.0	48.0	None

1 DEFAULT: You can reset safety parameters to their default values.

SAVE: Save button is password protected. Password: flexiv (lowercase).

2 Different safety limits can be configured for Normal Mode and Reduced Mode.

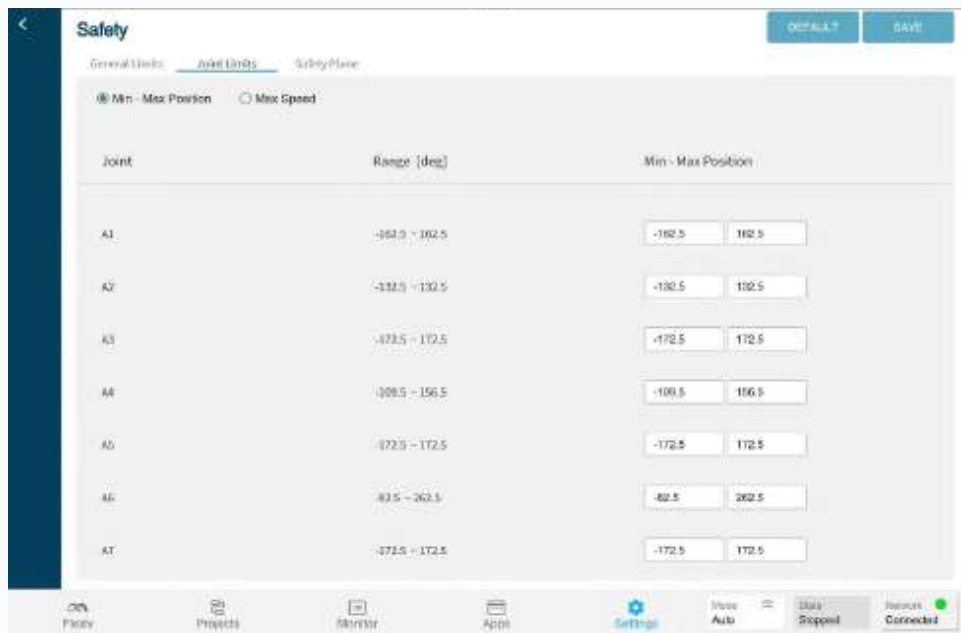
3 When the robot exceeds the safety limits, the system will trigger the corresponding stop category. You can change the stop category for certain safety limits as needed.

7.6 Safety

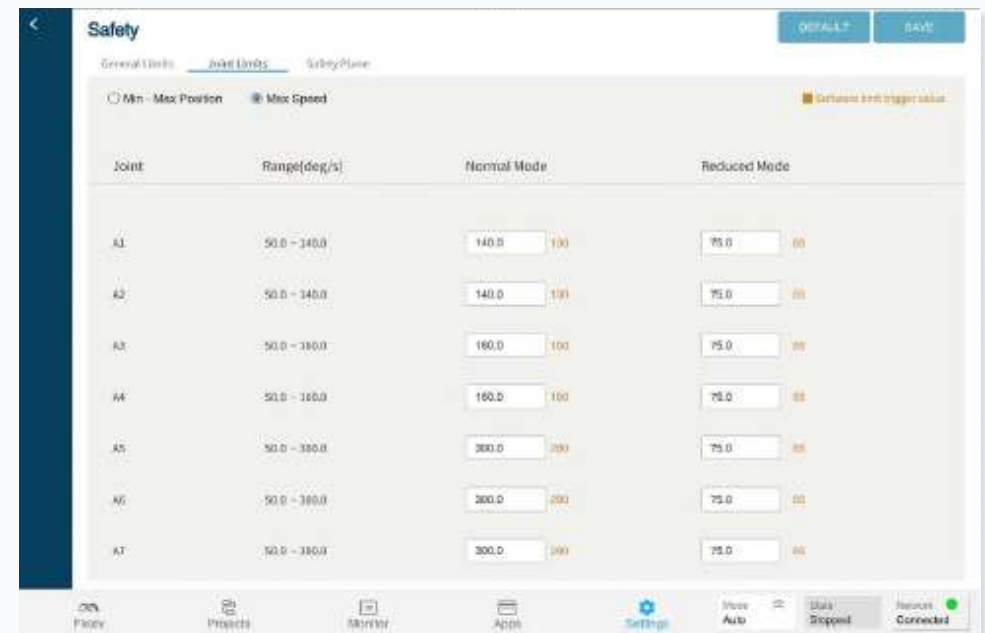
In this tab, you can configure safety-related parameters (password is required to save changes).

Tab 2/3 - Joint Limits

In the **Joint Limits** tab, you can configure joint angle limits and joint angular velocity limits for Normal Mode and Reduced Mode.



Joint	Range [deg]	Min - Max Position
A1	-162.5 ~ 162.5	-162.5 162.5
A2	-132.5 ~ 132.5	-132.5 132.5
A3	-172.5 ~ 172.5	-172.5 172.5
A4	-108.5 ~ 156.5	-108.5 156.5
A5	-172.5 ~ 172.5	-172.5 172.5
A6	-82.5 ~ 262.5	-82.5 262.5
A7	-172.5 ~ 172.5	-172.5 172.5



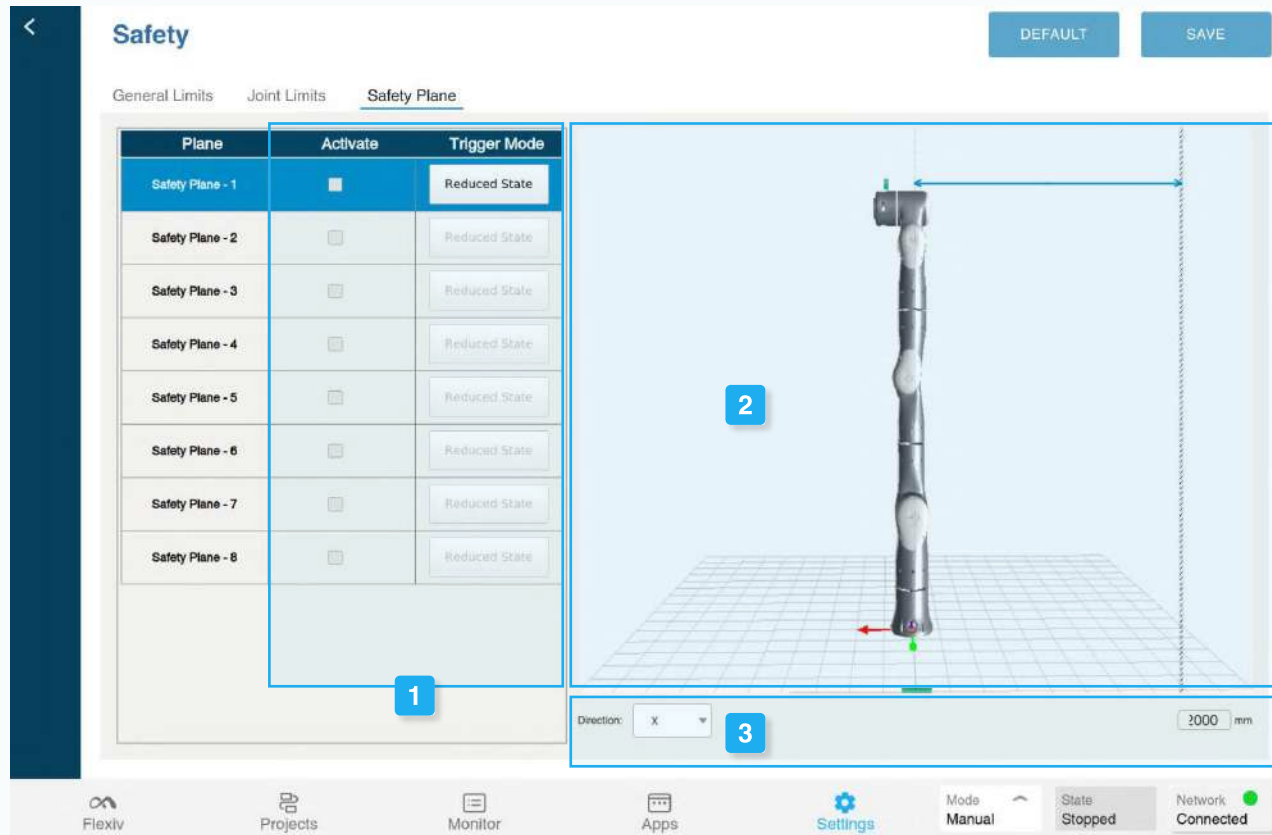
Joint	Range(deg/s)	Normal Mode	Reduced Mode
A1	50.0 ~ 140.0	140.0 130	75.0 60
A2	50.0 ~ 140.0	140.0 130	75.0 60
A3	50.0 ~ 180.0	180.0 100	75.0 60
A4	50.0 ~ 180.0	180.0 100	75.0 60
A5	50.0 ~ 380.0	380.0 200	75.0 60
A6	50.0 ~ 380.0	380.0 200	75.0 60
A7	50.0 ~ 380.0	380.0 200	75.0 60

7.6 Safety

You can configure safety-related parameters (password is required to save changes).

Tab 3/3 - Safety Plane

A safety plane is a flat, two-dimensional surface that constrains a safety area. In the **Safety Plane** tab, you can configure up to **eight** safety planes for Cartesian workspace.



The screenshot displays the 'Safety' configuration tab in the Flexiv software. It features a table for configuring safety planes and a 3D visualization of the robot arm and the safety plane.

Plane	Activate	Trigger Mode
Safety Plane - 1	☒	Reduced State
Safety Plane - 2	☐	Reduced State
Safety Plane - 3	☐	Reduced State
Safety Plane - 4	☐	Reduced State
Safety Plane - 5	☐	Reduced State
Safety Plane - 6	☐	Reduced State
Safety Plane - 7	☐	Reduced State
Safety Plane - 8	☐	Reduced State

The 3D visualization shows the robot arm and the safety plane. The safety plane is represented by a vertical line. The robot arm is shown in a vertical position. The safety plane is located at a distance of 2000 mm from the robot arm. The interface includes buttons for 'DEFAULT' and 'SAVE' at the top right. The bottom status bar shows 'Mode: Manual', 'State: Stopped', and 'Network: Connected'.

- 1 Activate:** The safety plane constraints can be activated or deactivated.

Trigger Mode: Configure the plane violation trigger types: Reduced State or Category 0. The robot will enter the corresponding state if the safety plane constraints are exceeded.

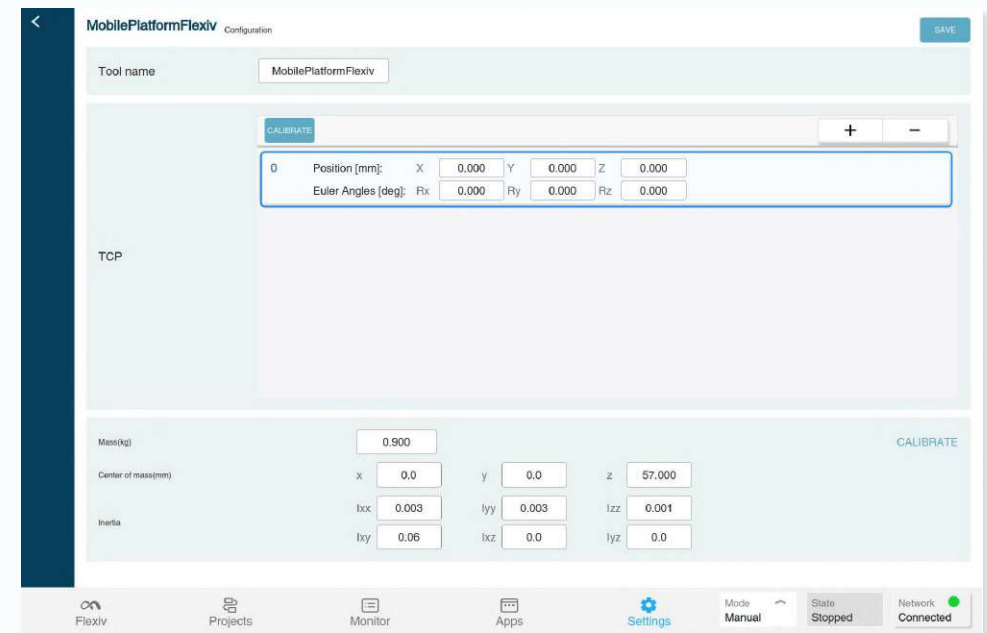
- 2 Visualize the safety plane.**
- 3 Configure the safety plane's key parameters (location).**

7.7 Tool

In this tab, you can add, configure, and enable the use of tools.



- 1 Add a new tool by tapping , or select an existing tool from the list.



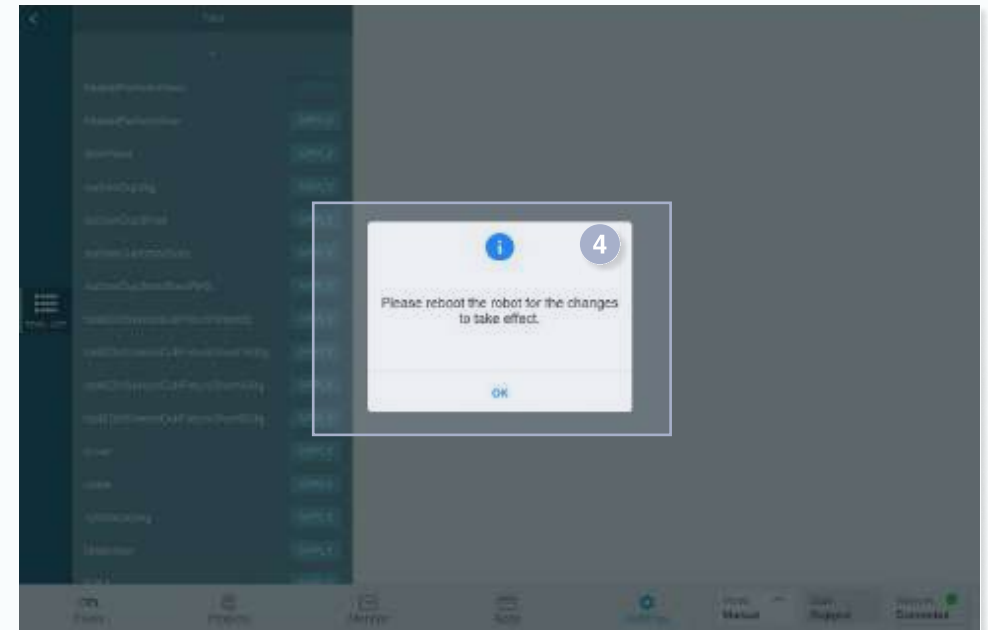
- 2 Configure the tool's name, its TCP, and its load data, then tap  to save the configuration.

7.7 Tool

In this tab, you can add, configure, and enable the use of tools.



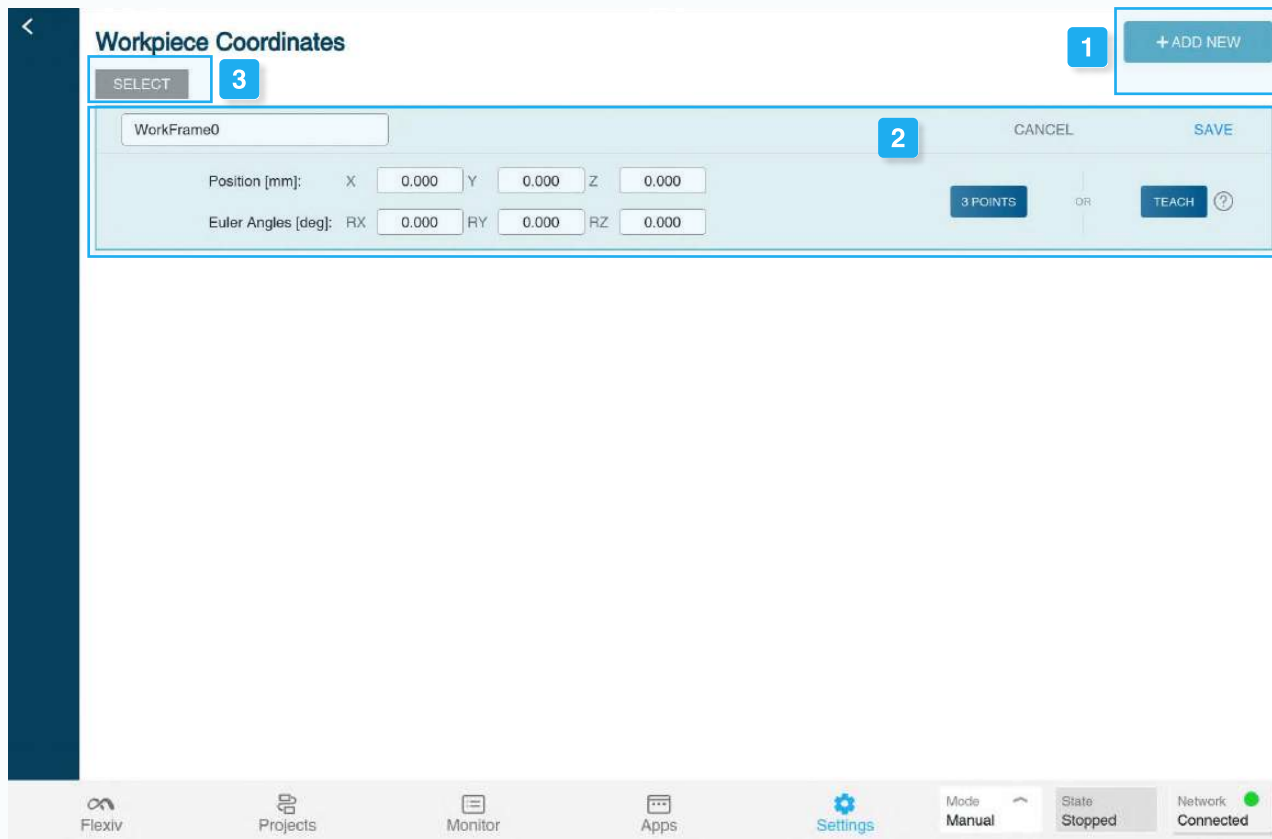
3 Tap **APPLY** to enable the use of the tool.



4 Reboot the robot for the configuration to take effect.

7.8 Workpiece Coordinates

In this tab, you can create, define, and manage workpiece coordinates.



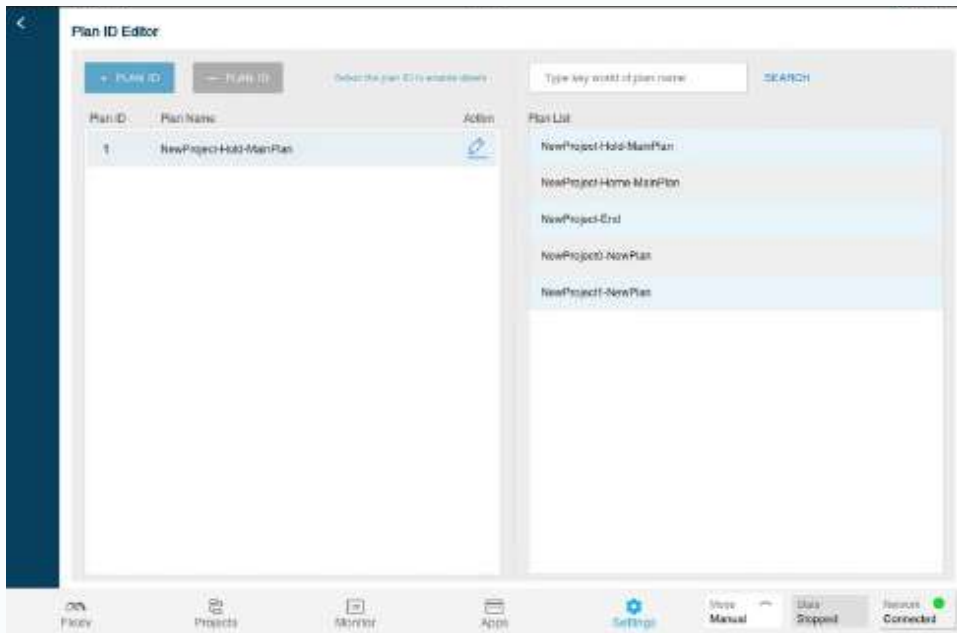
- 1 Tap **+ ADD NEW** to add a new workpiece coordinate.
- 2 Tap **EDIT** to define the coordinate. Two methods are supported: 1. **3 POINTS** (3 points method); 2. **TEACH**.
- 3 In the read-only mode, tap **SELECT** to select multiple coordinates and delete them all at once.

7.9 Plan ID Editor

In this tab, you can configure a unique plan index for a plan stored in the robot to allow the host to use that index to invoke the plan. The plan index/ID should conform to the decimal number system.

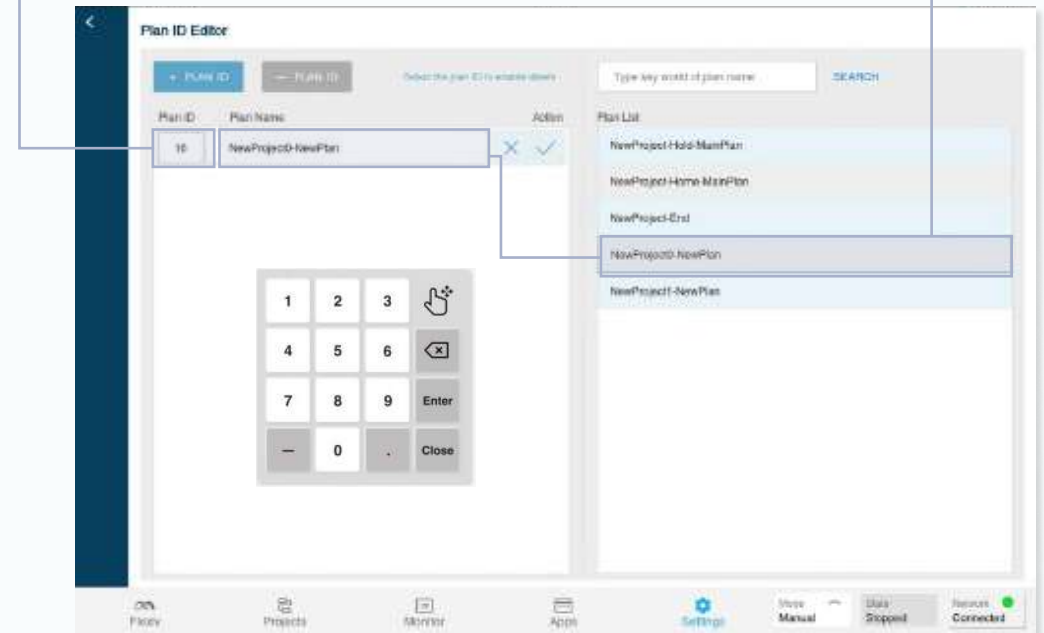
1 Tap **+ PLAN ID** to add a new plan index.

2 Tap  to enter edit mode.



3 Choose the plan by tapping the plan name from the **Plan List**.

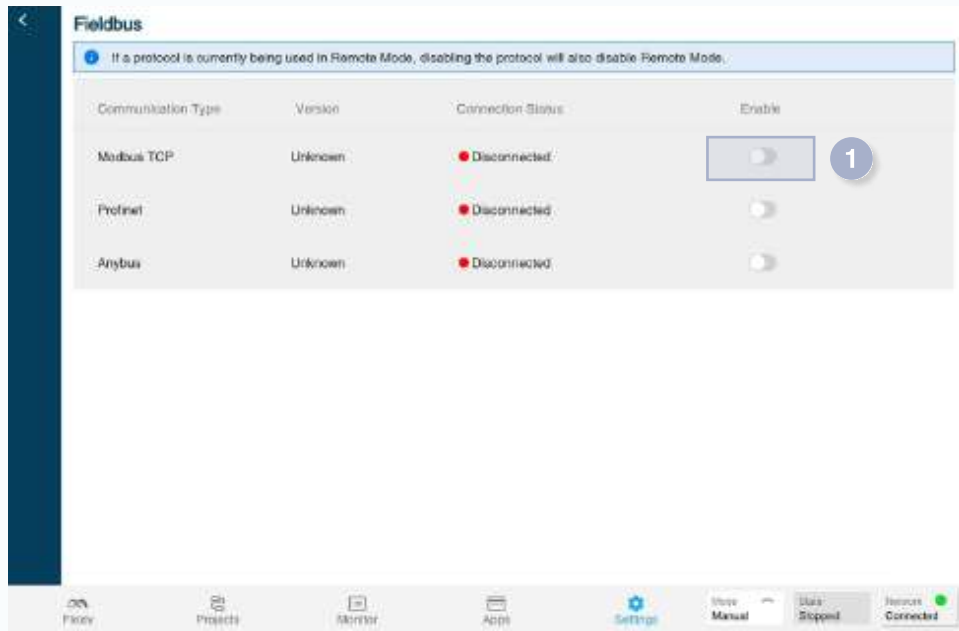
4 Enter a unique index for the selected plan.



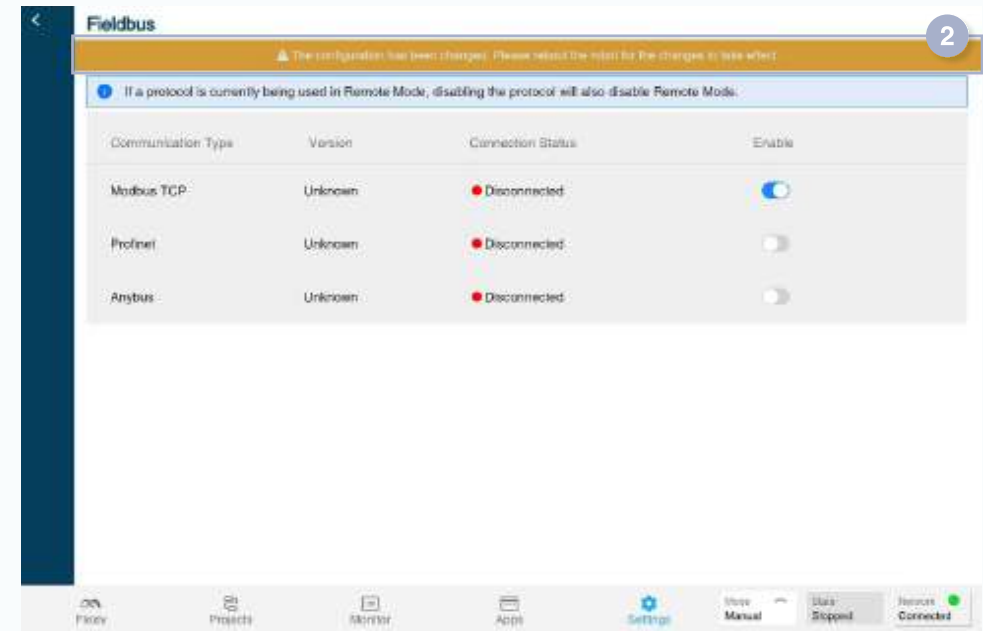
5 Tap  /  to confirm/discard the configuration.

7.10 Fieldbus

In this tab, you can configure industrial communication types, including **Modbus TCP**, **Profinet** (optional) and **Anybus** (optional). **Fieldbus** also displays the version and the connection status* of the communication type. Only one type can be enabled at a time.



1 To enable a communication type, toggle from  to .



2 Reboot the robot for the changes to take effect.

***Note:** Connection status:  Connected Robot is connected to the peripheral device
 Disconnected Robot is not connected to the peripheral device

Remote Mode enables the robot process to be controlled by a peripheral device (e.g. a PLC). Five types of remote modes are currently offered, namely, **I/O**, **Modbus TCP**, **Profinet**, **Anybus** and **RDK**.

<

Remote Mode

SAVE

Remote Mode: ☒

Select a Mode:

I/O

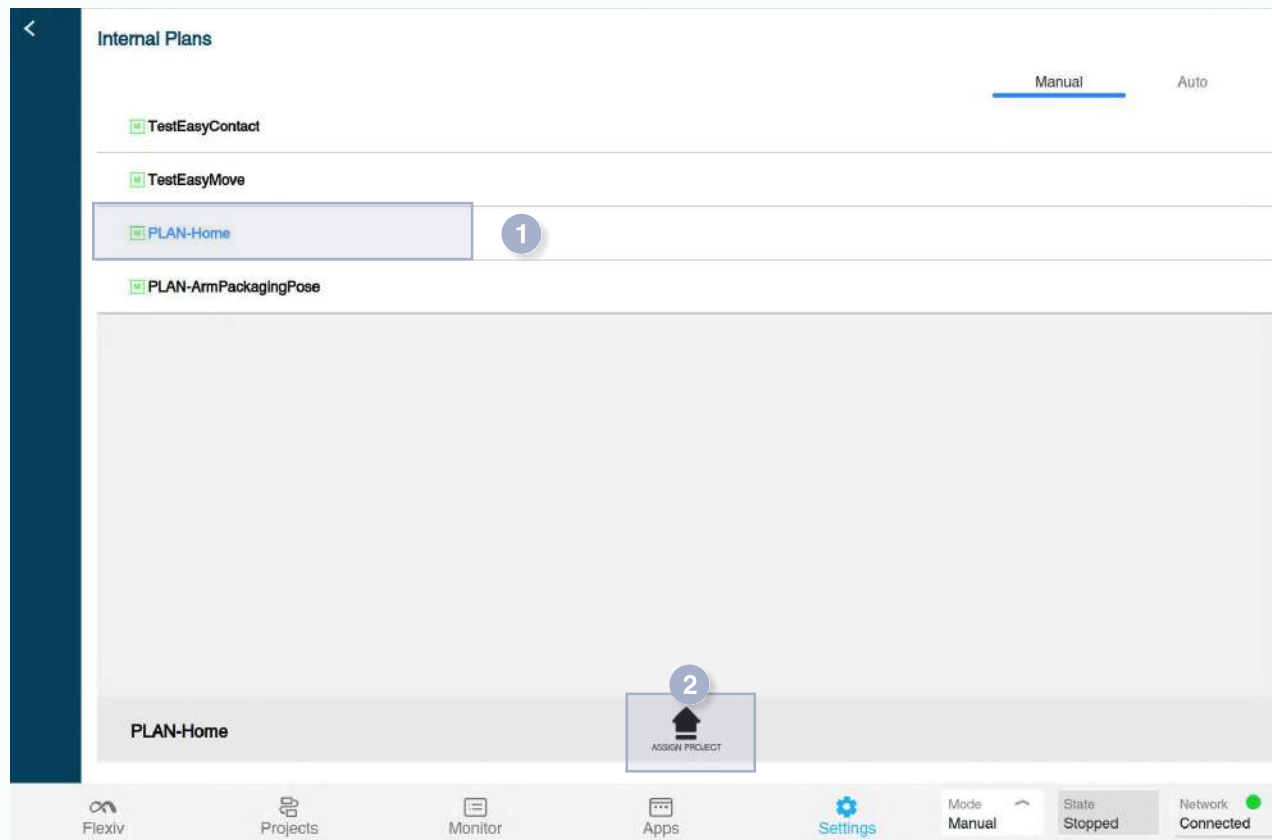
Input	Port	Output	Port
Start Program:	Input0	E-Stop:	Output0
Stop Program:	Input1	Remote Active:	Output1
Pause Program:	Input2	Program Request:	Output2
Enable Move:	Input3	Program Running:	Output3
Validate Program:	Input4	Fault State:	Output4
Clear Fault:	Input5	Motor On State:	Output5
Motor On:	Input6	Plan ID State:	6,7,8,9
Plan ID:	6,7,8,9	Program Paused:	Output6

Low bit -> High bit

Low bit -> High bit

7.12 Internal Plans

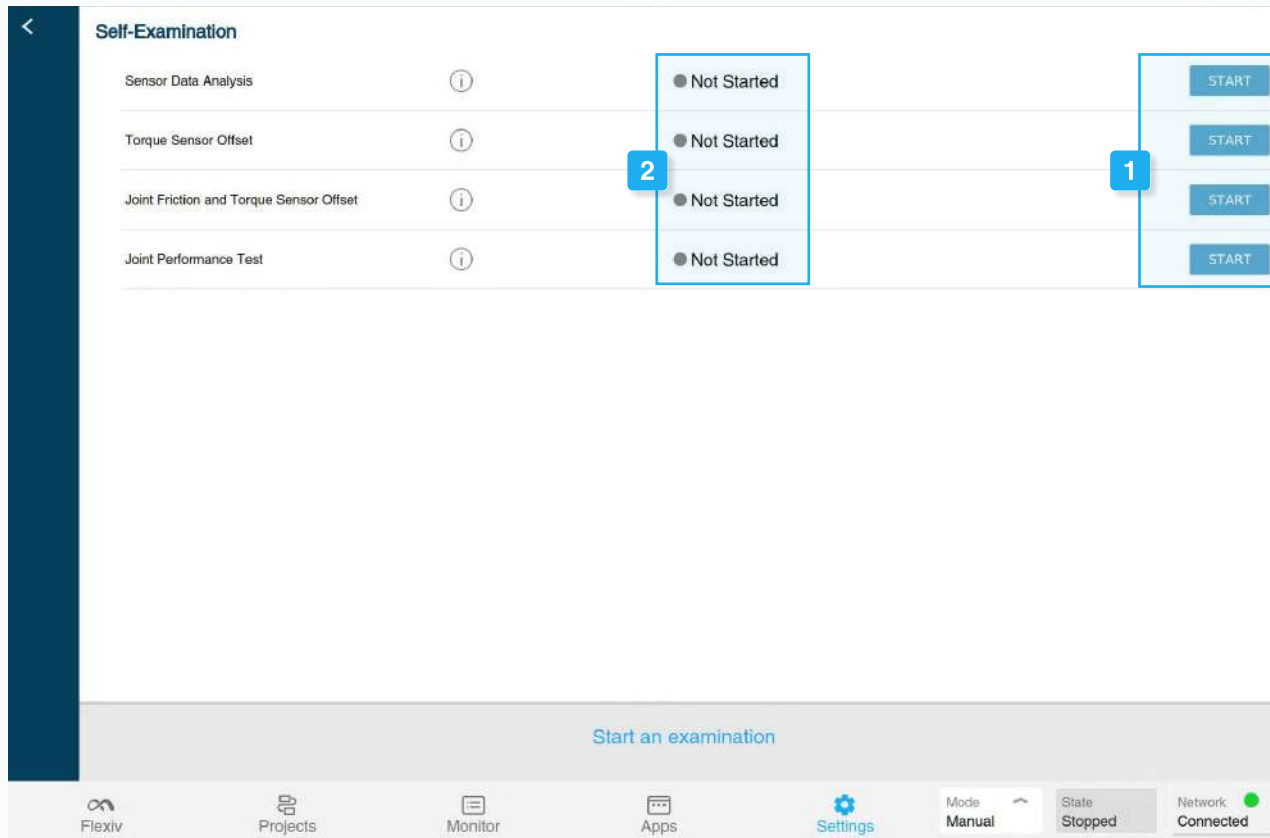
In this tab, you can assign and execute the internal plans (built-in plans on the Control Box).



- 1 Select a plan from the plan list (switch the Motion Bar to Manual Mode to enable selection).
- 2 Tap  to assign the selected internal plan.

7.13 Self-Examination

In this tab, you can check the robot's health condition by running self-examinations. Available examinations include **Sensor Data Analysis**, **Torque Sensor Offset**, **Joint Friction and Torque Sensor Offset**, and **Joint Performance Test**.



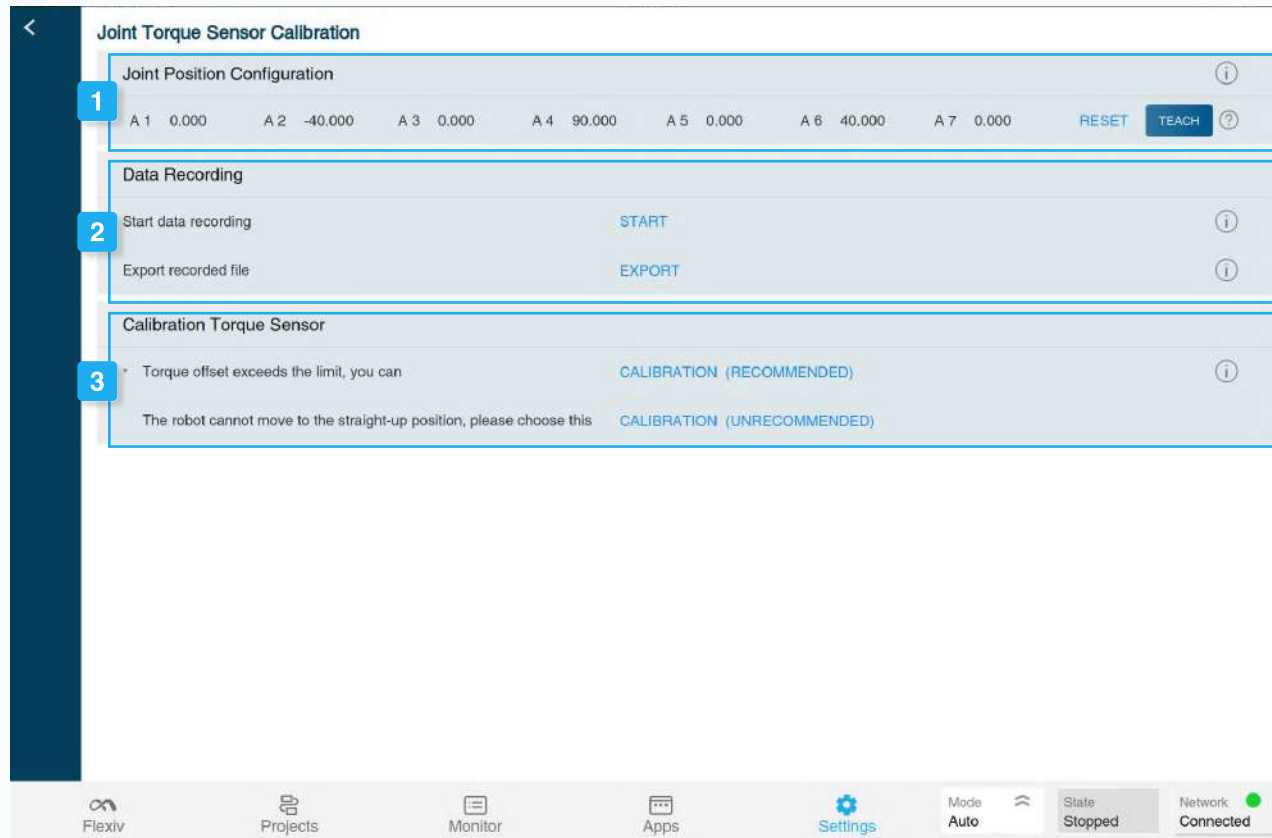
1 Tap **START** to run an examination.

2 Examination Status:

- 1. Not Started ● Not Started
- 2. In Progress ● In Progress
- 3. Pass ● Pass
- 4. Fail ● Fail

7.14 Joint Torque Sensor Calibration

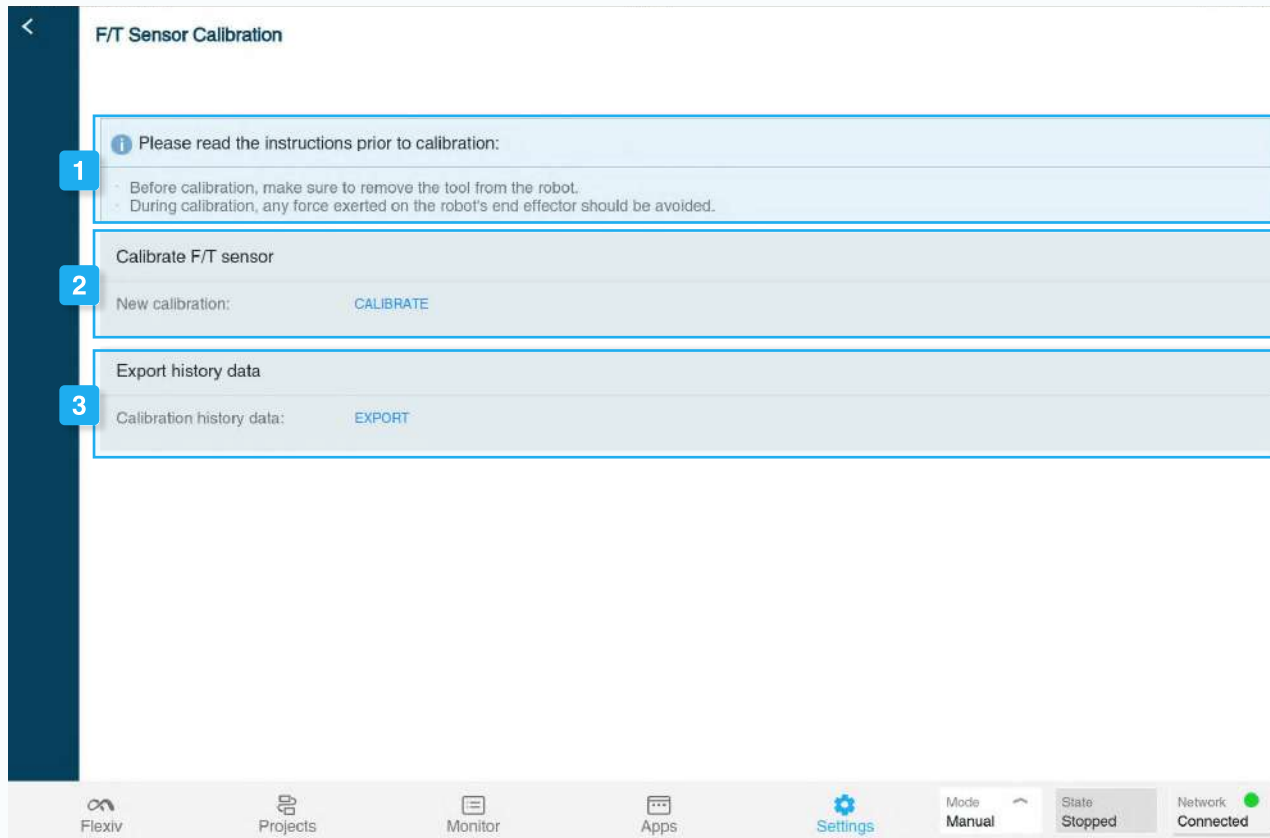
In this tab, you can calibrate joint torque sensors.



- 1 **Joint Position Configuration:** You can configure **joint position** to a user-defined one using **Freedrive** (TEACH) or tap **RESET** to set joint position to the straight-up position.
- 2 **Data Recording:** Tap **START** to execute torque sensor data recording. Recorded file can be exported for further analysis.
- 3 **Calibration Torque Sensor:** It is recommended to use the straight-up position for calibration. If the robot CANNOT move to the straight-up position due to environmental constraints, use **joint position** to execute the calibration.

7.15 F/T Sensor Calibration

In this tab, you can calibrate the Force/Torque (F/T) sensor and export calibration history data. F/T sensors are installed on Rizon 4S and Rizon 10S robots only.



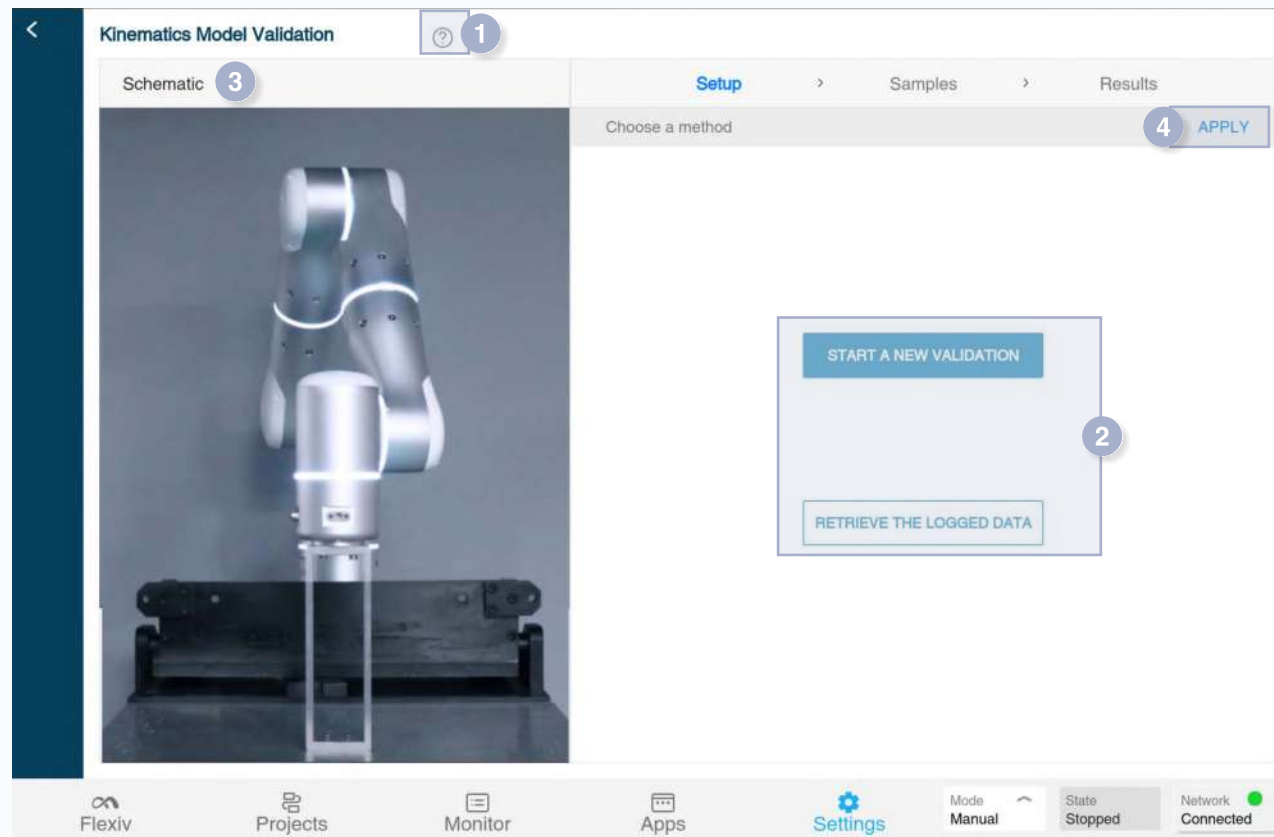
- 1 Instructions:** 1. Before calibration, remove the tool from the robot. 2. During calibration, avoid exerting any force on the end effector.
- 2 Calibrate F/T sensor:** Tap **CALIBRATE** to execute the calibration process. When the calibration is complete, you can view the calibration results and choose whether you want to apply the results.
- 3 Export history data:** Tap **EXPORT** to export history calibration data for further analysis.

7.16 Kinematics Model Validation

In this tab, you can validate the kinematics model by making the robot perform null space movement.

Step 1/3 - Setup

You can choose a method for collecting the validation samples, and then follow the schematic and text instructions to complete the installation of the robot before the null space movement.



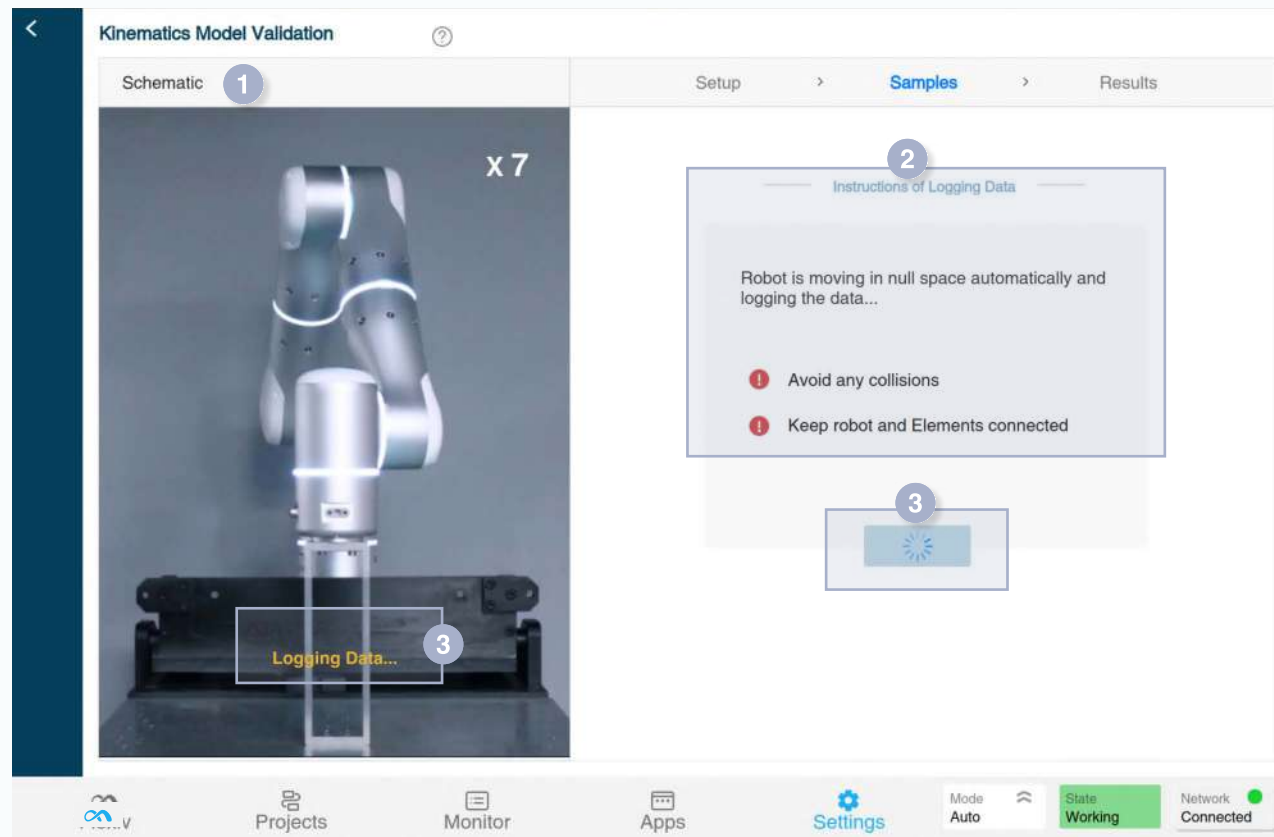
- 1 Tap  to see the explanation of the term.
- 2 Choose a validation sample collection method.
- 3 Demonstrates how the robot is set before the null space movement.
- 4 Tap **APPLY** to go to the samples collection section.

7.16 Kinematics Model Validation

In this tab, you can validate the kinematics model by making the robot perform null space movement.

Step 2/3 - Samples

You can monitor the data logging status during the null space movement.



1 Demonstrates the state of the robot during the null space movement.

2 Instructions for logging data

3 Data logging status

 In progress

↓
 Completed

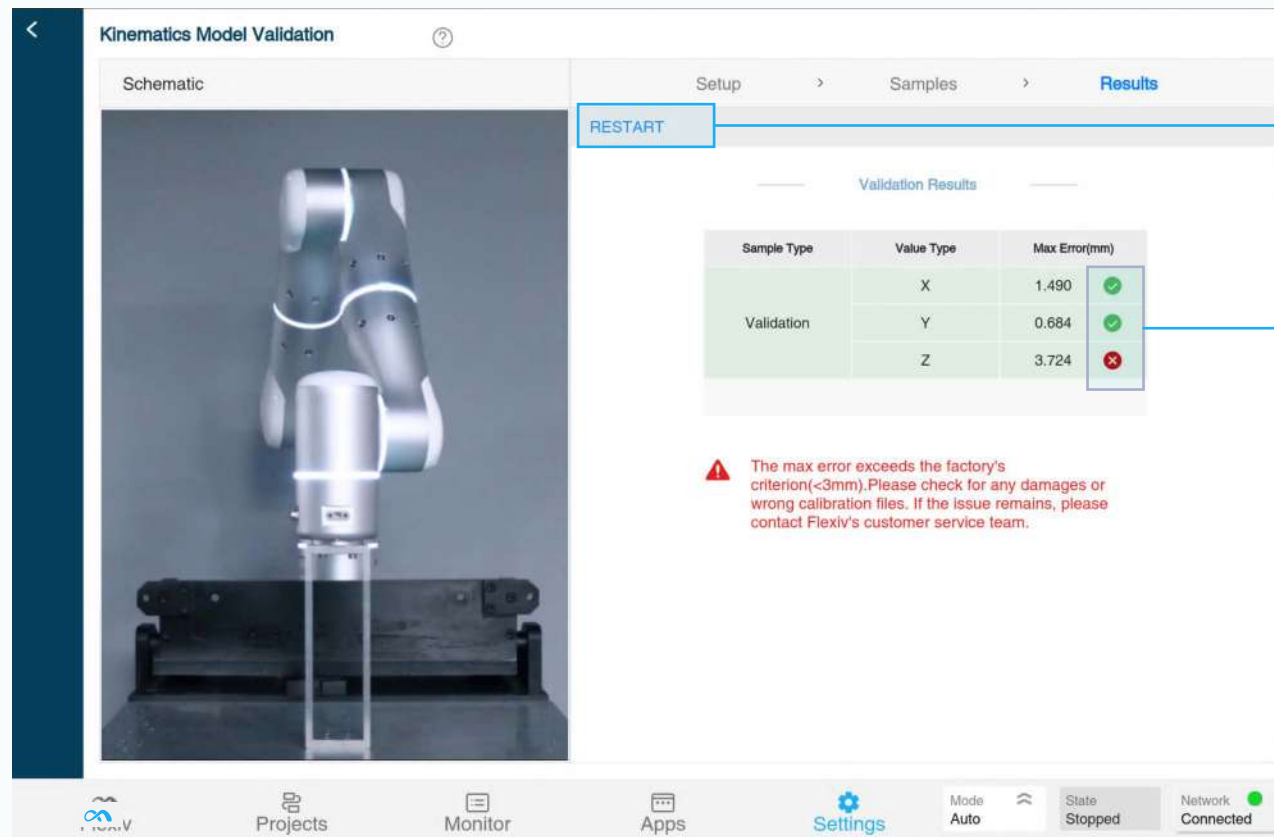
Tap **DONE** to calculate the errors based on logged data.

7.16 Kinematics Model Validation

In this tab, you can validate the kinematics model by making the robot perform null space movement.

Step 3/3 - Results

You can view the results of validation based on logged data generated by the null space movement.



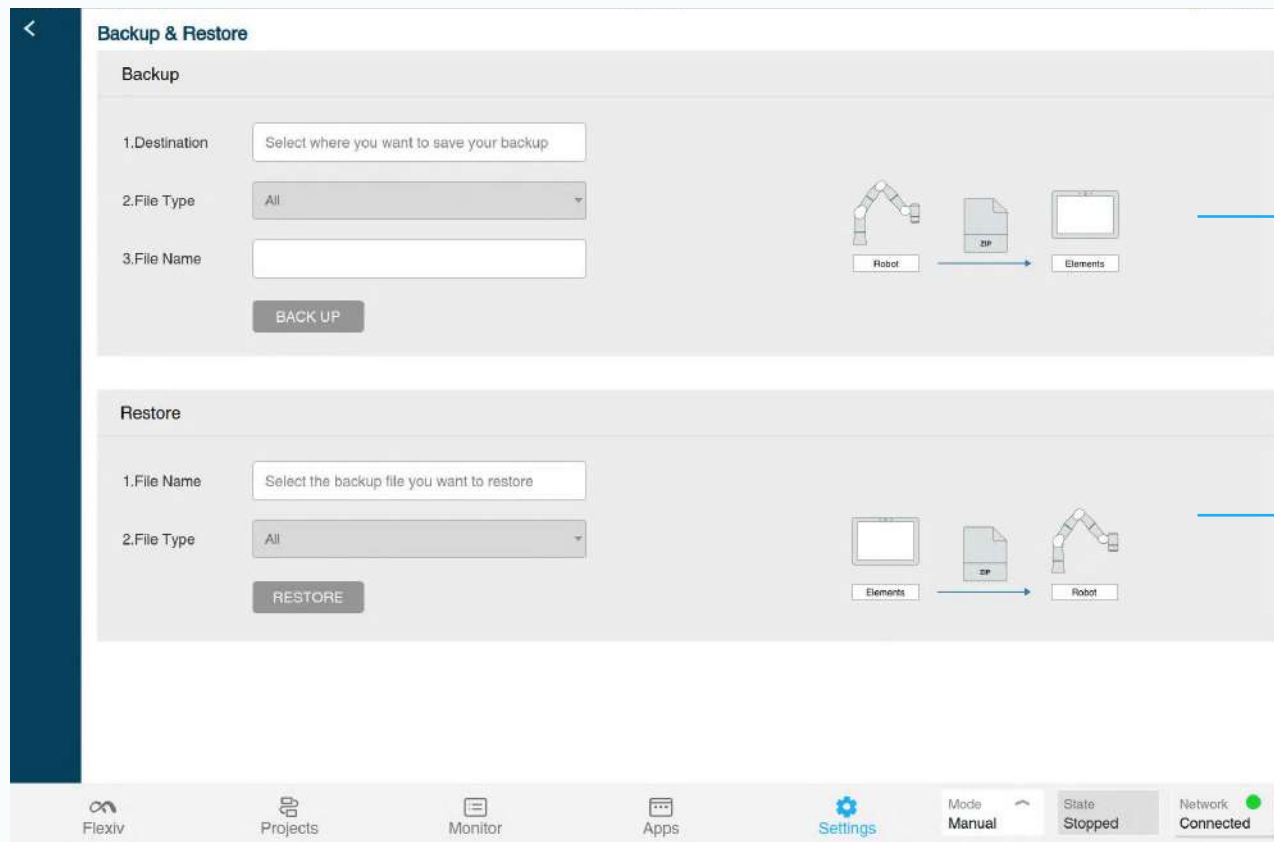
1 Tap **RESTART** to re-validate samples

2  Max error falls within the factory criteria

 Max error falls outside of the factory criteria

7.17 Backup & Restore

In this tab, you can back up user data (including projects, global variables, tools, and workpiece coordinates) from the robot to **Flexiv Elements** and restore your backed up data from **Flexiv Elements** to the same robot or to other robots.



1 Backup:

- Select the destination folder where you want to save your backup > select the file type > edit the file name > tap **BACK UP**.

2 Restore:

- Select the backup file you want to restore > select the file type > tap **RESTORE**.

***Caution:** The restore operation will overwrite all user data in the robot. Please back up your data first to prevent data loss.

8.1 Introduction

After establishing communication with an external device, you can use primitive commands to check the state of the device or send primitive commands to control the device.

Part 1/2 I/O, Tool, and AI State

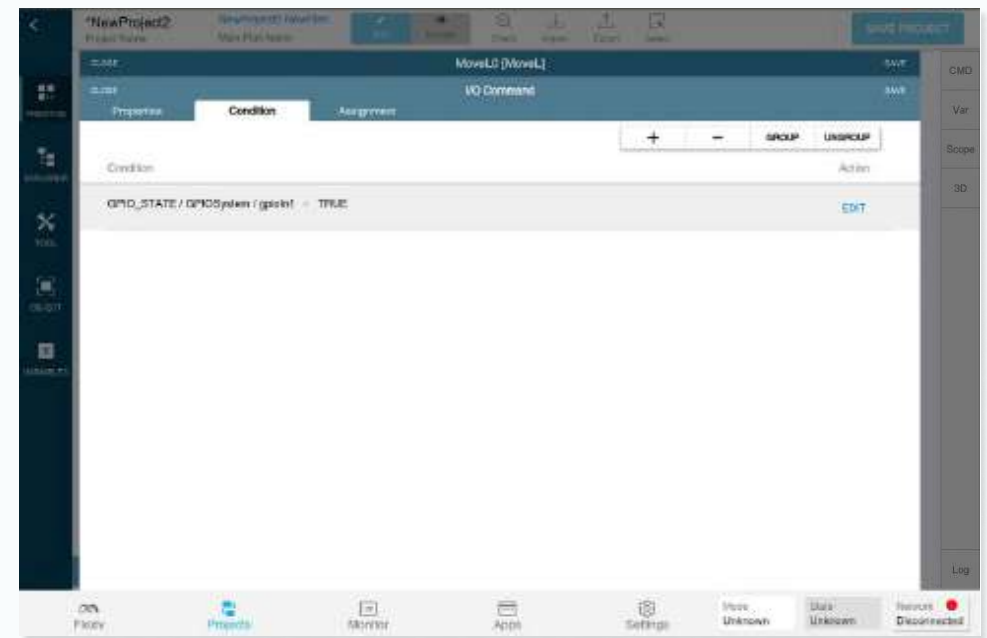
As inputs from an external device, **I/O**, **Tool**, and **AI** state can be used as **transition/primitive condition** in a plan. See the examples below.

1. As Transition:



E.g. Tool state **GripperFlexiv reachedForce=True** is used as the transition between **Home0** and **MoveL0**.

2. As Primitive Condition:



E.g. I/O state **gpioIn0=True** is used as the condition of **MoveL0**.

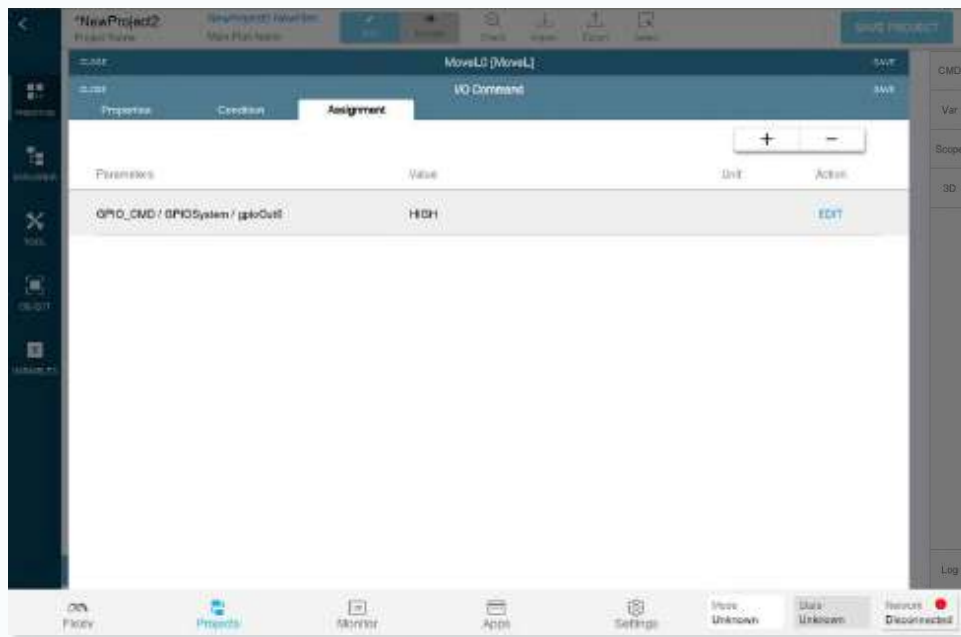
8.1 Introduction

After establishing communication with an external device, you can use primitive commands to check the state of the device or send primitive commands to control the device.

Part 2/2 I/O, Tool, and AI Command

As outputs to an external device, **I/O**, **Tool**, and **AI** command can be used as **primitive assignment** in a plan. See the example below.

As Primitive Assignment:

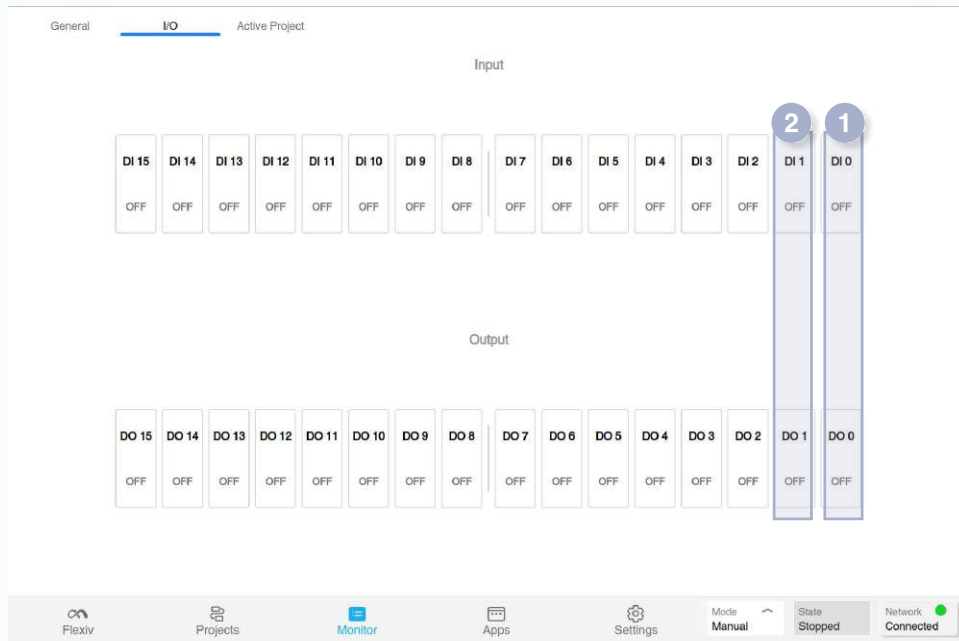


E.g. I/O assignment **gpioOut0=High** is used as the assignment of **MoveL0**.

8.2 I/O Command Instruction

8.2.1 Introduction

This section explains how to configure input signals and output states with the following examples.



Initial Settings in **Monitor** > I/O

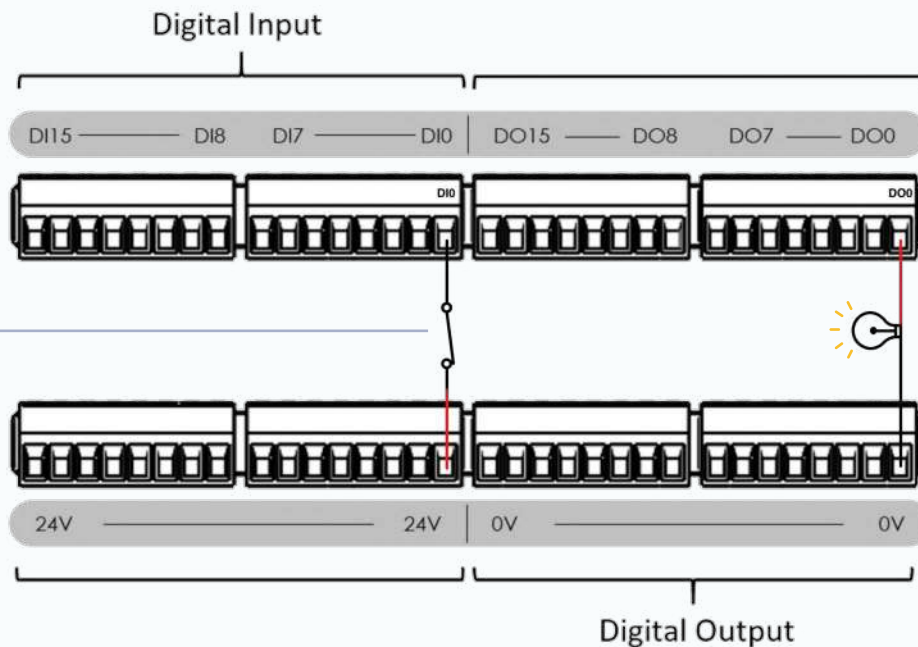
e.g.	Initial Settings	Final Settings	Processes
1	DI 0 OFF	DI 0 ON	Configure in the Main Plan 1. Use Input0 signal as condition of transition 2. Control Output0 status with primitive command
2	DI 1 OFF	DI 1 ON	Configure in the Subplan 1. Use Input1 signal as condition of transition 2. Control Output1 status with primitive command

Examples and Processes

8.2 I/O Command Instruction

8.2.2 Configuration Overview

Configure in the Main Plan



Step 1: Plan executes the **Home0** node

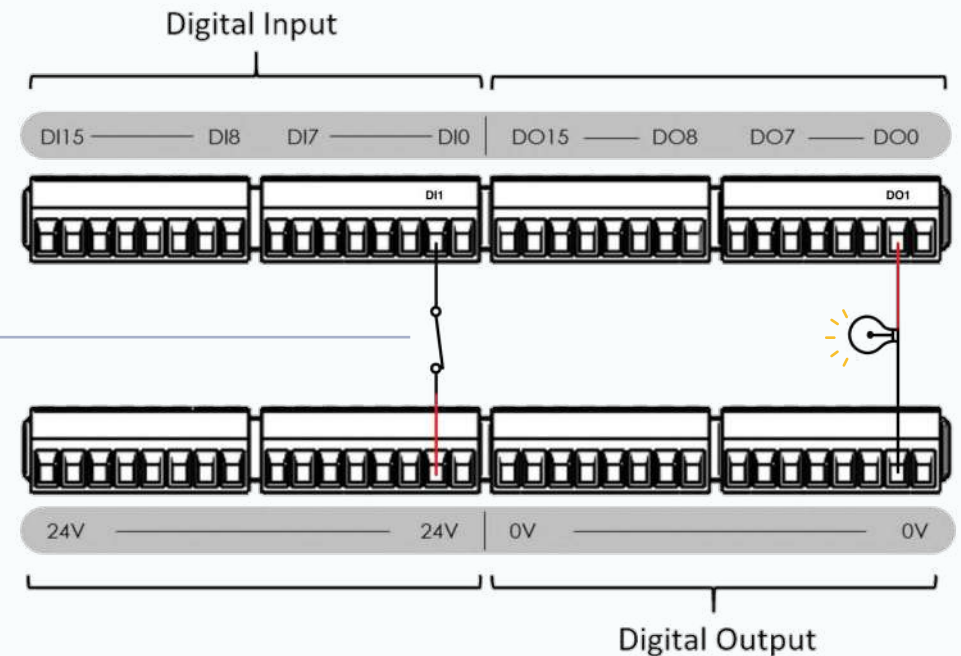
Condition: **Input0** signal is ON when **DI0** port is ON

Step 2: Plan executes the **MoveL0** node

Assignment: **DO0** port is ON when **Output0** signal is ON

(See page 71-77)

Configure in the Subplan



Step 1: Plan executes the **Plan0-Home0** node

Condition: **Input1** signal is on when **DI1** port is ON

Step 2: Plan executes the **MoveC0** node

Assignment: **DO1** port is ON when **Output1** signal is ON

(See page 78-84)

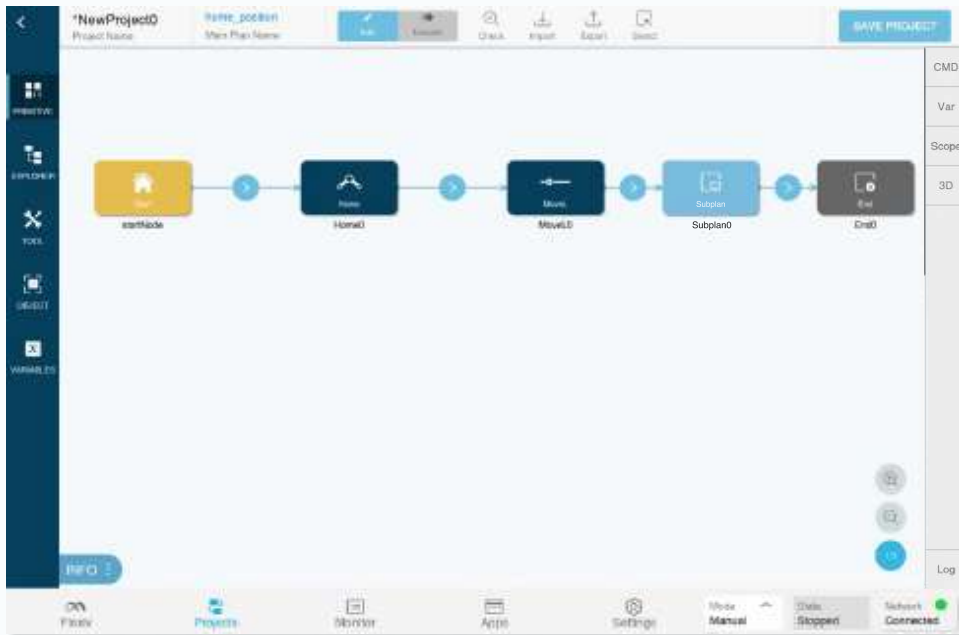
8.2 I/O Command Instruction

8.2.3 Configure in the Main Plan

Part 1/2 Use input signal as transition condition

Example: Add **Input 0=1** as the transition condition to run **MoveL0**

Step: 1 2 3 4 5 6 7 8 9 10



Main Plan Interface



Main Plan Interface

1 In the **Projects** tab, create a project as above

2 Tap > between **Home0** and **MoveL0**

3 Tap **Edit** , the edit screen will pop up

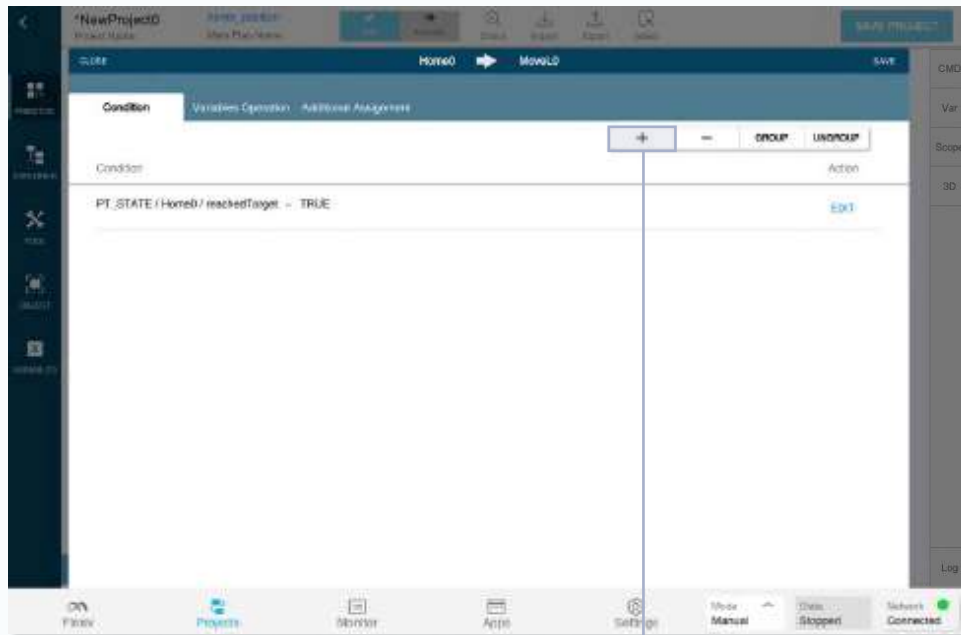
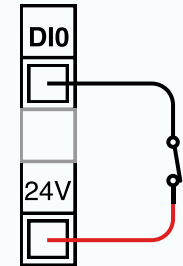
8.2 I/O Command Instruction

8.2.3 Configure in the Main Plan

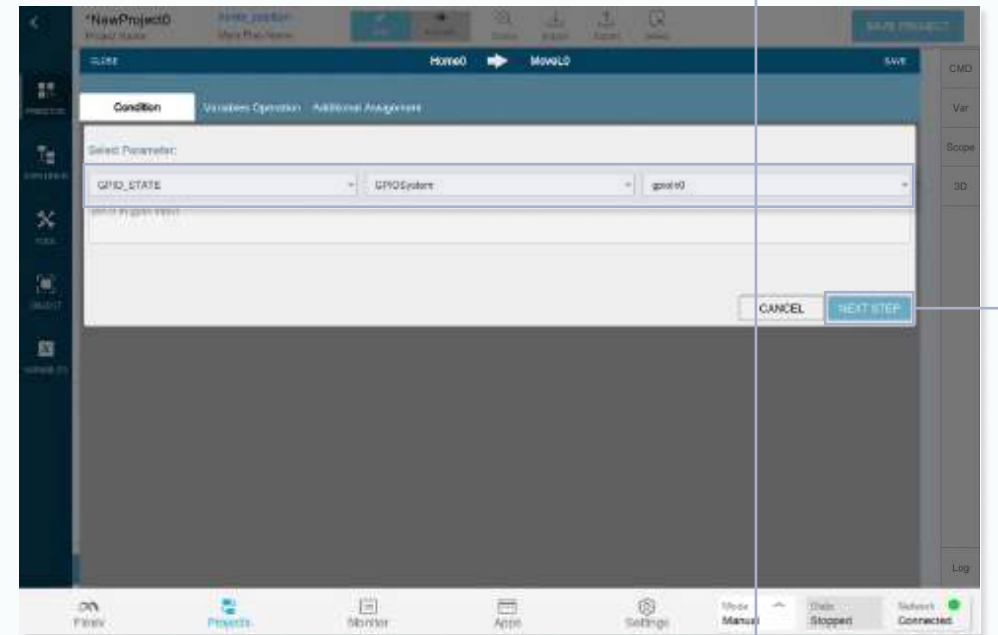
Part 1/2 Use input signal as transition condition

Example: Add **Input 0=1** as the transition condition to run **MoveL0**

Step: 1 2 3 4 5 6 7 8 9 10



4 Tap + to add a condition



5 Select **GPIO_STATE**, **GPIOSystem**, **gpioIn0** in turn

6 Tap **NEXT STEP**

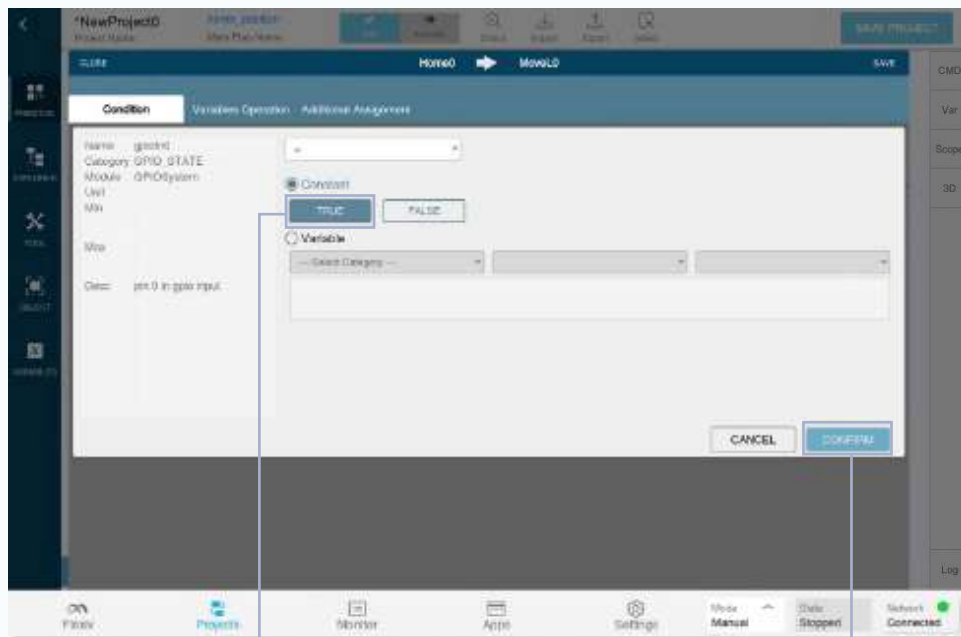
8.2 I/O Command Instruction

8.2.3 Configure in the Main Plan

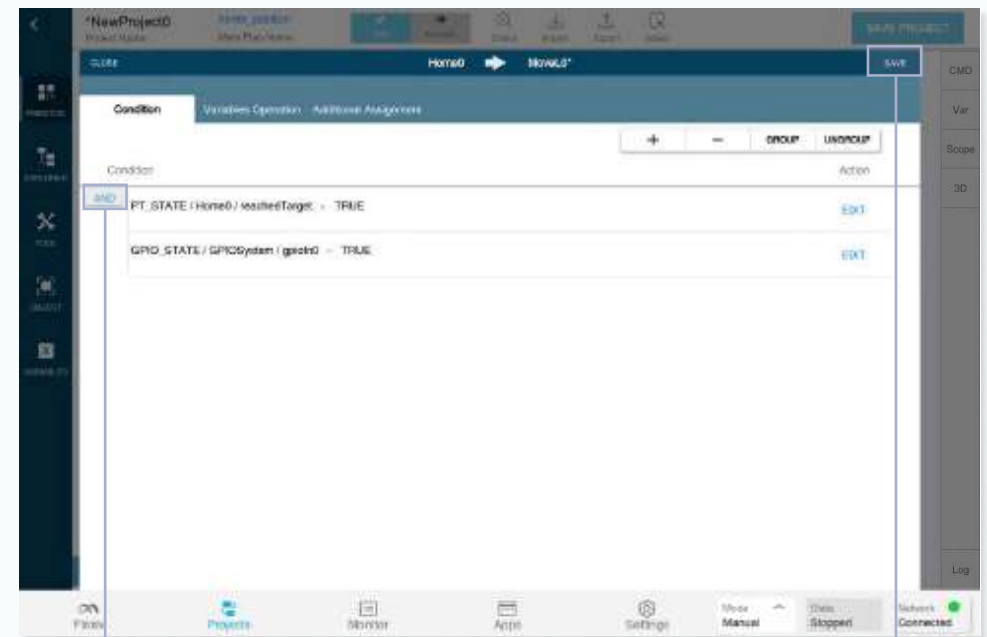
Part 1/2 Use input signal as transition condition

Example: Add **Input 0=1** as the transition condition to run **MoveL0**

Step: 1 2 3 4 5 6 7 8 9 10



- 7 Select **TRUE**
- 8 Tap **CONFIRM**



- 9 Change **OR** (default) to **AND**
- 10 Tap **SAVE**

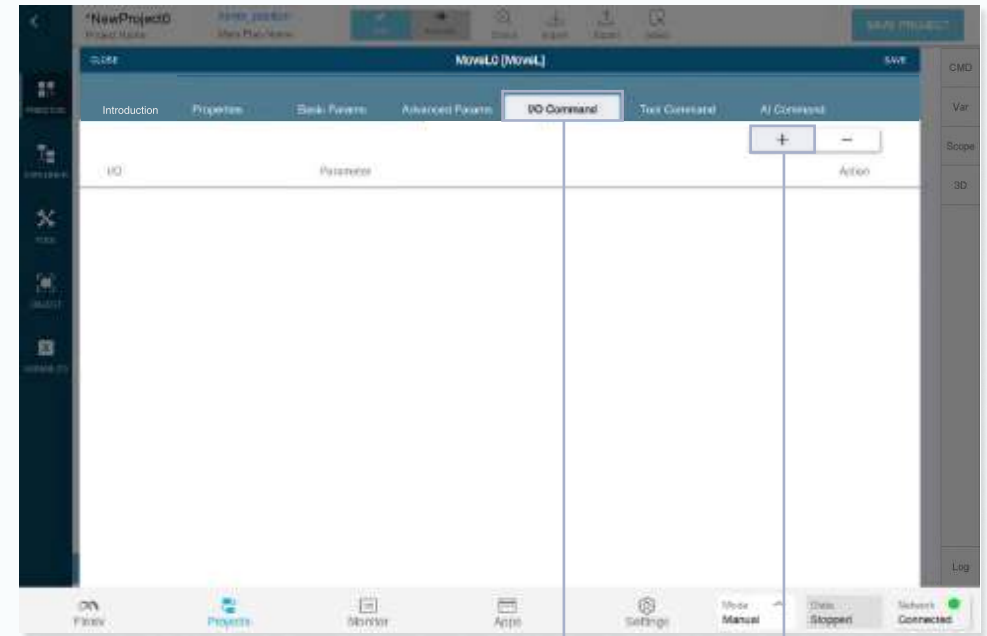
8.2 I/O Command Instruction

8.2.3 Configure in the Main Plan

Part 2/2 Change output status

Example: Change the status of **Output 0** by configuring the I/O command of **MoveL0**

Step: 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16



1 Tap the **MoveL0** node

2 Tap **Edit**

3 Select the **I/O Command** tab

4 Tap **+** to add I/O command

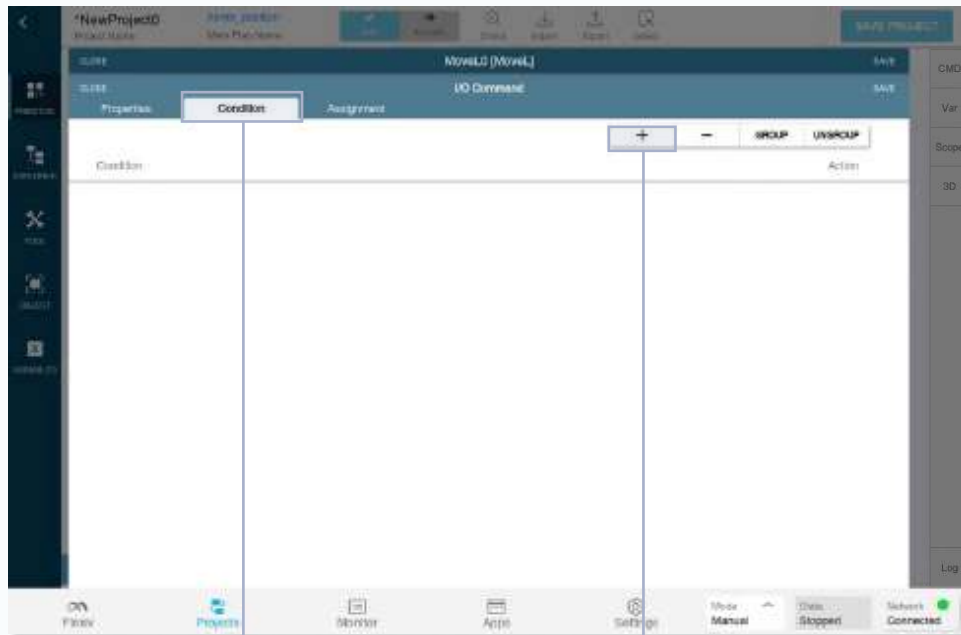
8.2 I/O Command Instruction

8.2.3 Configure in the Main Plan

Part 2/2 Change output status

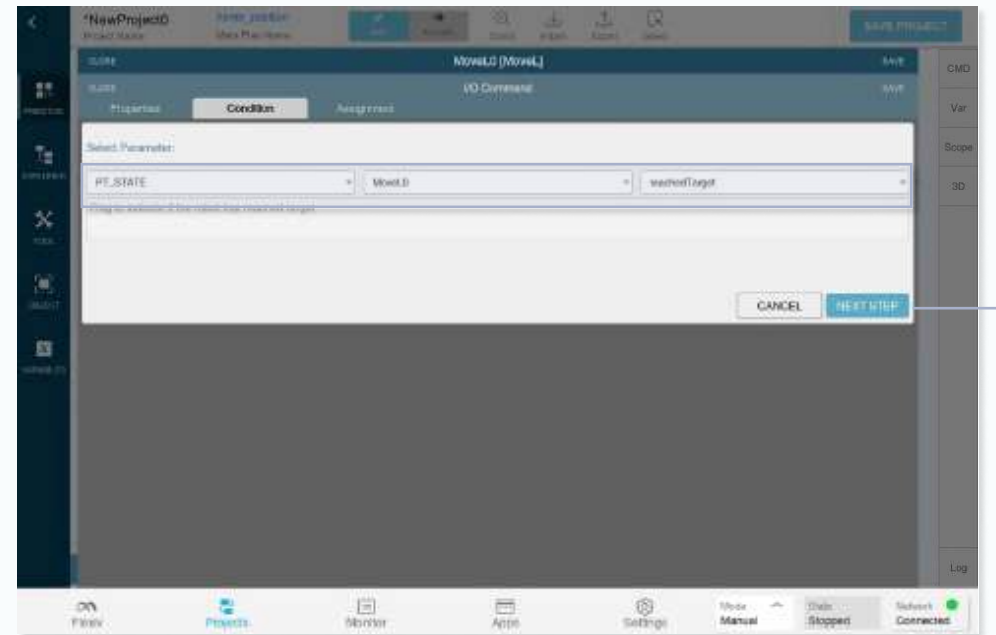
Example: Change the status of **Output 0** by configuring the I/O command of **MoveL0**

Step: 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16



5 Select the **Condition** tab

6 Tap **+** to add a condition



7 Select **PT_STATE**, **MoveL0**, **reachedtarget** in turn

8 Tap **NEXT STEP**

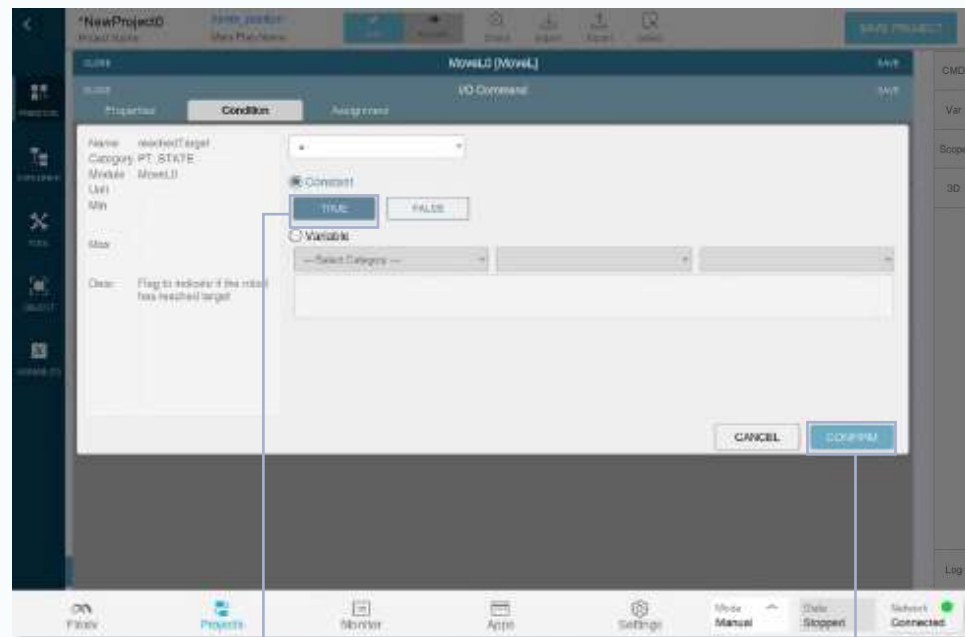
8.2 I/O Command Instruction

8.2.3 Configure in the Main Plan

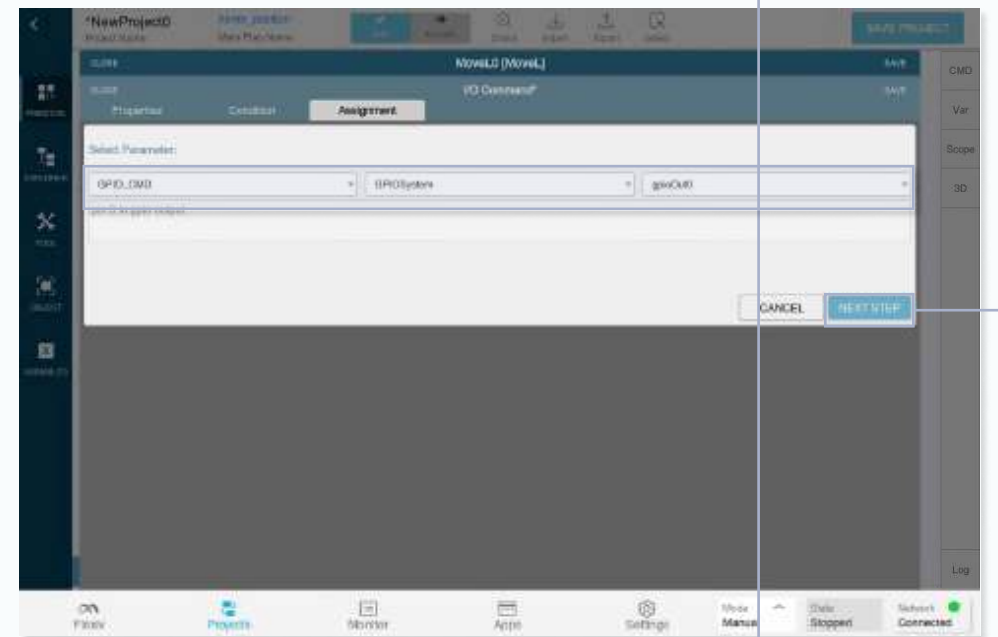
Part 2/2 Change output status

Example: Change the status of **Output 0** by configuring the I/O command of **MoveL0**

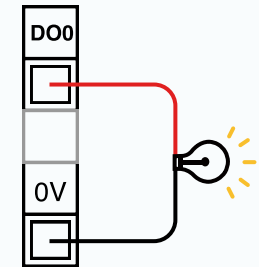
Step: 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16



- 9 Select **TRUE**
- 10 Tap **CONFIRM**



- 11 Select the **Assignment** tab
- 12 Select **GPIO_CMD**, **GPIOSystem**, **gpioOut0** in turn
- 13 Tap **NEXT STEP**



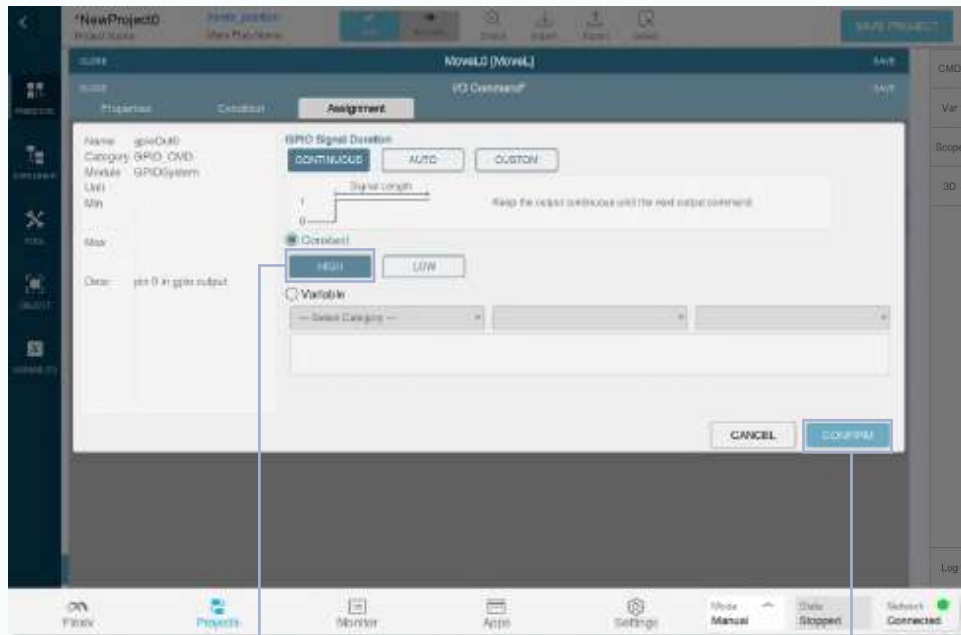
8.2 I/O Command Instruction

8.2.3 Configure in the Main Plan

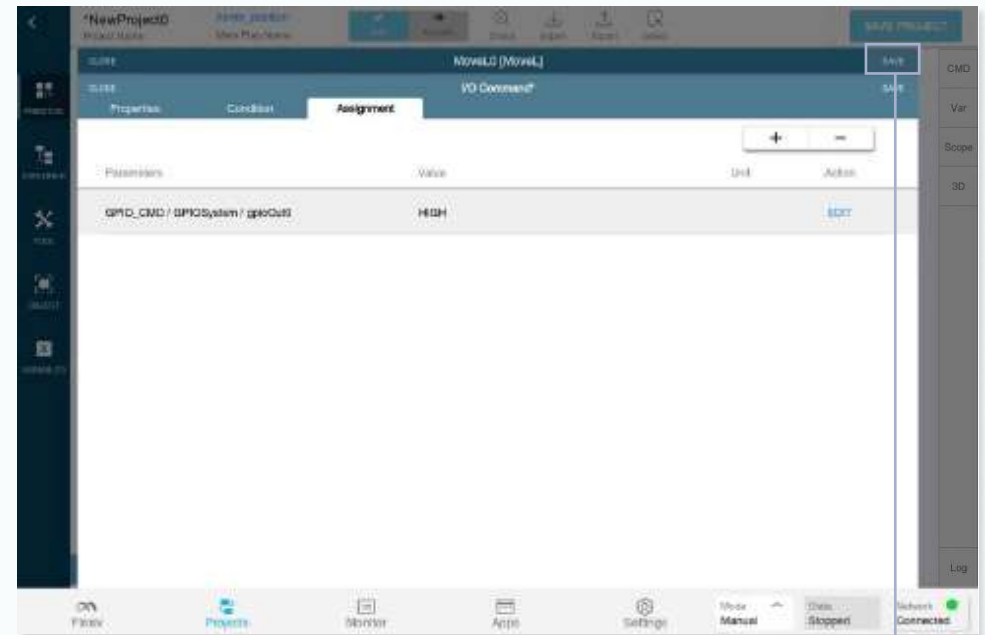
Part 2/2 Change output status

Example: Change the status of **Output 0** by configuring the I/O command of **MoveL0**

Step: 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16



- 14 Select **HIGH**
- 15 Tap **CONFIRM**



- 16 Tap **SAVE**

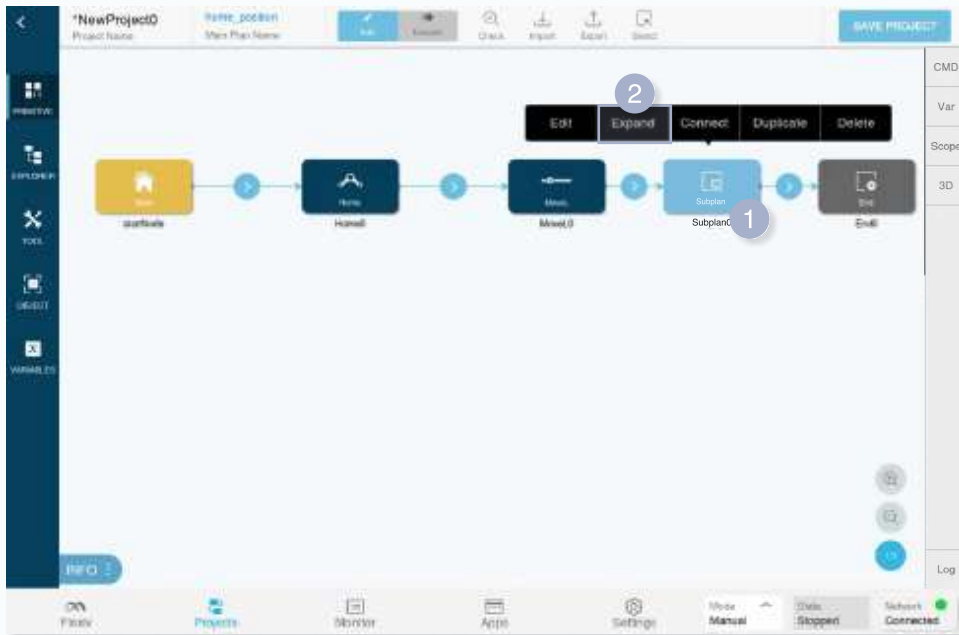
8.2 I/O Command Instruction

8.2.4 Configure in the Subplan

Part 1/2 Use input signal as transition condition

Example: Add **Input 1=1** as the transition condition to run **MoveC0** in **Subplan0**

Step: 1 2 3 4 5 6 7 8 9 10 11



Main Plan Interface



Subplan Interface

1 Tap the **Subplan0** node

2 Tap **Expand**

3 Tap **>** between **Home0** and **MoveC0**

4 Tap **Edit** , the edit screen will pop up

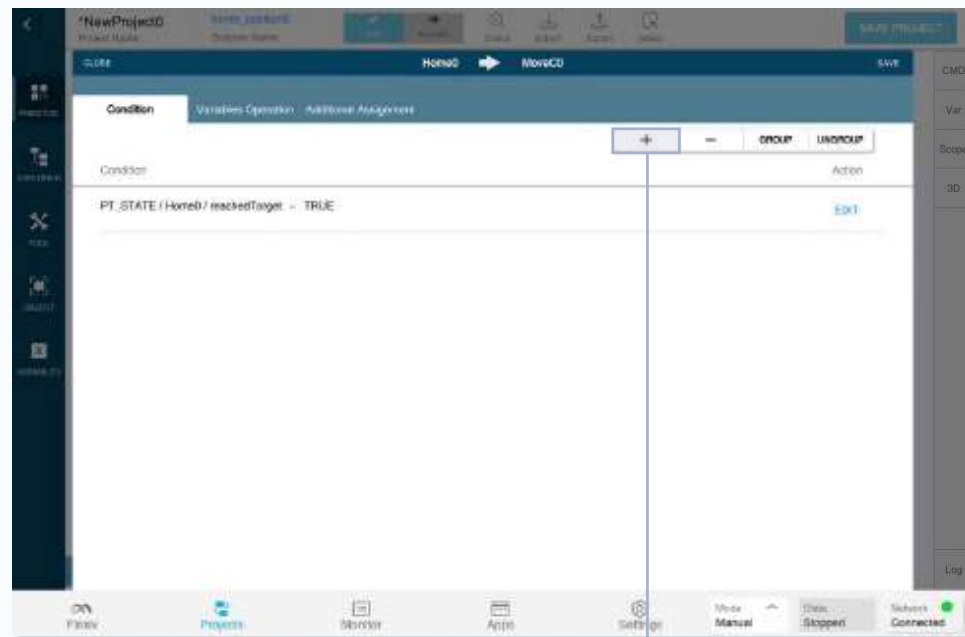
8.2 I/O Command Instruction

8.2.4 Configure in the Subplan

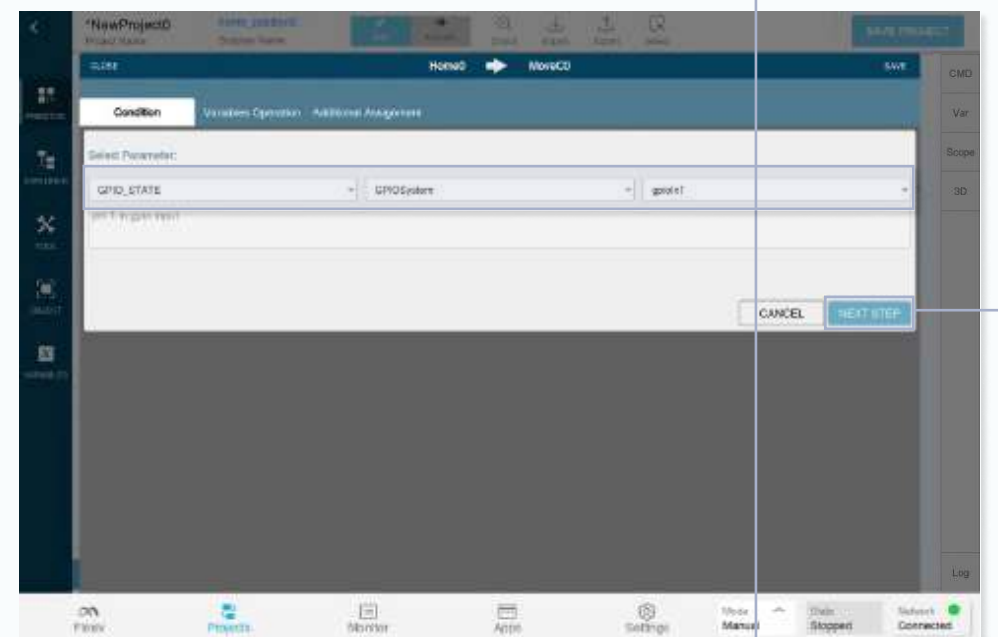
Part 1/2 Use input signal as transition condition

Example: Add **Input 1=1** as the transition condition to run **MoveC0** in **Subplan0**

Step: 1 2 3 4 5 6 7 8 9 10 11

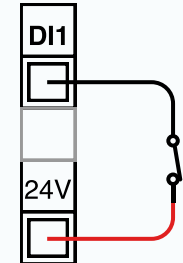


5 Tap + to add a condition



6 Select **GPIO_STATE**, **GPiOSystem**, **gpioIn1** in turn

7 Tap **NEXT STEP**



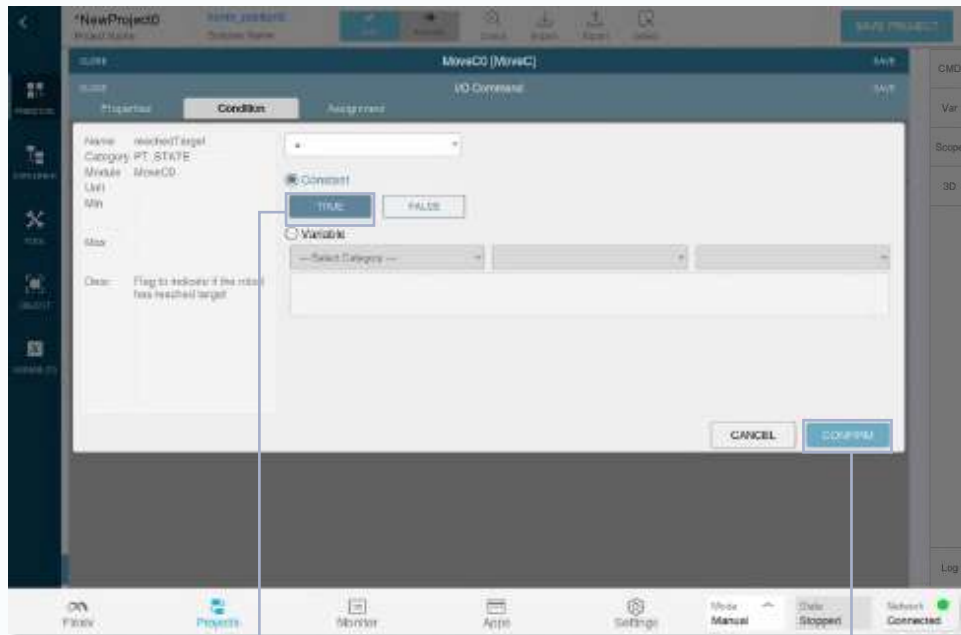
8.2 I/O Command Instruction

8.2.4 Configure in the Subplan

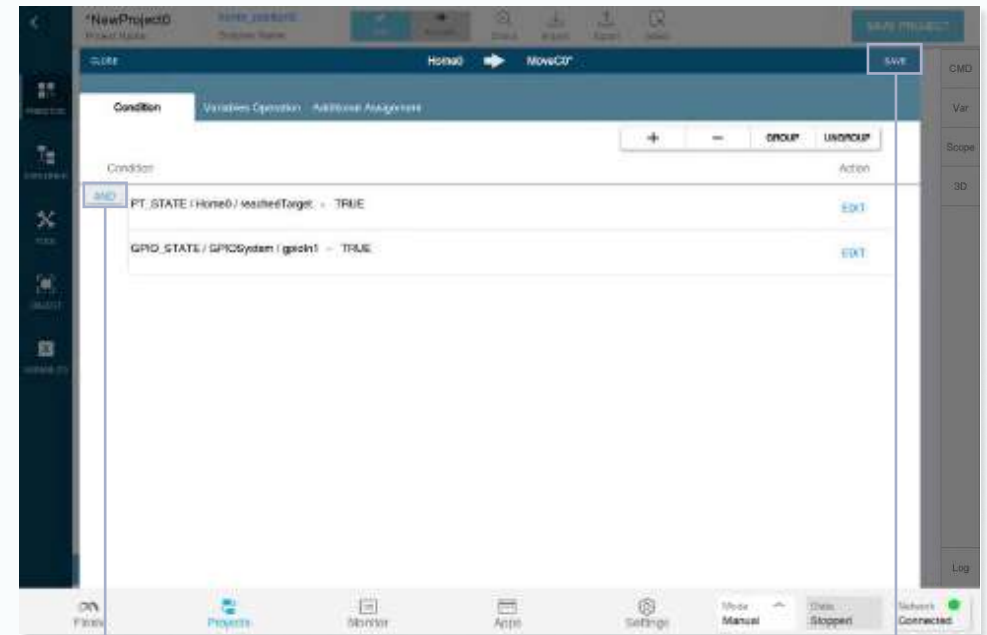
Part 1/2 Use input signal as transition condition

Example: Add **Input 1=1** as the transition condition to run **MoveC0** in **Subplan0**

Step: 1 2 3 4 5 6 7 8 9 10 11



- 8 Select **TRUE**
- 9 Tap **CONFIRM**



- 10 Change **OR** (default) to **AND**
- 11 Tap **SAVE**

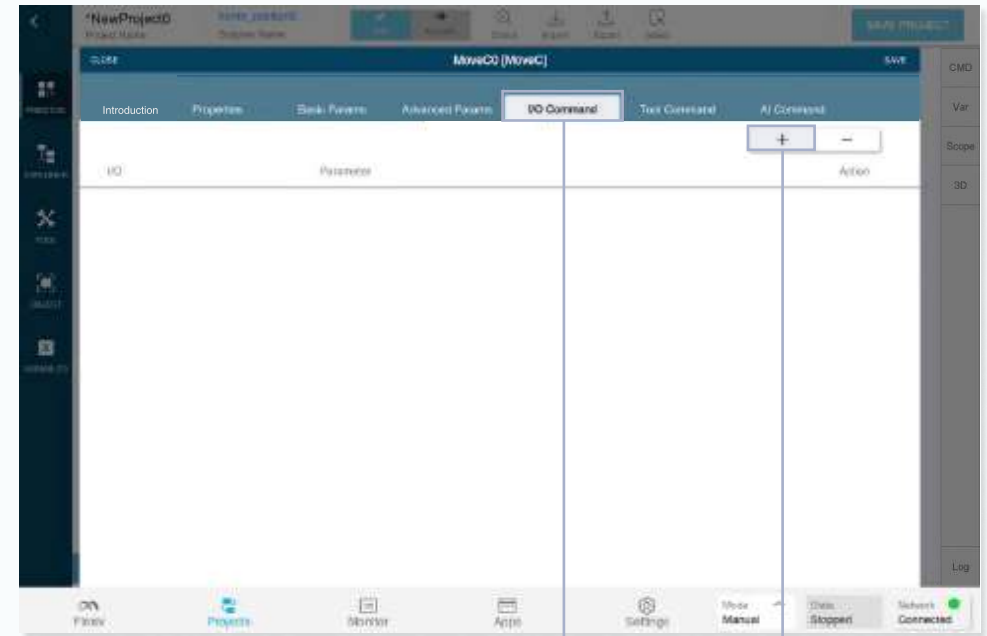
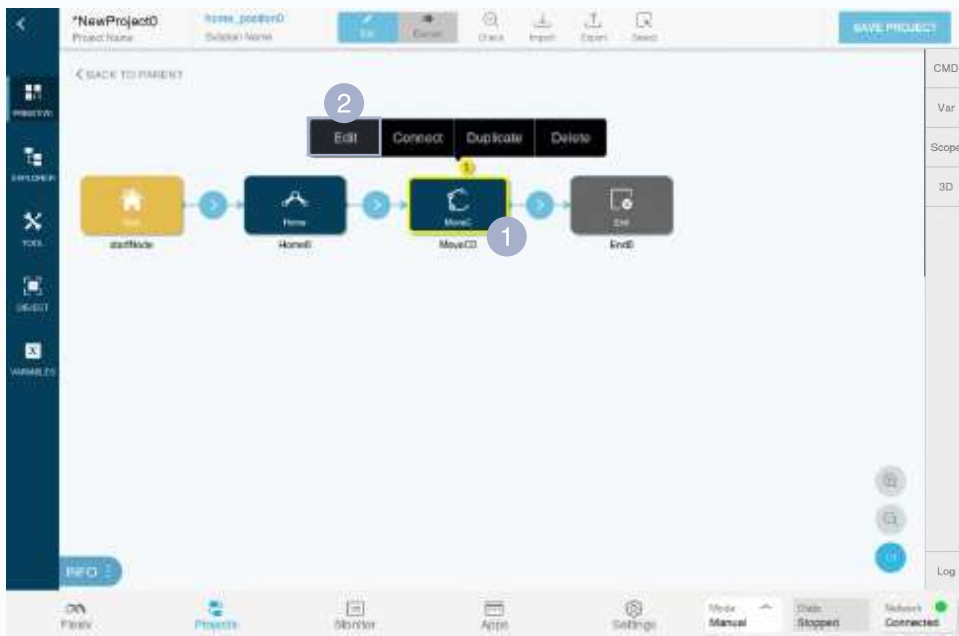
8.2 I/O Command Instruction

8.2.4 Configure in the Subplan

Part 2/2 Change output status

Example: Change the status of **Output 1** by configuring the I/O command of **MoveC0**

Step: 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16



1 Tap the **MoveC0** node

2 Tap **Edit**

3 Select the **I/O Command** tab

4 Tap **+** to add I/O Command

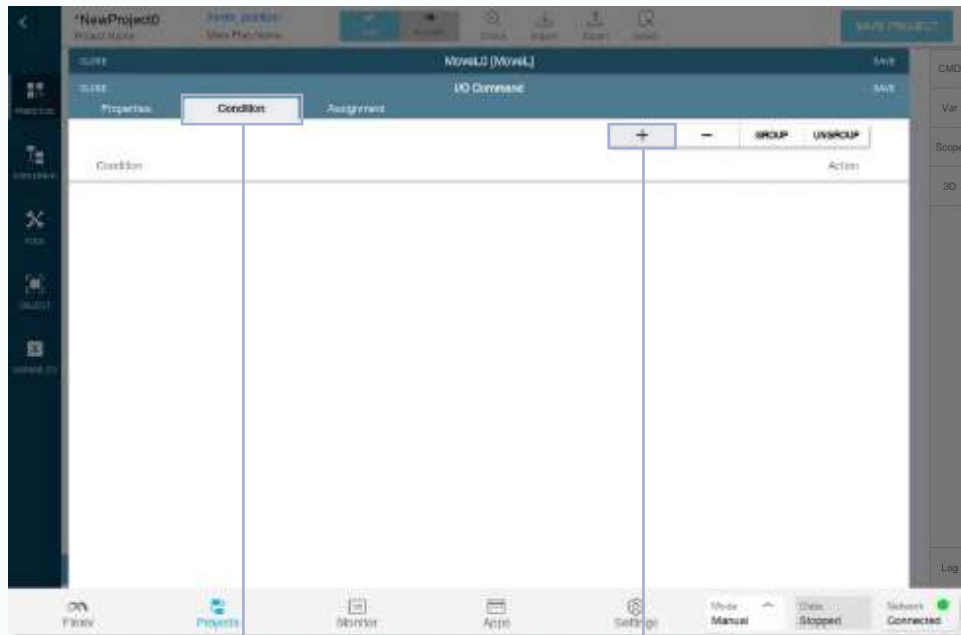
8.2 I/O Command Instruction

8.2.4 Configure in the Subplan

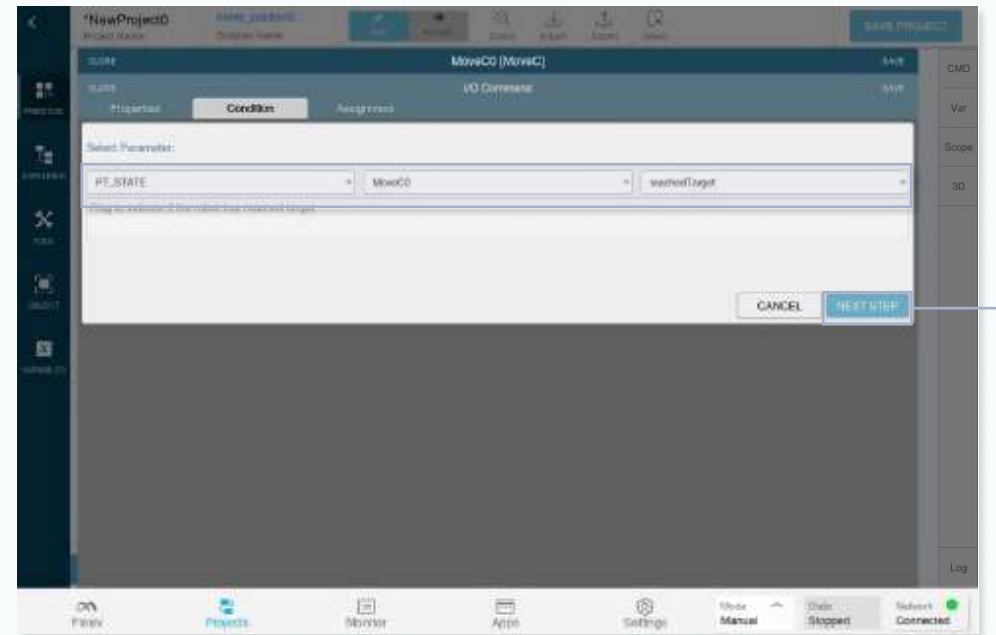
Part 2/2 Change output status

Example: Change the status of **Output 1** by configuring the I/O command of **MoveC0**

Step: 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16



- 5 Select the **Condition** tab
- 6 Tap **+** to add a condition



- 7 Select **PT_STATE**, **MoveC0**, **reachedtarget** in turn
- 8 Tap **NEXT STEP**

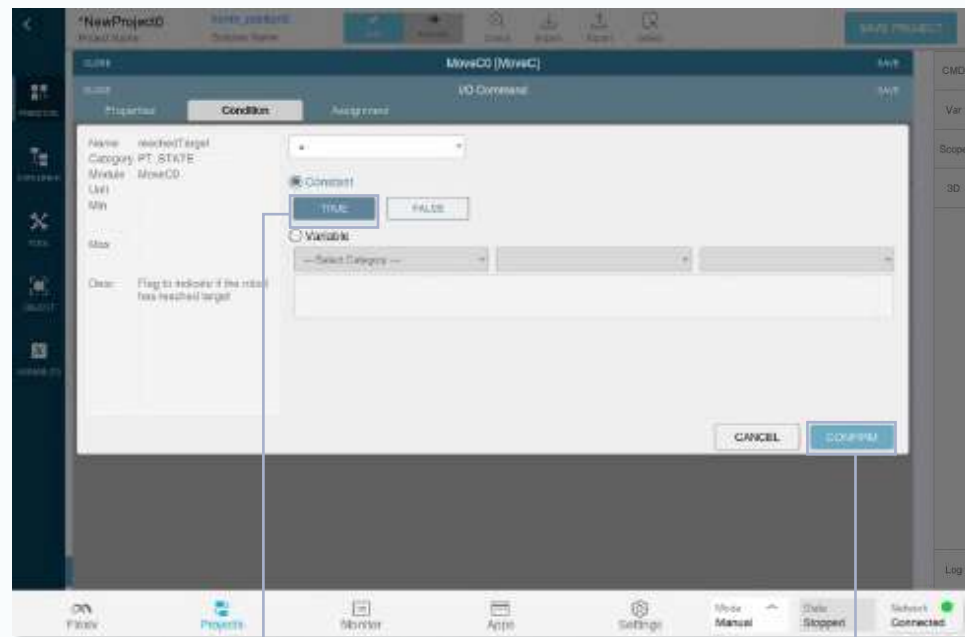
8.2 I/O Command Instruction

8.2.4 Configure in the Subplan

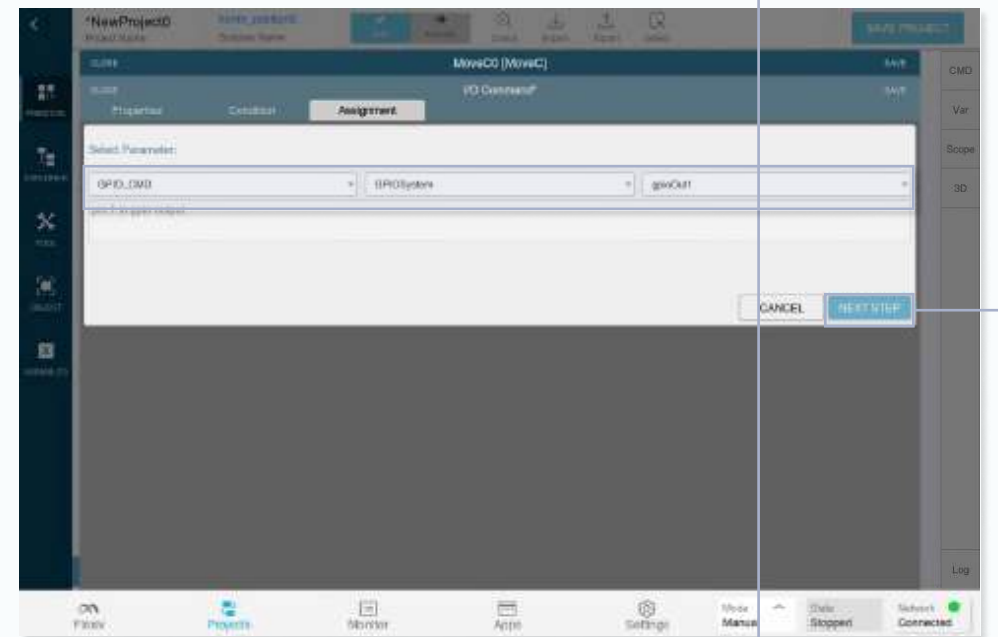
Part 2/2 Change output status

Example: Change the status of **Output 1** by configuring the I/O command of **MoveC0**

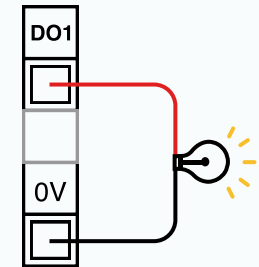
Step: 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16



- 9 Select **TRUE**
- 10 Tap **CONFIRM**



- 11 Select the **Assignment** tab
- 12 Select **GPIO_CMD**, **GPIOSystem**, **gpioOut1** in turn
- 13 Tap **NEXT STEP**



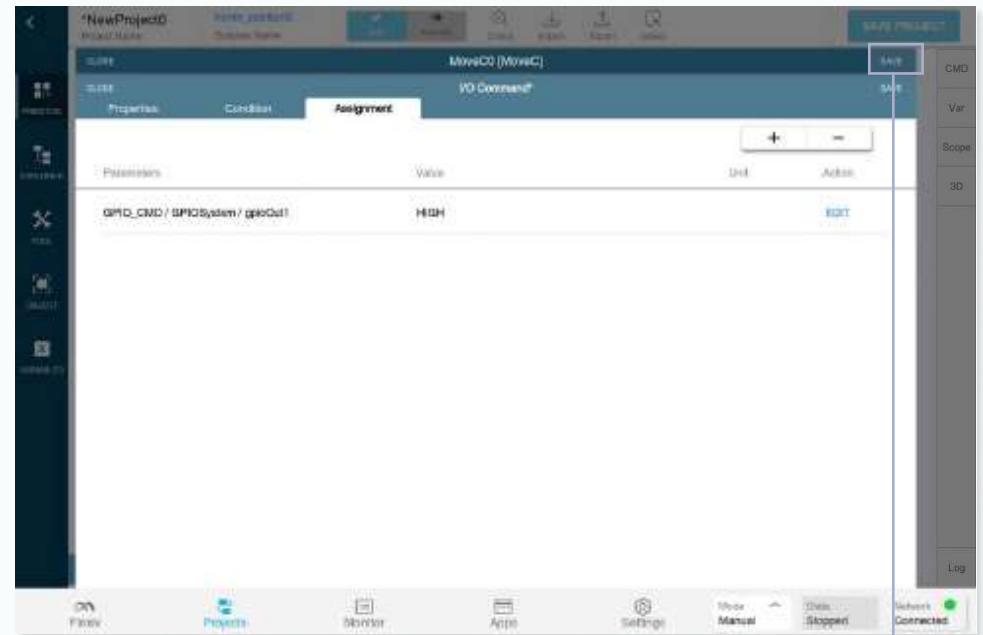
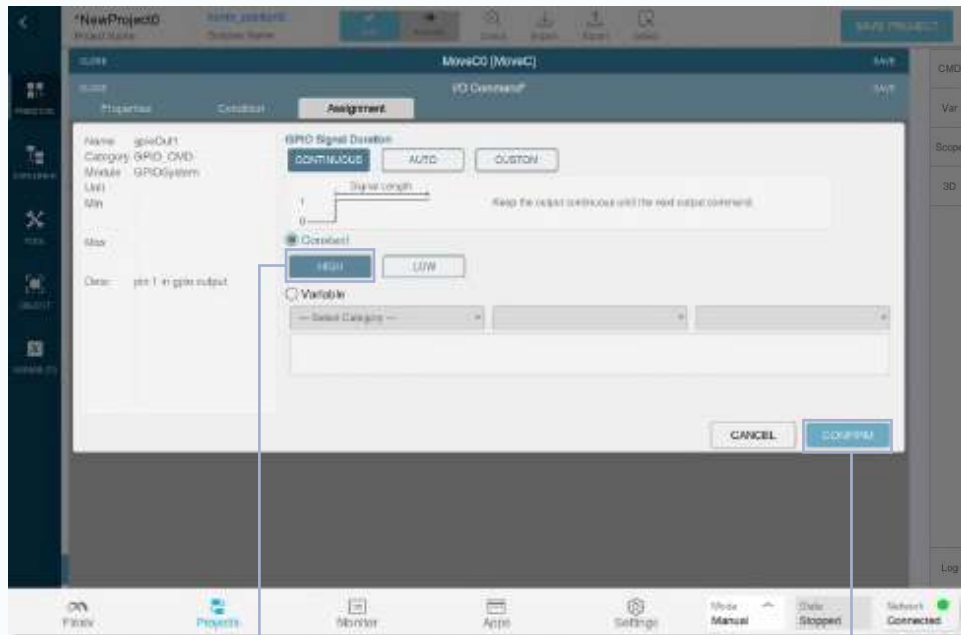
8.2 I/O Command Instruction

8.2.4 Configure in the Subplan

Part 2/2 Change output status

Example: Change the status of **Output 1** by configuring the I/O command of **MoveC0**

Step: 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16



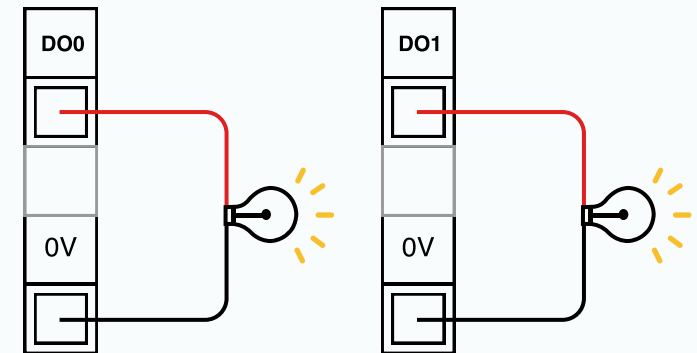
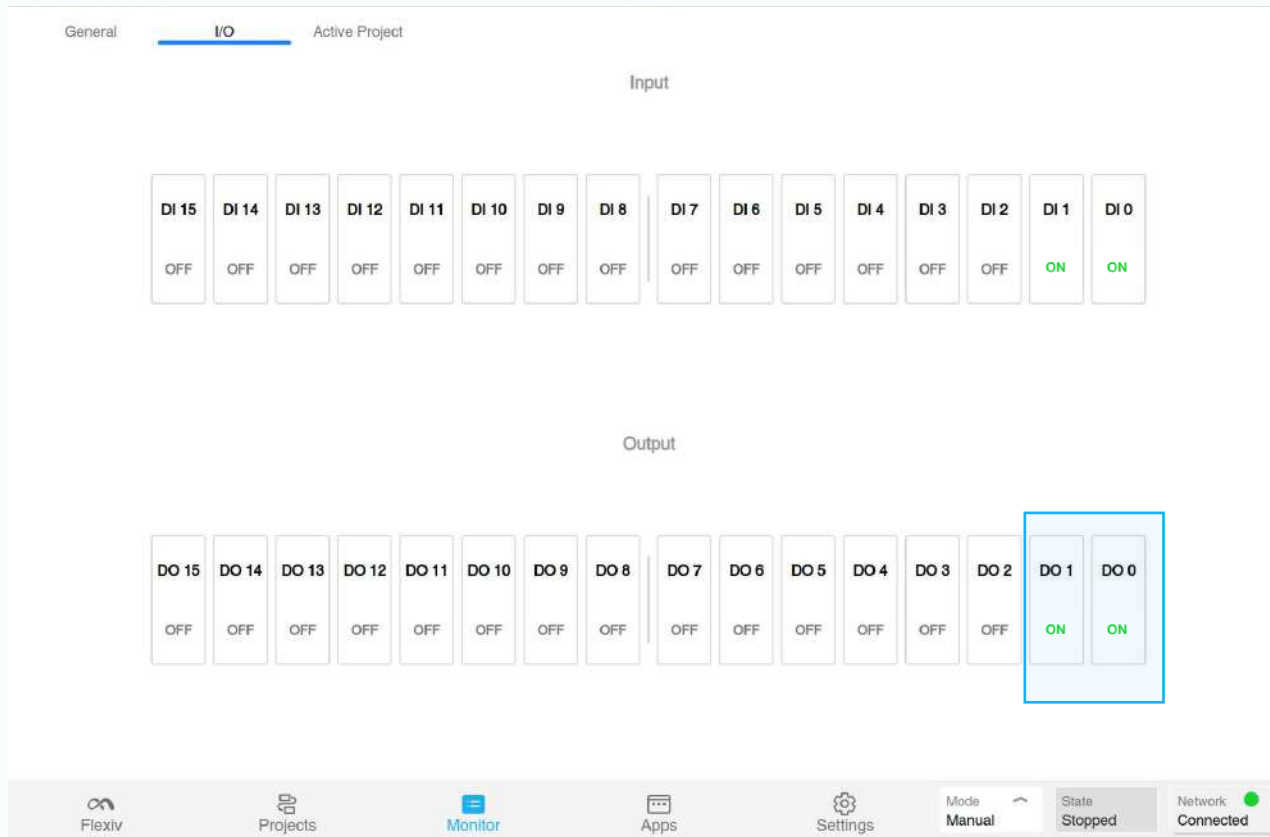
- 14 Select **HIGH**
- 15 Tap **CONFIRM**

- 16 Tap **SAVE**

8.2 I/O Command Instruction

8.2.5 Configuration Result

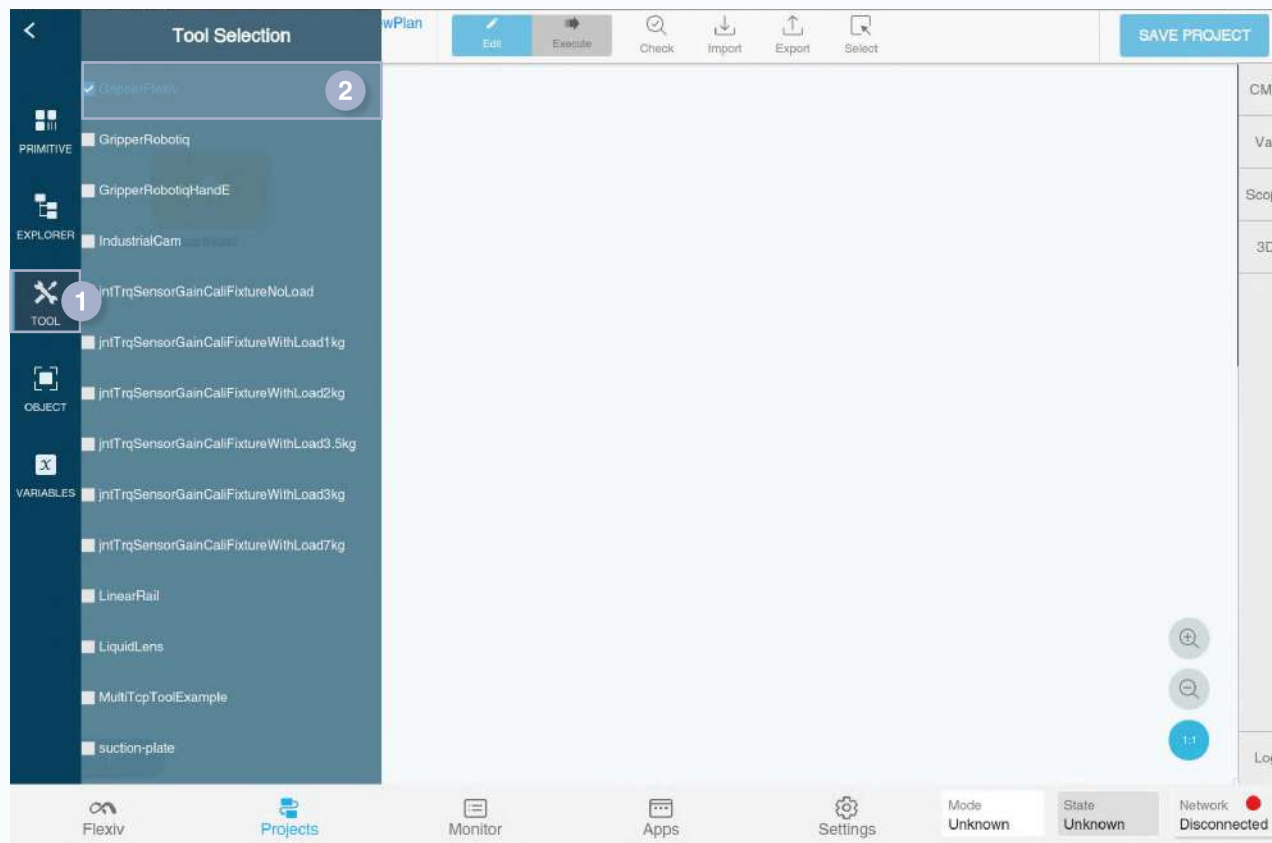
At the end of the plan, **DO0** and **DO1** are switched to **ON** in **Monitor** > **I/O** tab and on the Control Box.



8.3 Tool Command Instruction

8.3.1 Select the tool

This section explains how to configure tool commands with the following example.



1 In the **Projects** tab, tap



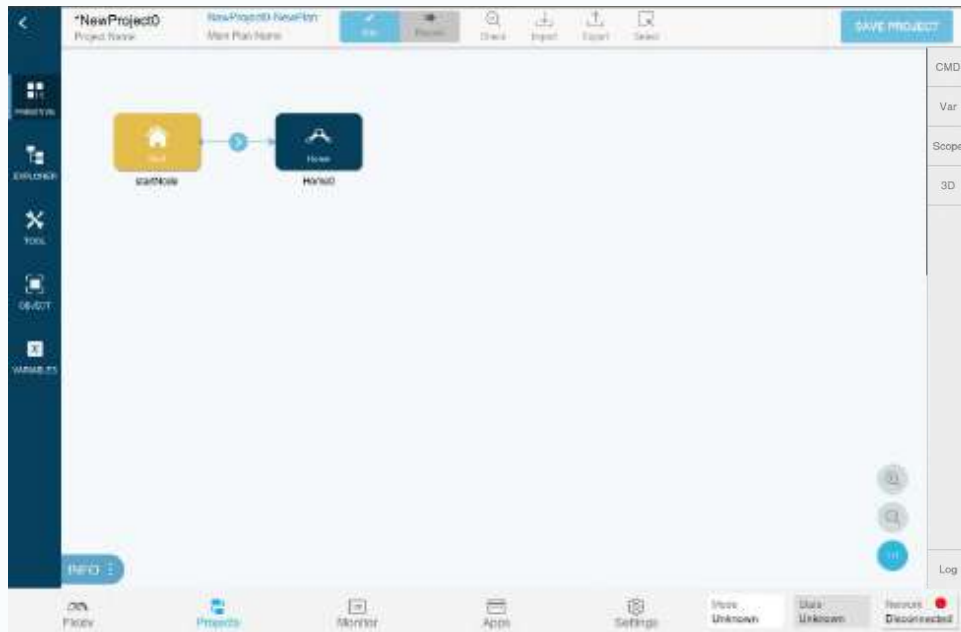
2 Select the tool that is currently being used

8.3 Tool Command Instruction

8.3.2 Move to the "home position" with the tool

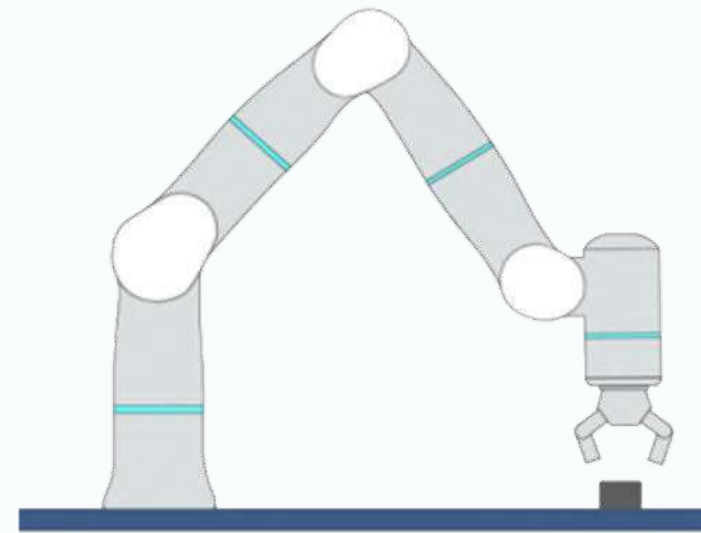
User Operation

- 1 Drag out **Home** from the **PRIMITIVE** list
- 2 Connect **startNode** and **Home0**



Robot Operation

Move to the "home position" with the tool

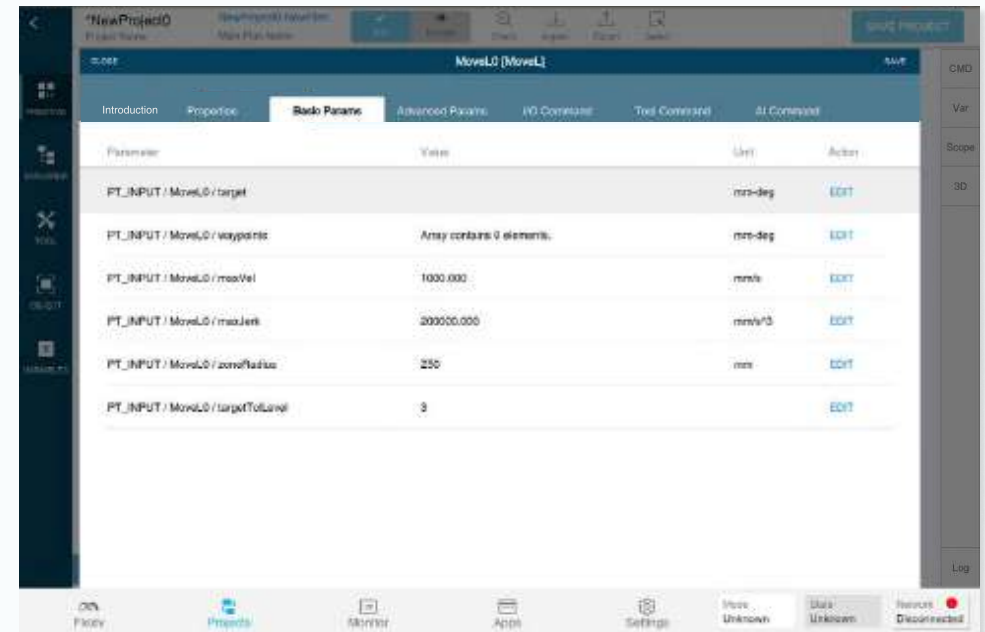


8.3 Tool Command Instruction

8.3.3 Move to a desired gripping pose

User Operation

- 1 Drag out **MoveL** from the **PRIMITIVE** list
- 2 Connect **Home0** and **MoveL0**
- 3 Tap **MoveL0**
- 4 Tap **Edit**
- 5 In **Basic Params** > **EDIT** > **PT_INPUT/MoveL0/target**, edit the parameter to teach a desired gripping pose
- 6 Tap **SAVE**, then tap **CLOSE**



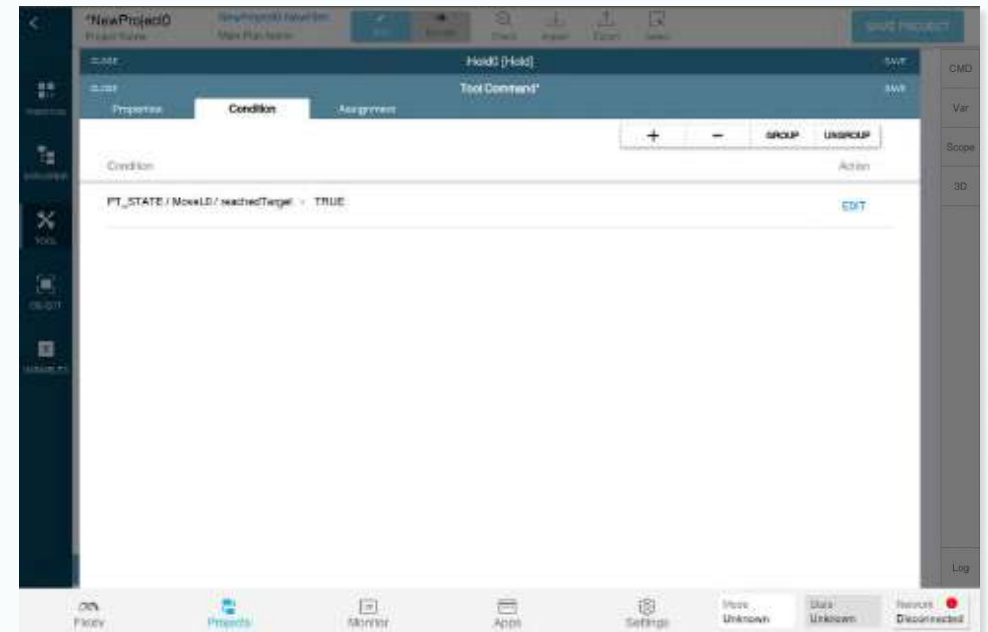
8.3 Tool Command Instruction

8.3.4 Pick up the workpiece by configuring Tool Command

Part 1/3

User Operation - 1. Edit Condition

- 1 Drag out **Hold** from the **PRIMITIVE** list
- 2 Connect **MoveL0** and **Hold0**
- 3 Tap **Hold0**
- 4 Tap **Edit**
- 5 In the **Tool Command** tab, tap **+**
- 6 Edit **Condition** as below



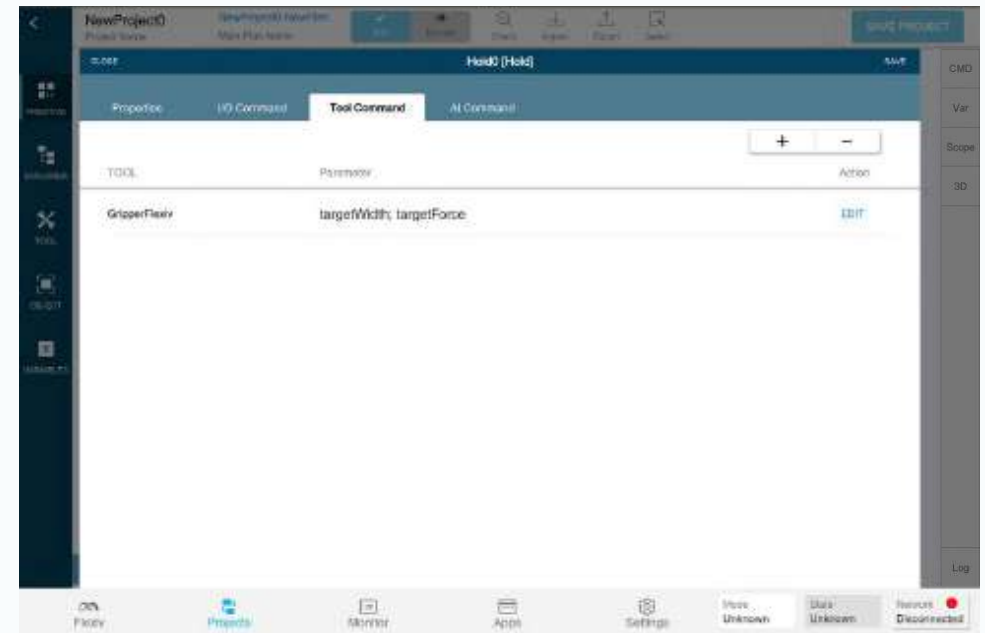
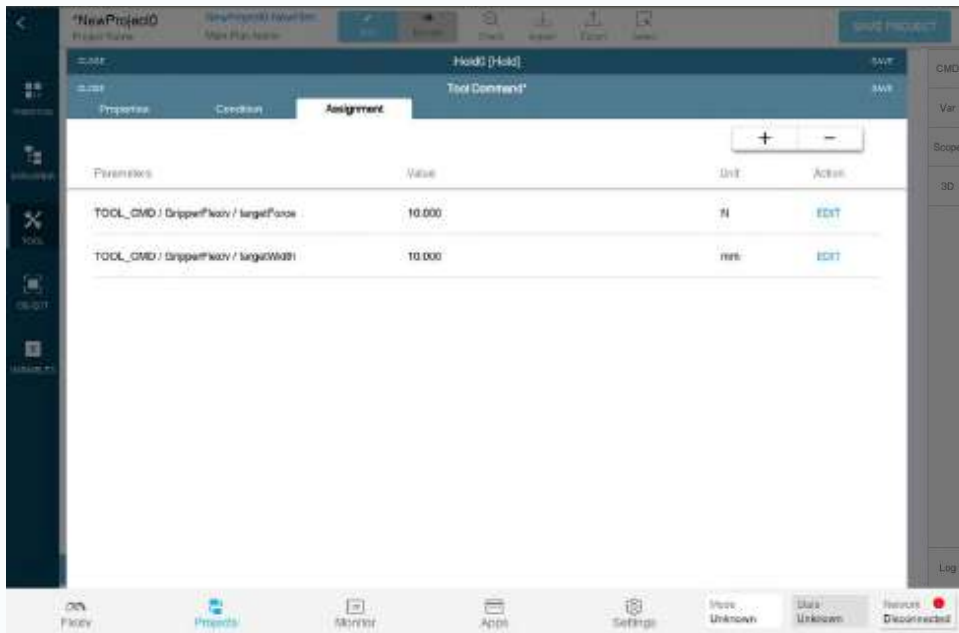
8.3 Tool Command Instruction

8.3.4 Pick up the workpiece by configuring Tool Command

Part 2/3

User Operation - 2. Edit Assignment

- 7 Select the **Assignment** tab
- 8 Tap **+**
- 9 Edit **Assignment** as below
- 10 Tap **SAVE**, then tap **CLOSE**
- 11 In the **Tool Command** tab, your last edited tool command is shown below
- 12 Tap **SAVE**, then tap **CLOSE**



8.3 Tool Command Instruction

8.3.4 Pick up the workpiece by configuring Tool Command

Part 3/3

When you finish configuring the tool command, the status of the

Hold0 node in the canvas is shown below.



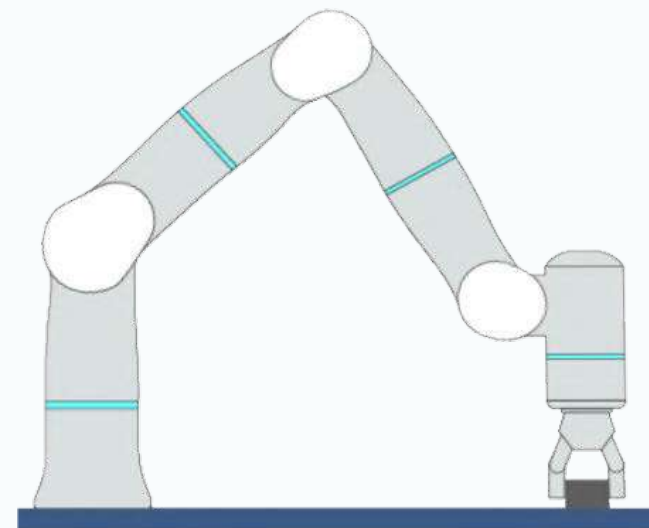
Hold0

1 : Number of tool command added to the node

● : Green is the representative color of tool command

Robot Operation

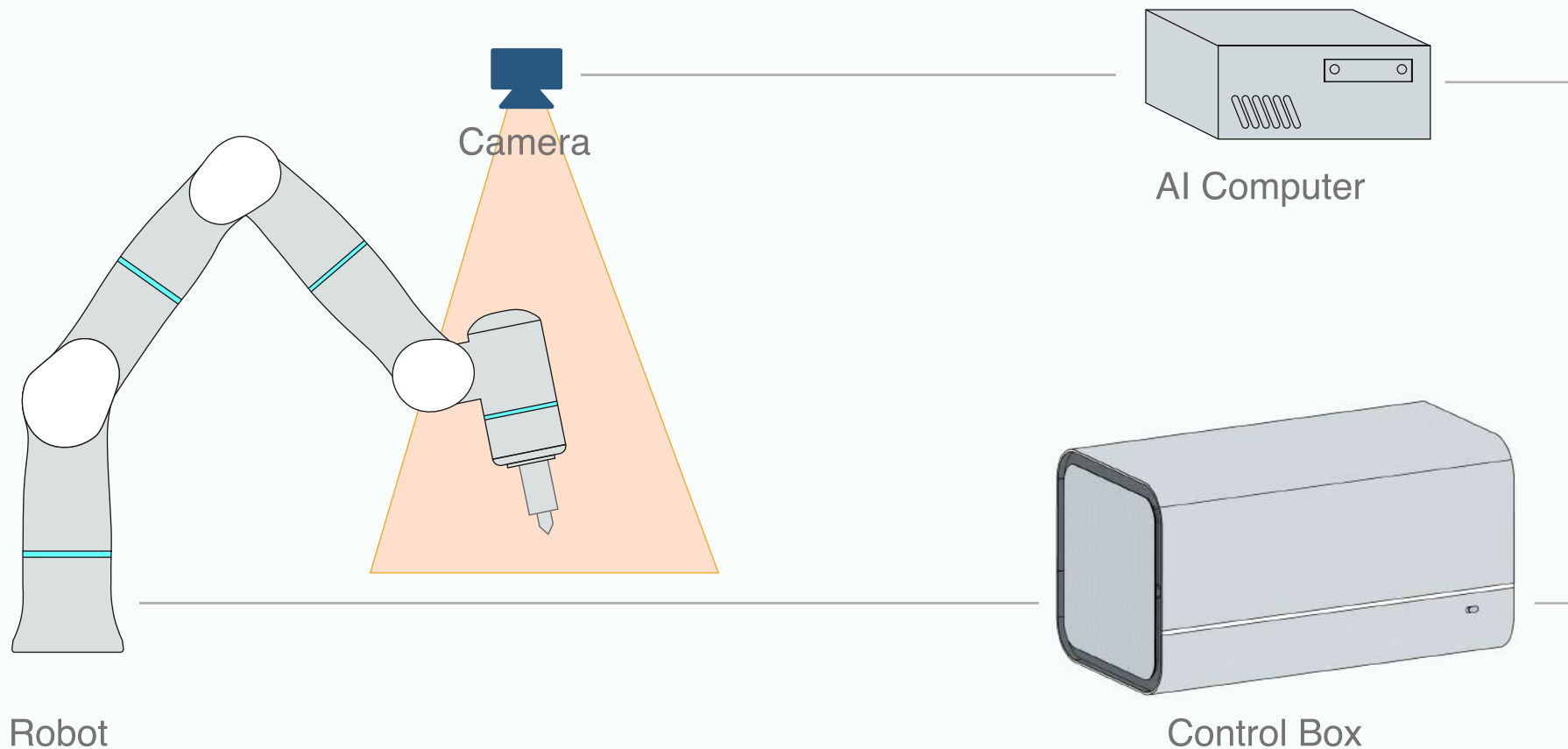
Pick up the workpiece with the tool



8.4 AI Command Instruction

8.4.1 Introduction

This section explains through an example how to configure AI commands. The connection between the robot and the AI system is shown in the figure below.

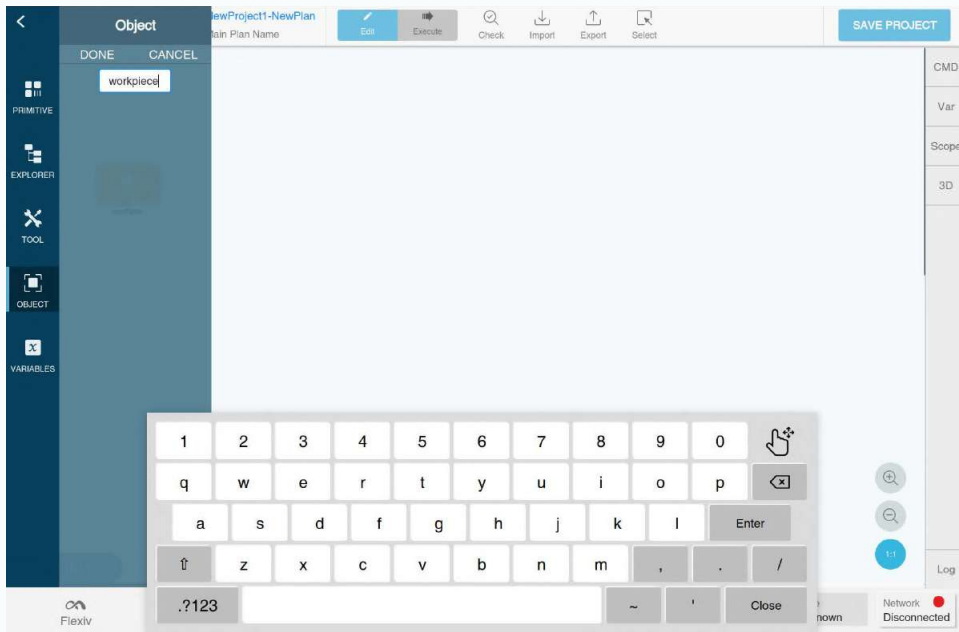


8.4 AI Command Instruction

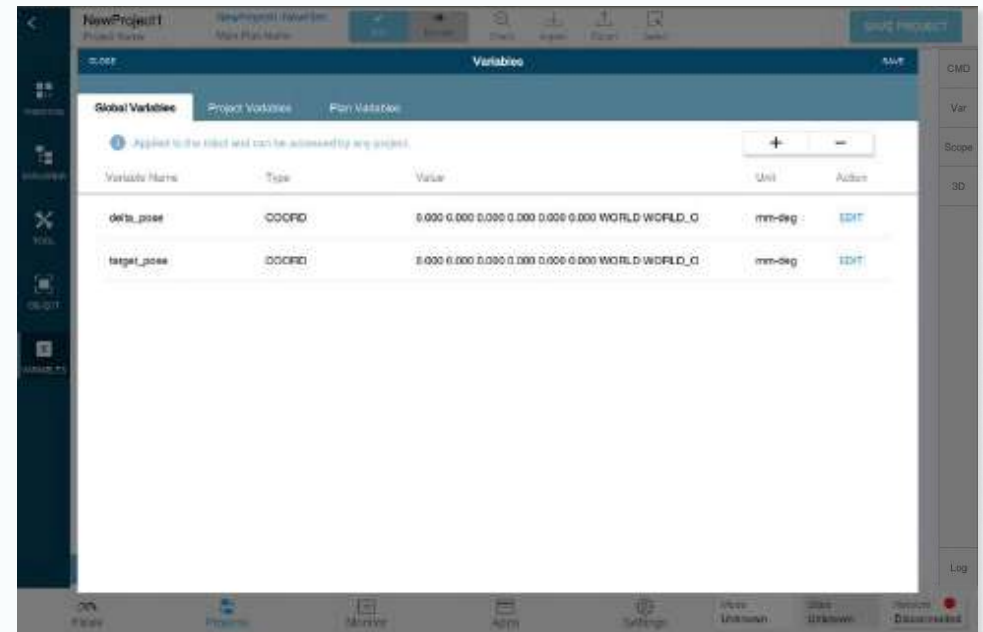
8.4.2 Add the object

User Operation

- 1 In the **Projects** tab, tap 
- 2 Enter a name (e.g. **workpiece**) for the object to be photographed by the camera



- 3 Tap 
- 4 In the **Global Variables** tab, edit the variables as below. These two variables are used to calculate the robot's pose during the use of the AI command



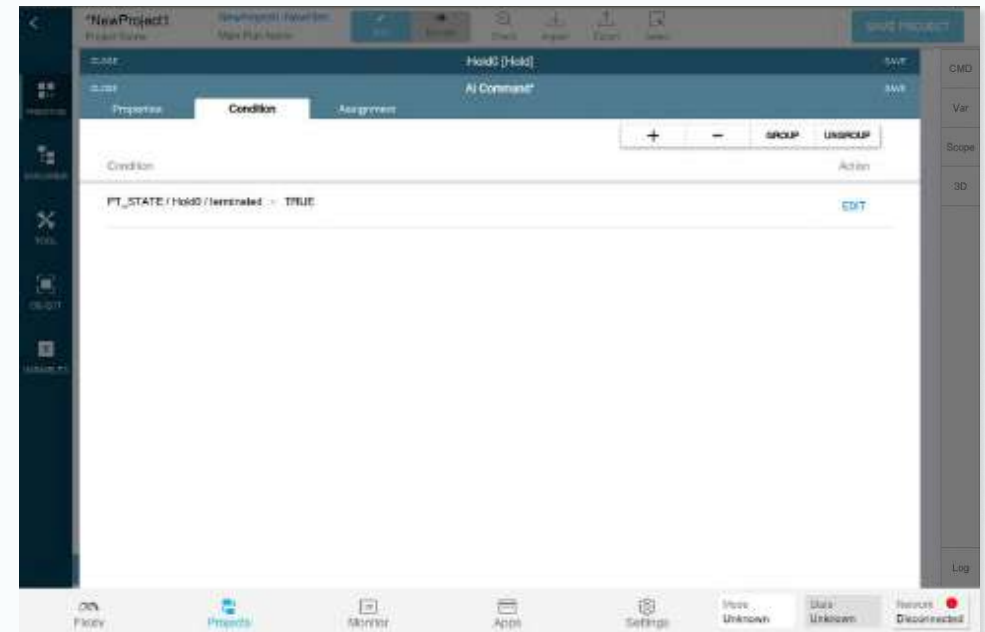
8.4 AI Command Instruction

8.4.3 Clear the original information of AI Computer

Part 1/2

User Operation

- 1 Drag out **Hold** from the **PRIMITIVE** list
- 2 Connect **startNode** and **Hold0**
- 3 Tap **Hold0**, then tap **Edit**
- 4 Select the **AI Command** tab, tap **+**
- 5 In the **Condition** tab, edit the condition as below



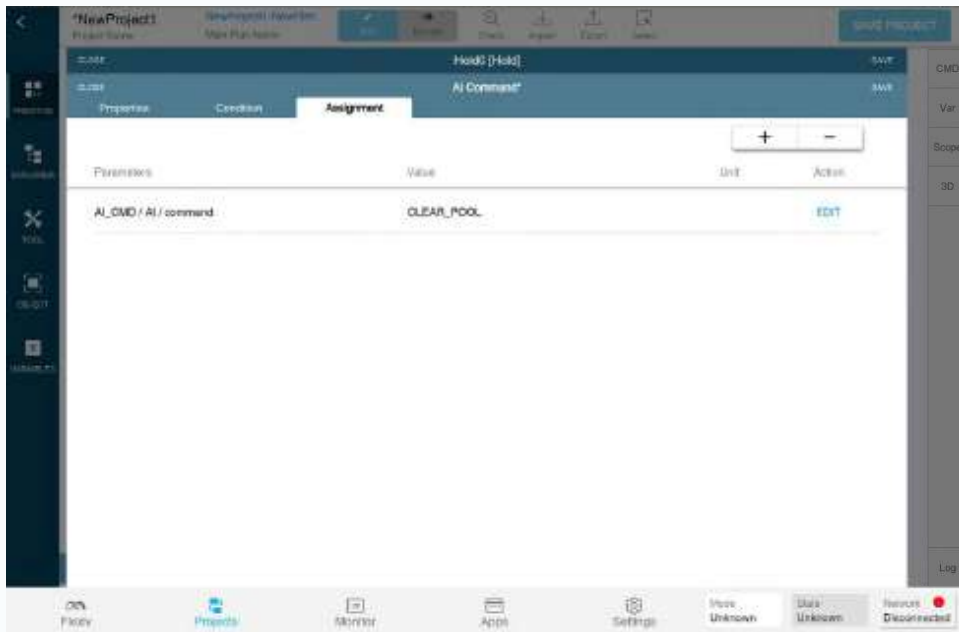
8.4 AI Command Instruction

8.4.3 Clear the original information of AI Computer

Part 2/2

User Operation

- 6 Select the **Assignment** tab, tap 
- 7 Edit **Assignment** as below
- 8 Tap **SAVE**, then tap **CLOSE**



When you finish editing an AI command, the status of **Hold0** in the canvas is shown below.



1 : Number of AI command added to the node

● : Red is the representative color of AI command



8.4 AI Command Instruction

8.4.4 Move to the original pose

User Operation

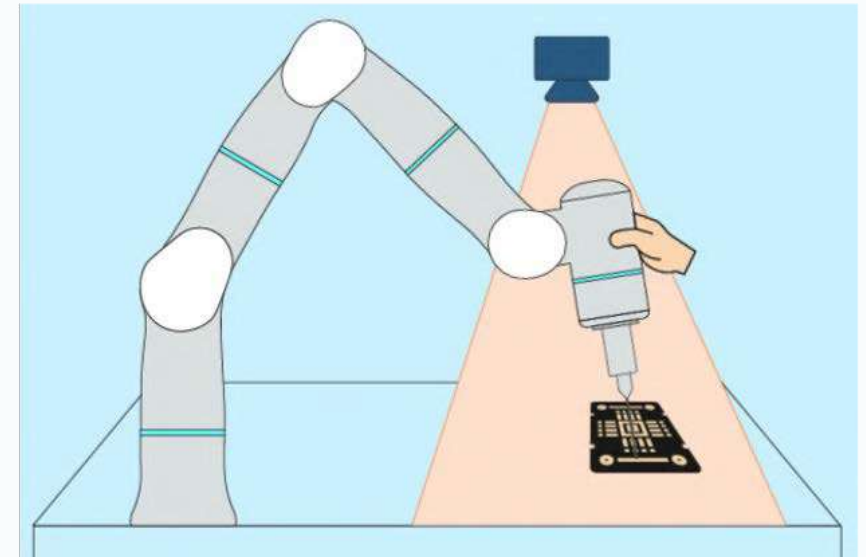
- 1 Drag out **MoveL** from the **PRIMITIVE** list
- 2 Connect **Hold0** and **MoveL0**
- 3 Tap **MoveL0**
- 4 Tap **Edit**



- 5 Select the **Basic Params** tab, tap **EDIT** to teach **PT_INPUT/MoveL0/target** to a target pose
- 6 Tap **SAVE**, then tap **CLOSE**

Robot Operation

Move to the target pose



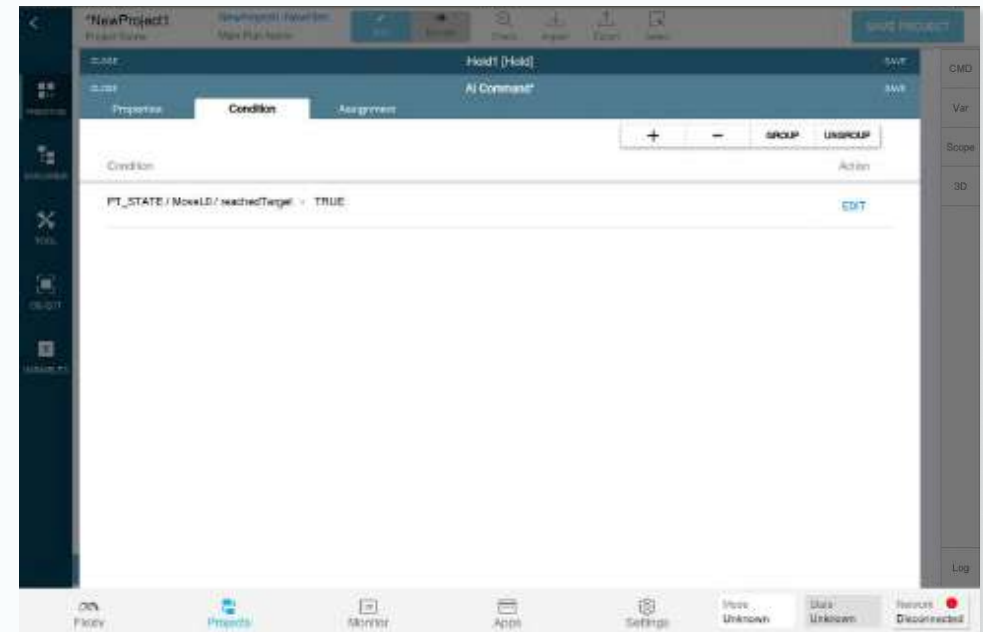
8.4 AI Command Instruction

8.4.5 Take the picture for the target point with AI Command

Part 1/2

User Operation

- 1 Drag out **Hold** from the **PRIMITIVE** list
- 2 Connect **MoveL0** and **Hold1**
- 3 Tap **Hold1**
- 4 Tap **Edit**
- 5 Select the **AI Command** tab, tap **+**
- 6 In the **Condition** tab, edit the condition as below



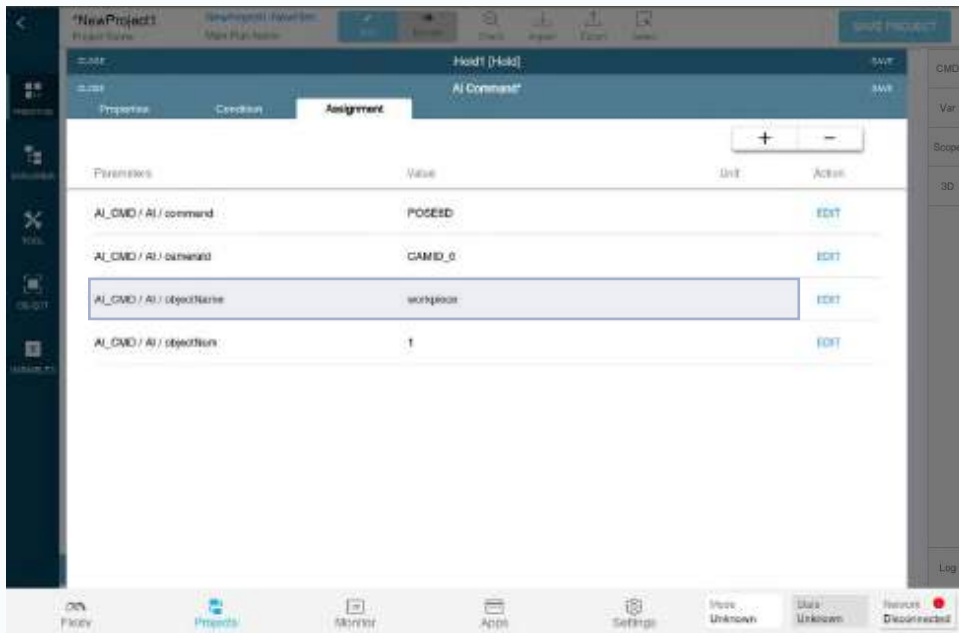
8.4 AI Command Instruction

8.4.5 Take the picture for the target point with AI Command

Part 2/2

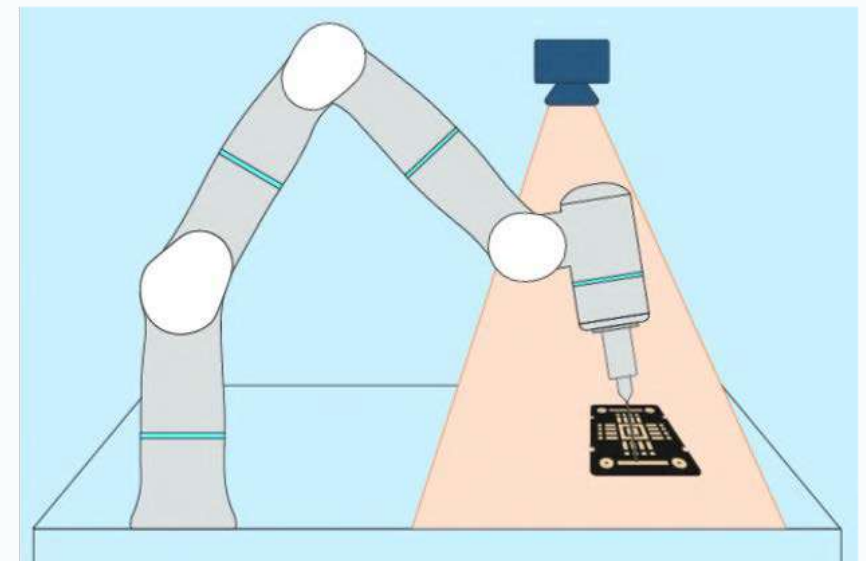
User Operation

- 7 Select the **Assignment** tab, tap **+**
- 8 Edit **Assignment** as below (the object name here is the variable created in 8.4.2)
- 9 Tap **SAVE**, then tap **CLOSE**



Robot Operation

Take picture to get target point of object named “workpiece” with AI command



8.4 AI Command Instruction

8.4.6 Move to target point of the workpiece

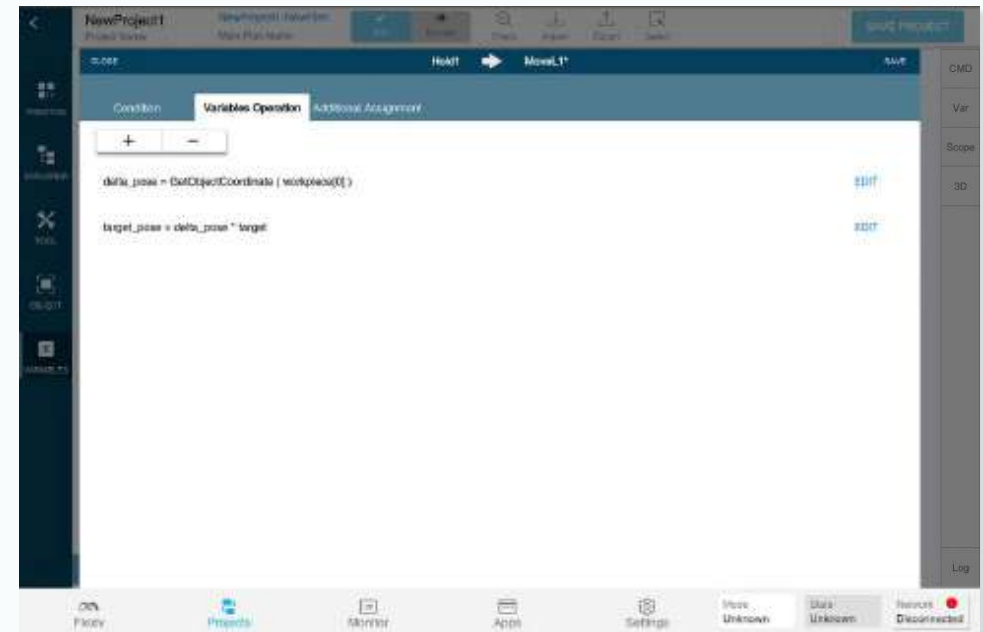
Part 1/2

User Operation

- 1 Drag out **MoveL** from the **PRIMITIVE** list
- 2 Connect **Hold1** and **MoveL1**
- 3 Tap **>** between **Hold1** and **MoveL1**
- 4 Tap **Edit**



- 5 Select the **Variables Operation** tab, tap **+**
Add the operations as below
- 6 Tap **SAVE**, then tap **CLOSE**



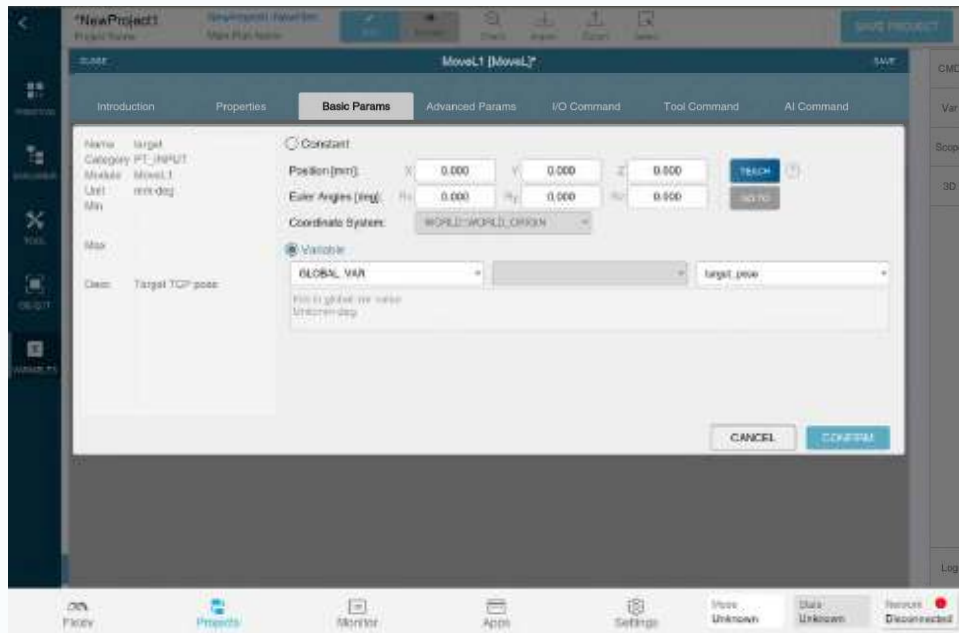
8.4 AI Command Instruction

8.4.6 Move to target point of the workpiece

Part 2/2

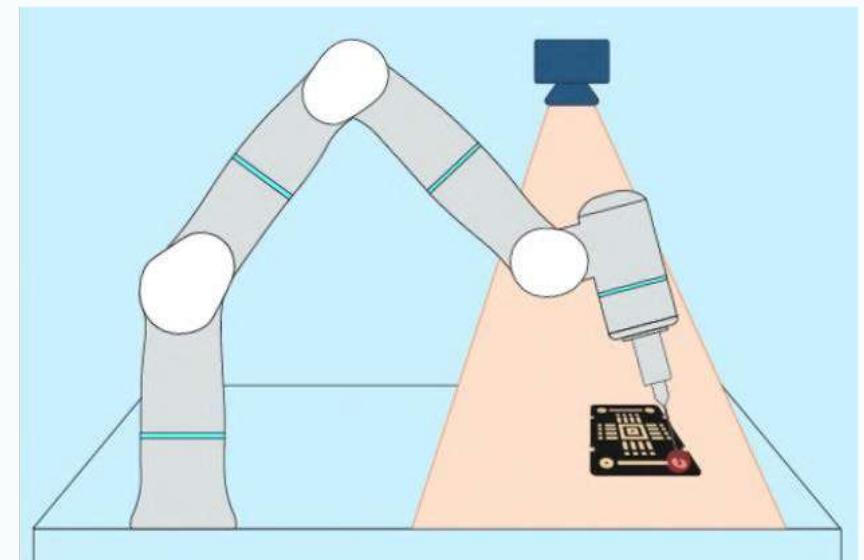
User Operation

- 7 Tap **MoveL1**, then tap **Edit**
- 8 Select the **Basic Params** tab, tap **EDIT** to edit **PT_INPUT/MoveL0/target** to a target pose
- 9 Tap **SAVE**, then tap **CLOSE**



Robot Operation

Move to target point of the workpiece



9.1 Remote Mode Configuration Instruction

9.1.1 Introduction

This section explains how to configure **Remote Mode**, so as to control the robot via peripheral devices.

When Remote Mode is enabled, the peripheral device sends the relevant robot process signals (such as move enabling, fault confirmation, program startup, etc.) to the robot through the active external interface. The robot sends information about the running state and fault state to the peripheral device.

9.1 Remote Mode Configuration Instruction

9.1.2 List of I/O Signals

#	Input Signal	Signal Type	Description
1	Start Program	Rising edge	Execute the selected plan when this signal changes from OFF to ON then to OFF .
2	Stop Program	Rising edge	Stop the plan running when this signal changes from OFF to ON then to OFF .
3	Pause Program	Rising edge	Pause the selected plan when this signal changes from OFF to ON then to OFF .
4	Enable Move	High discrete signal	Robot can move only when this signal is ON ; once it's OFF , robot can no longer move.
5	Validate Program	Rising edge	Notify the robot to set the received Plan ID when this signal changes from OFF to ON then to OFF .
6	Clear Fault	Rising edge	Clear minor errors of the robot when this signal changes from OFF to ON then to OFF .
7	Motor On	Rising edge	Switch on the motor and release the brakes when this signal changes from OFF to ON then to OFF .
8	Plan ID	Bit list (I/O)	Plan ID received from host and will be applied to the robot control box when the input signal Validate Program is switched ON .
		Integer (Modbus TCP, Profinet, Anybus)	
9	Speed Ratio*	Integer	Speed ratio received from host, range [1,100], unit %.

*Note: Speed Ratio is only available for Modbus TCP, Profinet, and Anybus.

9.1 Remote Mode Configuration Instruction

9.1.2 List of I/O Signals

#	Output Signal	Signal Type	Description
1	E-Stop	Low discrete signal	If the robot is in emergency stop state, this output signal is OFF , otherwise, it is ON .
2	Remote Active	High discrete signal	If Remote Mode is enabled, this output signal is ON , otherwise, it is OFF .
3	Program Request	High discrete signal	When the robot is ready to receive Plan ID, this output signal is ON , otherwise, it is OFF . * ON Condition: FaultState = OFF , ProgramRunning = OFF , MotorOnState = ON
4	Program Running	High discrete signal	If a plan is running, this output signal is ON , otherwise, it is OFF .
5	Program Paused	High discrete signal	If a running plan is paused, this output signal is ON , otherwise, this signal is OFF .
6	Fault State	High discrete signal	If the robot is in fault state, this output signal is ON , otherwise, it is OFF .
7	Motor On State	High discrete signal	If robot motor is ON , this output signal is ON , otherwise, it is OFF .
8	Plan ID State	Bit list (I/O)	Specify the current activated Plan ID
		Integer (Modbus TCP, Profinet, Anybus)	

9.1 Remote Mode Configuration Instruction

9.1.3 Remote Mode - I/O

A robot can be controlled externally via GPIOs on the Control Box.

This instruction consists of five parts:

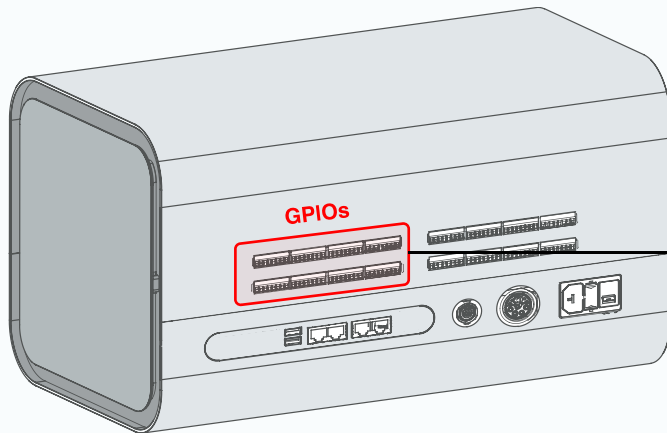
Part	Instruction	Page
1	Physical connection between the robot and a peripheral device	Page 105
2	Configuration in Settings > Remote Mode	Page 106 ~ 107
3	An example of how to use the input signal of Plan ID	Page 108
4	Steps to start a plan via I/O signals	Page 109
5	Steps to pause/resume and stop/restart a plan via I/O signals	Page 110

9.1 Remote Mode Configuration Instruction

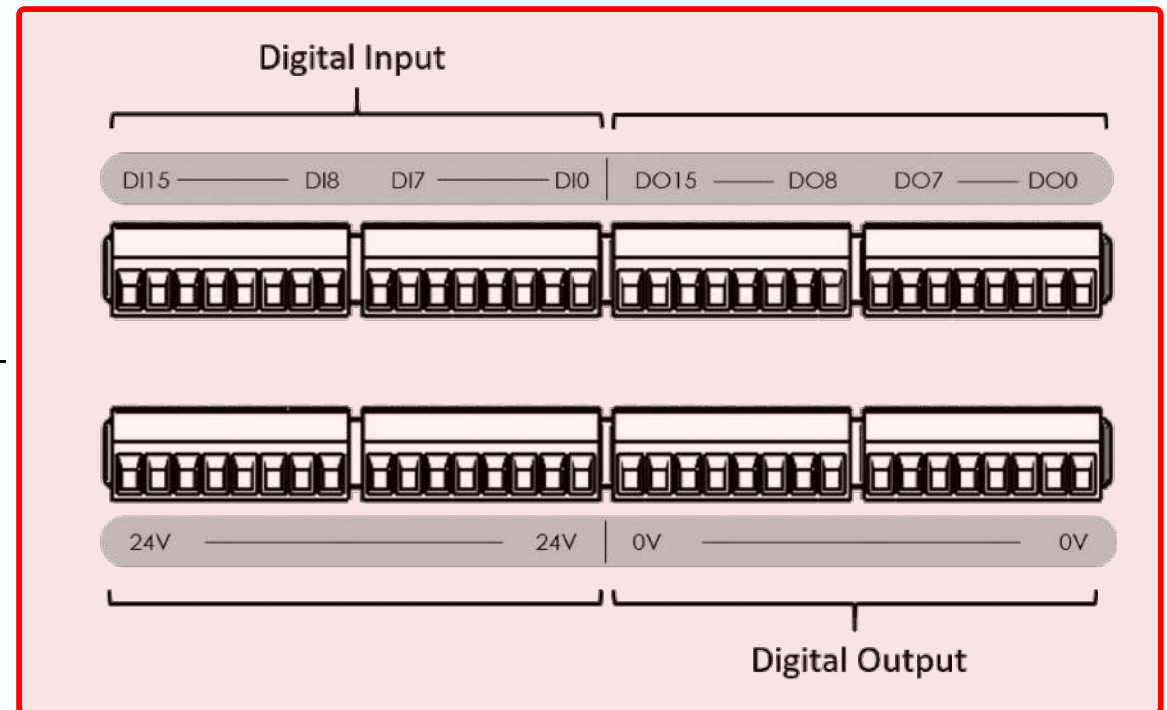
9.1.3 Remote Mode - I/O

Part 1 - Physical connection between the robot and a peripheral device

A robot's GPIOs are located on the Control Box, as the one shown in the illustration below. Use those ports for the physical connection between the robot and a peripheral device.



Control Box



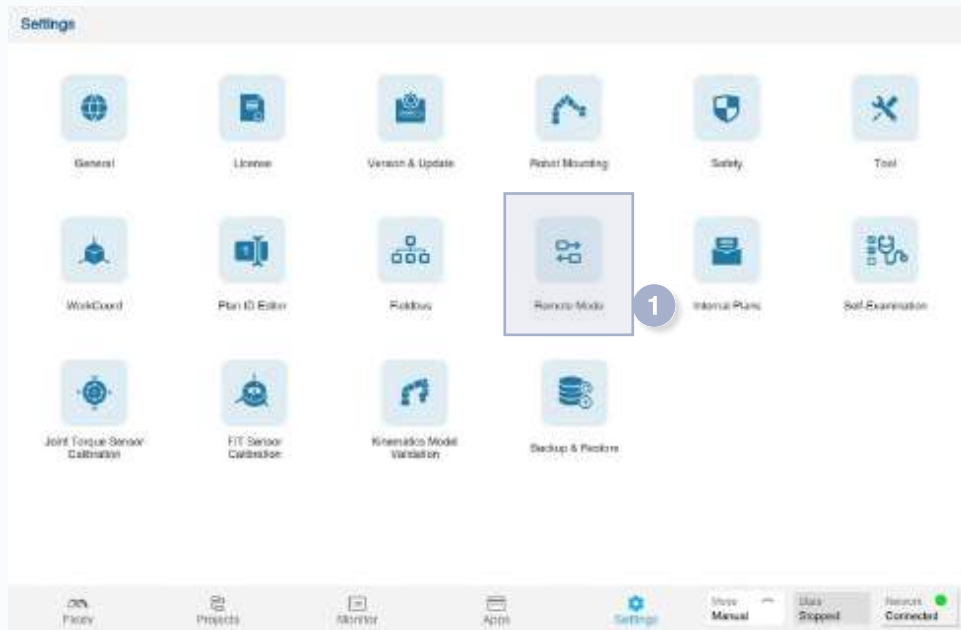
GPIOs on the Control Box

9.1 Remote Mode Configuration Instruction

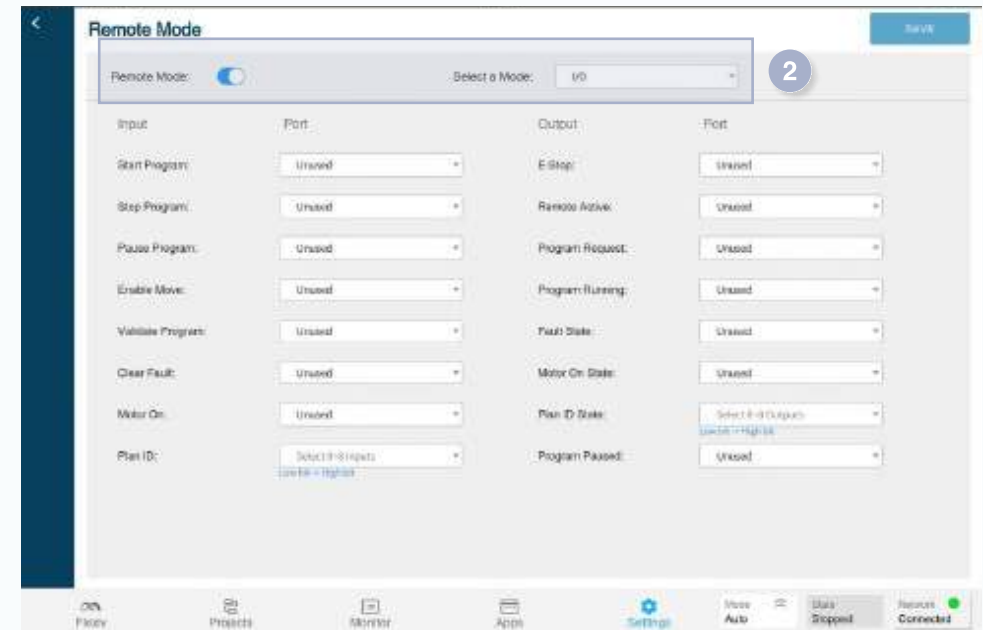
9.1.3 Remote Mode - I/O

Part 2 - Configuration in Settings > Remote Mode

Steps: 1 2 3 4



1 Go to **Settings > Remote Mode**.



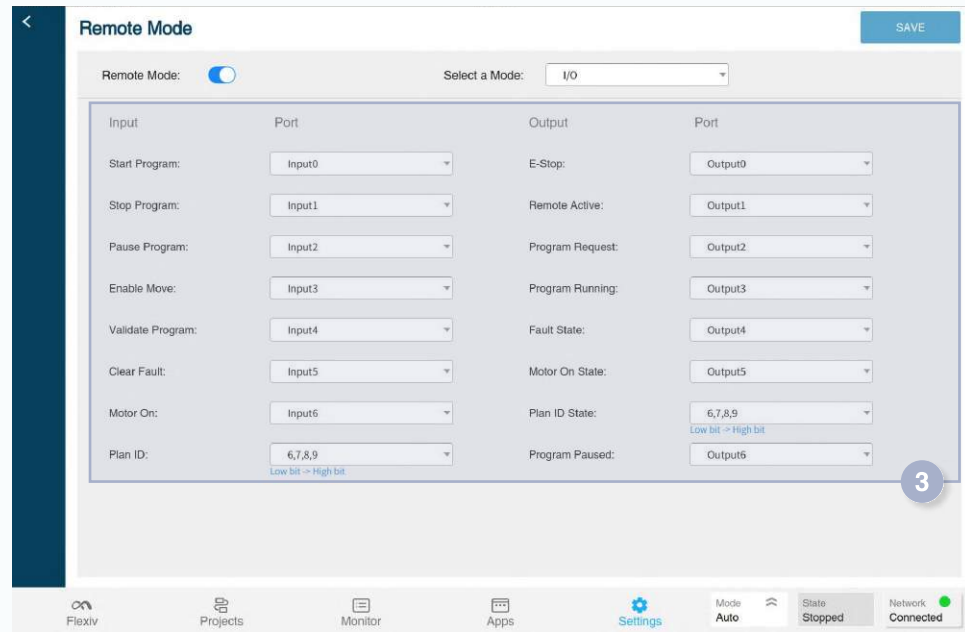
2 To enable **Remote Mode**, switch the toggle from  to  , then select **I/O** from the **Select a Mode** drop-down menu.

9.1 Remote Mode Configuration Instruction

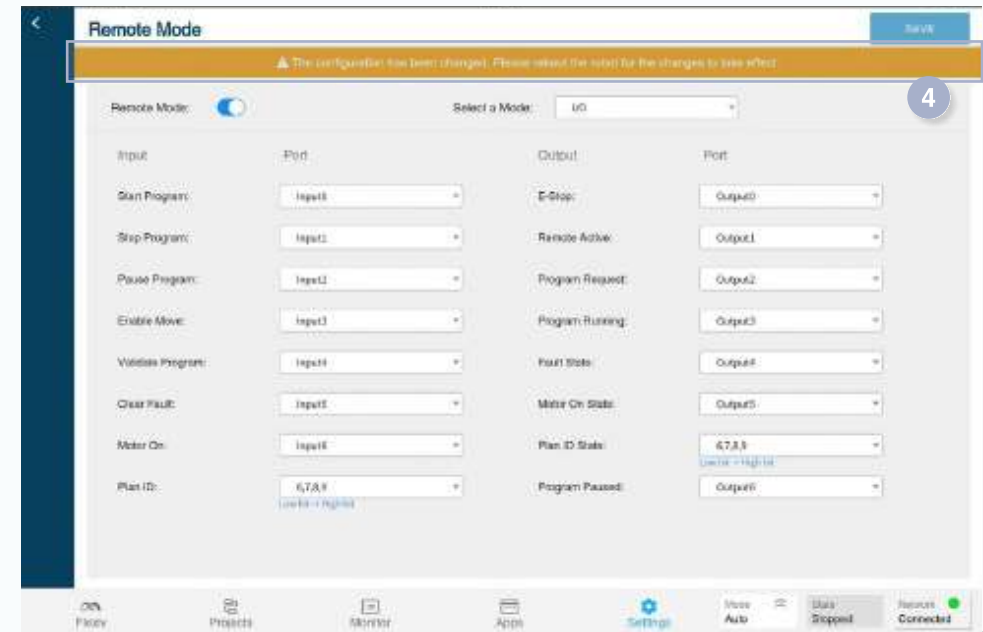
9.1.3 Remote Mode - I/O

Part 2 - Configuration in Settings > Remote Mode

Steps: 1 2 3 4



3 Configure input and output signals, and tap to save the changes.



4 Reboot the robot for the changes to take effect.

9.1 Remote Mode Configuration Instruction

9.1.3 Remote Mode - I/O

Part 3 - An example of how to use the input signal of Plan ID

In this example, the goal is to assign the plan whose ID is 10 using the input signal of Plan ID. A plan's ID can be set and edited in **Settings > Plan ID Editor** and conforms to the decimal number system. On the contrary, the input signal of Plan ID conforms to the binary number system. See the instructions below for details.

Step	Calculation	Reference																				
1	The input signal of Plan ID conforms to the binary number system. To be converted to 10, it needs at least 4 ports ($2^4 = 16 > 10$). In this example, assign ports of DI6, DI7, DI8, DI9 to the input signal of Plan ID.	Plan ID: Select 0~8 Inputs Low bit --> High bit Plan ID: 6,7,8,9 Low bit --> High bit																				
2	The binary equivalent for the decimal number 10_{10} is 1010_2 . Assign the rightmost calculated binary value [0] to the lowest input port DI6, and the leftmost calculated binary value [1] to the highest input port DI9. The remaining part is [1] to DI7, and [0] to DI8.	<table><tr><td>Input port</td><td>DI9</td><td>DI8</td><td>DI7</td><td>DI6</td></tr><tr><td>Bit order</td><td>3</td><td>2</td><td>1</td><td>0</td></tr><tr><td>Binary value</td><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>Plan ID</td><td colspan="4">$10 = 1 \cdot 2^3 + 0 \cdot 2^2 + 1 \cdot 2^1 + 0 \cdot 2^0$</td></tr></table>	Input port	DI9	DI8	DI7	DI6	Bit order	3	2	1	0	Binary value	1	0	1	0	Plan ID	$10 = 1 \cdot 2^3 + 0 \cdot 2^2 + 1 \cdot 2^1 + 0 \cdot 2^0$			
Input port	DI9	DI8	DI7	DI6																		
Bit order	3	2	1	0																		
Binary value	1	0	1	0																		
Plan ID	$10 = 1 \cdot 2^3 + 0 \cdot 2^2 + 1 \cdot 2^1 + 0 \cdot 2^0$																					

9.1 Remote Mode Configuration Instruction

9.1.3 Remote Mode - I/O

Part 4 - Steps to start a plan via I/O signals

Step	Signal Direction	Action	I/O Behavior
1	PLC -> Robot	Clear robot fault	Set a rising edge to [IN] Clear Fault
2	PLC -> Robot	Switch on the motor	Set a rising edge to [IN] Motor On
3	Robot -> PLC	Ready to receive Plan ID	[OUT] Program Request is ON
4	PLC -> Robot	Send Plan ID to robot	Set a value to [IN] Plan ID
5	PLC -> Robot	Assign Plan ID	Set a rising edge to [IN] Validate Program
6	Robot -> PLC	Check Plan ID	[OUT] Plan ID State changes to the same value as [IN] Plan ID
7	PLC -> Robot	Enable robot motion	Set [IN] Enable Move to ON
8	PLC -> Robot	Execute the plan	Set a rising edge to [IN] Start Program

9.1 Remote Mode Configuration Instruction

9.1.3 Remote Mode - I/O

Part 5 - Steps to pause/resume and stop/restart a plan via I/O signals

Step	Signal Direction	Action	I/O Behavior
1	Robot -> PLC	Inform that plan is running	[OUT] Program Running is ON
2	PLC -> Robot	Pause plan	Set a rising edge to [IN] Pause Program
3	Robot -> PLC	Inform that plan is paused	[OUT] Program Paused is ON
4	PLC -> Robot	Resume plan	Set a rising edge to [IN] Start Program
5	Robot -> PLC	Inform that plan is running	[OUT] Program Running is ON
6	PLC -> Robot	Stop plan	Set a rising edge to [IN] Stop Program
7	Robot -> PLC	Inform that plan is stopped	[OUT] Program Running is OFF
8	Robot -> PLC	Ready to receive Plan ID	[OUT] Program Request is ON
9	PLC -> Robot	Activate the current plan	Set a rising edge to [IN] Validate Program
10	PLC -> Robot	Restart robot program	Set a rising edge to [IN] Start Program

9.1 Remote Mode Configuration Instruction

9.1.4 Remote Mode - Fieldbus

A robot can be controlled externally via a Fieldbus protocol. Currently, **Modbus TCP**, **Profinet**, and **Anybus** are supported. However, only one communication type will be available in the drop-down menu in **Settings > Remote Mode** according to your configuration in **Settings > Fieldbus**.

This instruction consists of four parts:

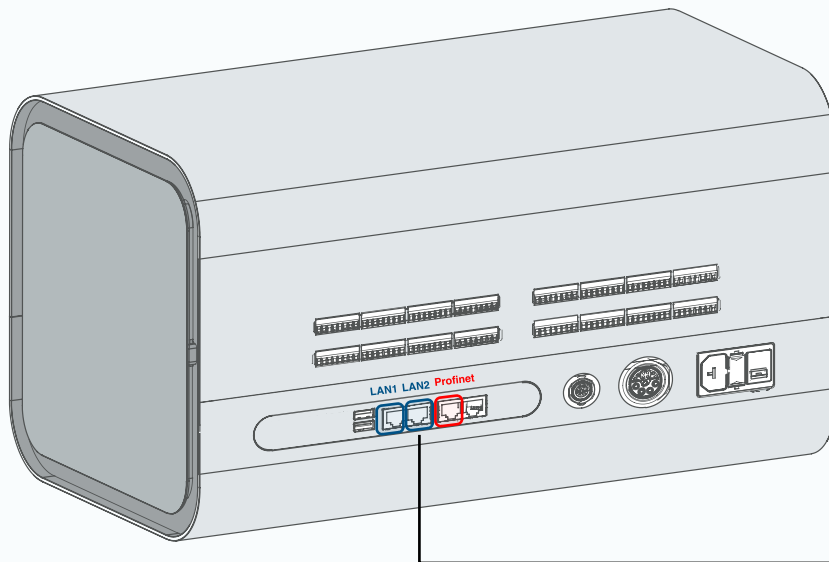
Part	Instruction	Page
1	Physical connection between the robot and a peripheral device	Page 112
2	Configuration in Settings > Fieldbus and Settings > Remote Mode	Page 113 ~ 115
3	Steps to start a plan via I/O signals	Page 116
4	Steps to pause/resume and stop/restart a plan via I/O signals	Page 117

9.1 Remote Mode Configuration Instruction

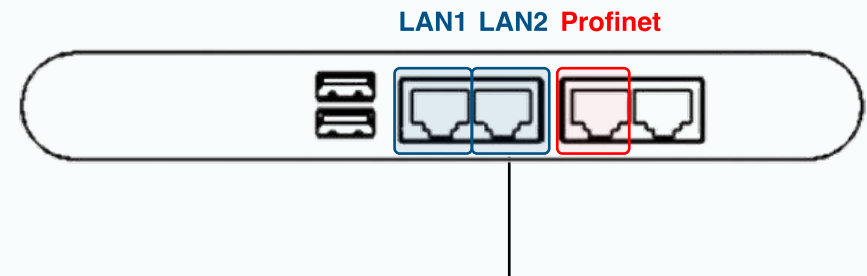
9.1.4 Remote Mode - Fieldbus

Part 1 - Physical connection between the robot and a peripheral device

For communicating via Profinet, use **Profinet**, and for communicating via Modbus TCP / Anybus*, use **LAN1** or **LAN2**.



Control Box



USB and Ethernet connections on the Control Box

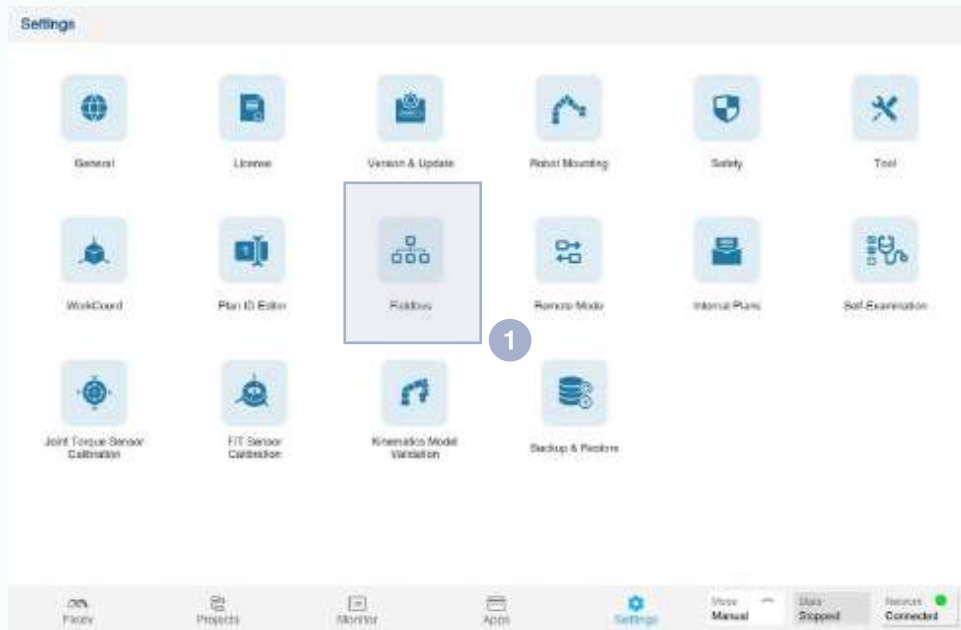
***Note:** For detailed configuration of Anybus, please refer to *Anybus Gateway User Manual*.

9.1 Remote Mode Configuration Instruction

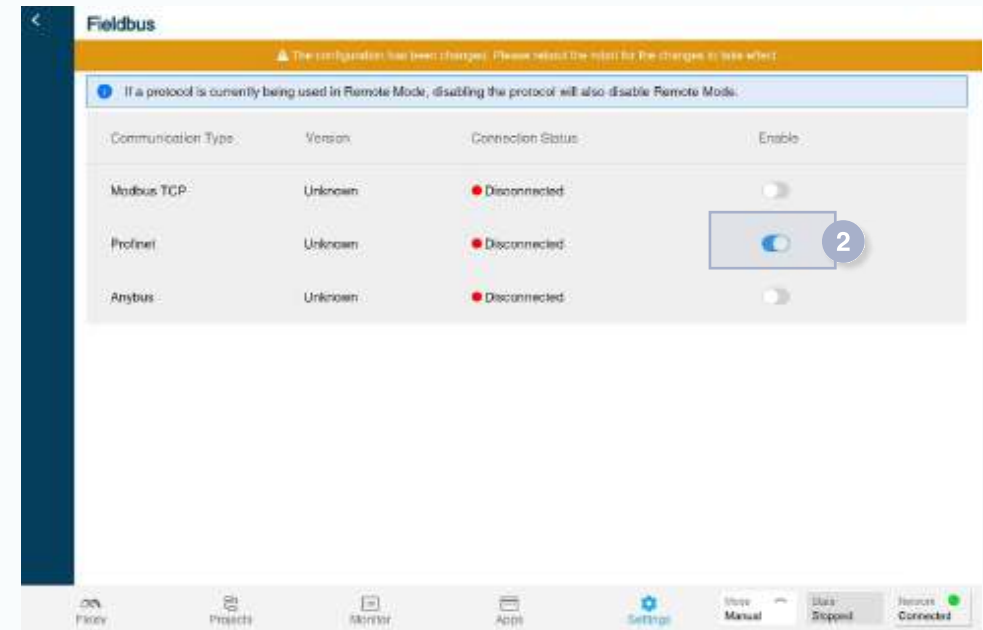
9.1.4 Remote Mode - Fieldbus

Part 2 - Configuration in Settings > Fieldbus and Settings > Remote Mode

Example: Profinet 1 2 3 4 5 6



1 Go to **Settings > Fieldbus**.



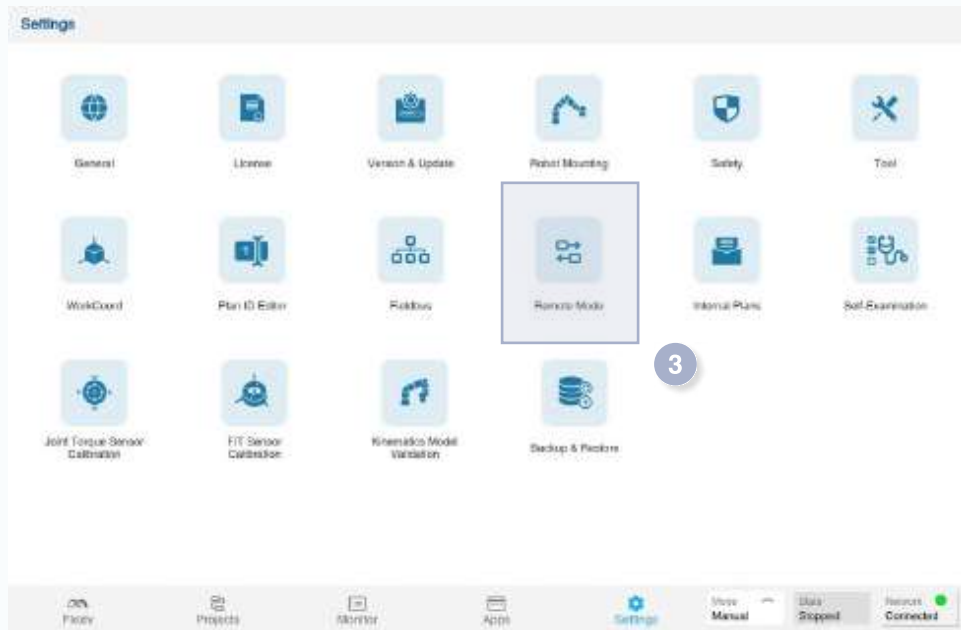
2 To enable **Profinet**, switch the toggle from  to .

9.1 Remote Mode Configuration Instruction

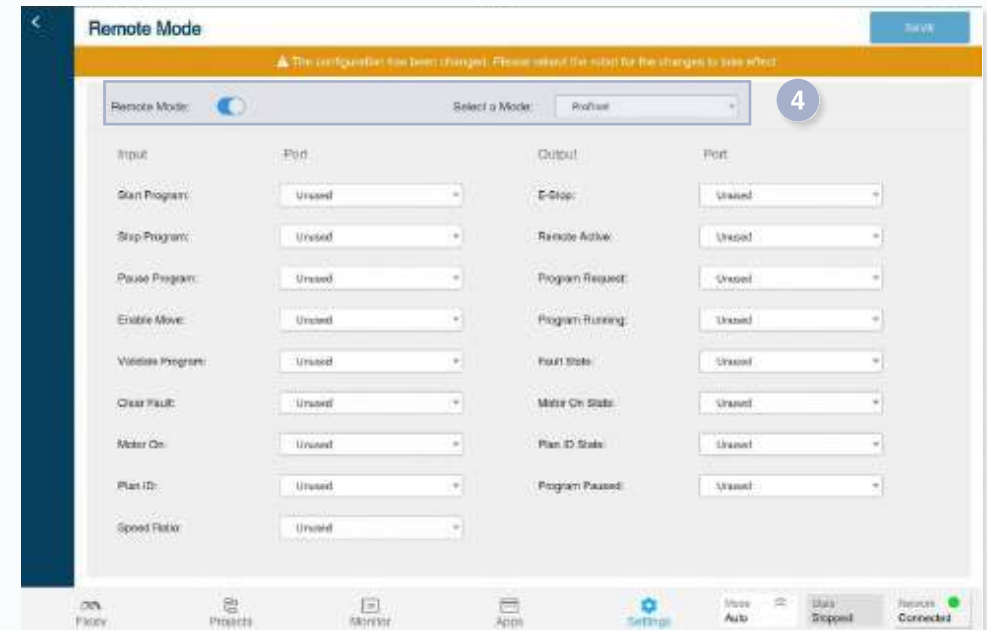
9.1.4 Remote Mode - Fieldbus

Part 2 - Configuration in Settings > Fieldbus and Settings > Remote Mode

Example: Profinet 1 2 3 4 5 6



3 Go to **Settings > Remote Mode**.



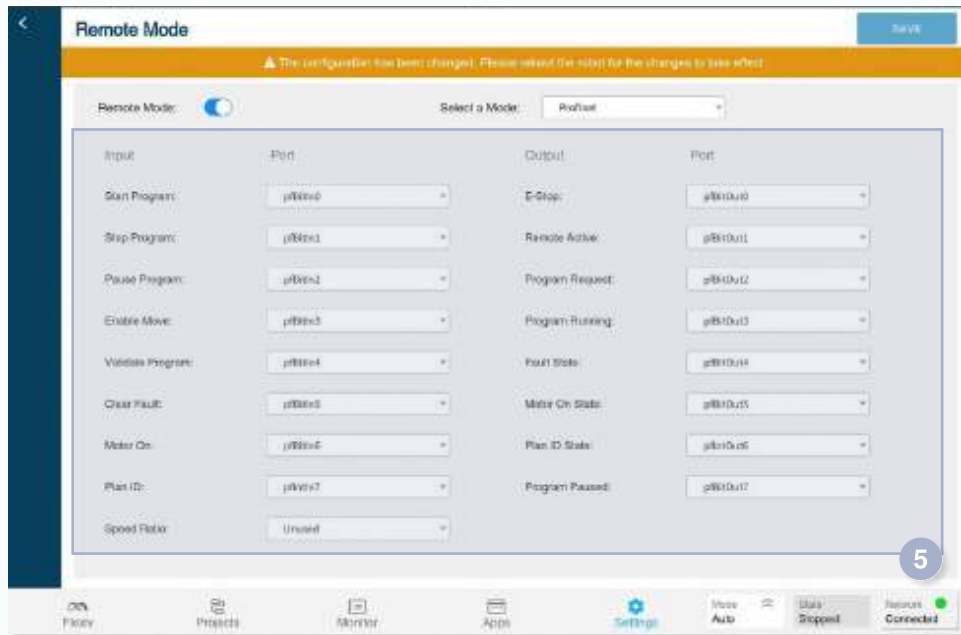
4 To enable **Remote Mode**, switch the toggle from to  , then select **Profinet** from the **Select a Mode** drop-down menu.

9.1 Remote Mode Configuration Instruction

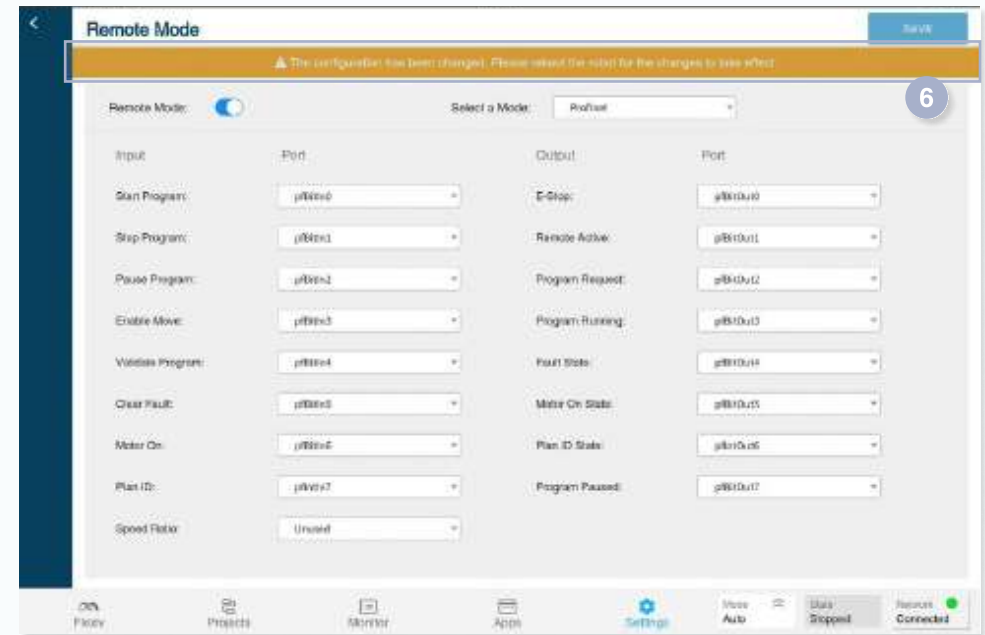
9.1.4 Remote Mode - Fieldbus

Part 2 - Configuration in Settings > Fieldbus and Settings > Remote Mode

Example: Profinet 1 2 3 4 5 6



5 Configure input and output signals, and tap to save the changes.



6 Reboot the robot for the changes (both **Fieldbus** and **Remote Mode** configuration) to take effect.

9.1 Remote Mode Configuration Instruction

9.1.4 Remote Mode - Fieldbus

Part 3 - Steps to start a plan via I/O signals

Step	Signal Direction	Action	I/O Behavior
1	PLC -> Robot	Clear robot fault	Set a rising edge to [IN] Clear Fault
2	PLC -> Robot	Switch on the motor	Set a rising edge to [IN] Motor On
3	Robot -> PLC	Ready to receive Plan ID	[OUT] Program Request is ON
4	PLC -> Robot	Send Plan ID to robot	Set an integer value to [IN] Plan ID
5	PLC -> Robot	Assign Plan ID	Set a rising edge to [IN] Validate Program
6	Robot -> PLC	Check Plan ID	[OUT] Plan ID State changes to the same value as [IN] Plan ID
7	PLC -> Robot	Enable robot motion	Set [IN] Enable Move to ON
8	PLC -> Robot	Execute the plan	Set a rising edge to [IN] Start Program
9 (Optional)	PLC -> Robot	Set robot moving speed	Set an integer value to [IN] Speed Ratio

9.1 Remote Mode Configuration Instruction

9.1.4 Remote Mode - Fieldbus

Part 4 - Steps to pause/resume and stop/restart a plan via I/O signals

Step	Signal Direction	Action	I/O Behavior
1	Robot -> PLC	Inform that plan is running	[OUT] Program Running is ON
2	PLC -> Robot	Pause plan	Set a rising edge to [IN] Pause Program
3	Robot -> PLC	Inform that plan is paused	[OUT] Program Paused is ON
4	PLC -> Robot	Resume plan	Set a rising edge to [IN] Start Program
5	Robot -> PLC	Inform that plan is running	[OUT] Program Running is ON
6	PLC -> Robot	Stop plan	Set a rising edge to [IN] Stop Program
7	Robot -> PLC	Inform that plan is stopped	[OUT] Program Running is OFF
8	Robot -> PLC	Ready to receive Plan ID	[OUT] Program Request is ON
9	PLC -> Robot	Activate the current plan	Set a rising edge to [IN] Validate Program
10	PLC -> Robot	Restart robot program	Set a rising edge to [IN] Start Program

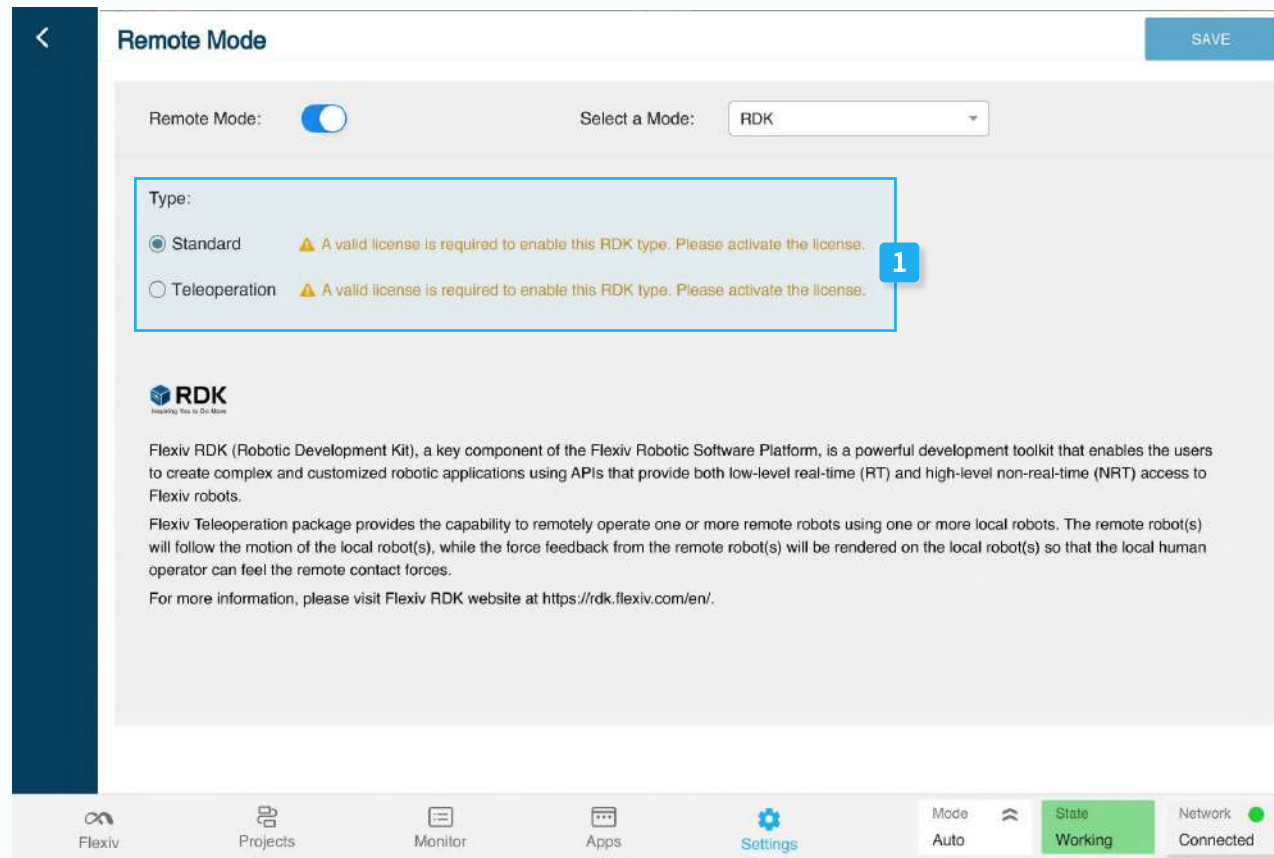
9.1 Remote Mode Configuration Instruction

9.1.5 Remote Mode - RDK

Flexiv RDK enables users to connect their own workstation PC to one or more Flexiv robots via Ethernet (TCP/IP) connection and gain real-time or non-real-time access to the robot(s).

RDK Teleoperation enables the users to operate remote robot(s) at a distance using local robot(s) while feeling the force feedback from the remote robot(s).

You can refer to **Flexiv RDK Manual** at <https://rdk.flexiv.com/manual/index.html> for detailed instructions.



1 Switch the RDK type between **Standard** and **Teleoperation**

10.1 Troubleshooting

10.1.1 Summary of Common Issues

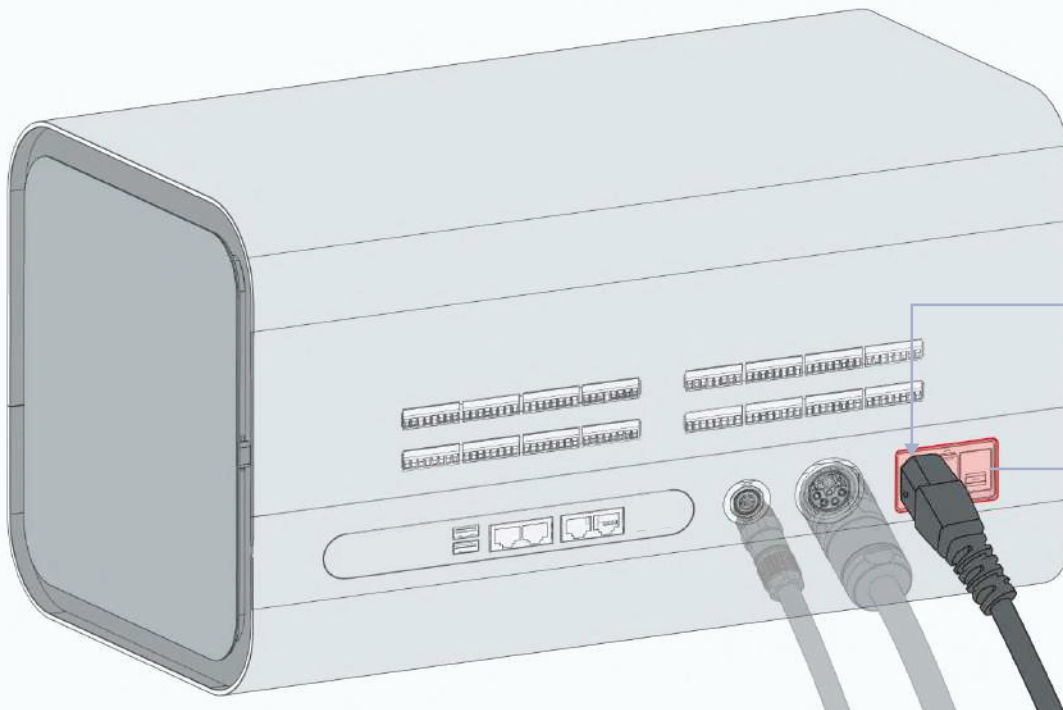
This appendix section contains troubleshooting information to help you identify and resolve the problems commonly encountered during the first power up.

#	Problem		Possible Cause	Solution
1	No sound from Control Box		Not powered on	Check the AC power cable connection and the Control Box switch
2	The light strips on Robot / Control Box / Motion Bar are not lit		Robot-Control Box cable / Motion Bar cable is not connected	Check the Robot-Control Box cable / Motion Bar cable connection
3	Network is not connected	Wi-Fi is not connected	Wi-Fi switch of the Control Box is off or router/IP is not configured correctly	Toggle the Wi-Fi switch of the Control Box on; reconfigure the router/IP
		Network cable is not connected	Network cable is not connected properly	Check network cable connection
4	Unable to servo on the robot		Jumper wires are not connected to pins properly	Check jumper wires connection
5	Robot state is displayed as Fault on Flexiv Elements		A fault or violation is detected by the system	If there is a fault clearing notification, tap it to clear fault; otherwise, manually reboot the Control Box

10.1 Troubleshooting

10.1.2 Control Box Power On Issue

Problem: There is no sound from the Control Box (you can hear the sound of a fan running in normal working state).



Ensure that both of the following conditions are met:

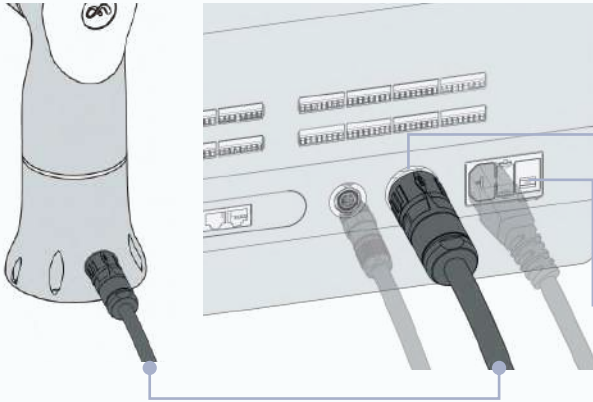
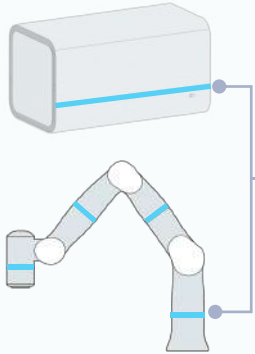
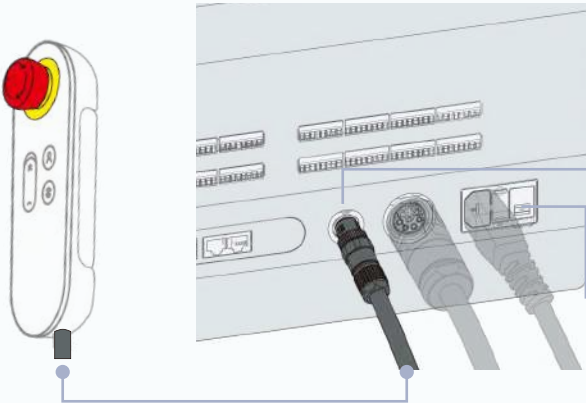
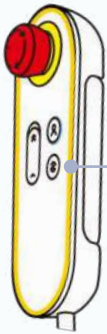
- 1 Connect the AC power cable to a 100-240 VAC power socket
- 2 Toggle the switch on the Control Box

***Note:** The robot can only start when AC power cable, Robot-Control Box cable, and Motion Bar cable are correctly connected and the Control Box switch is toggled.

10.1 Troubleshooting

10.1.3 Light Strip Issue

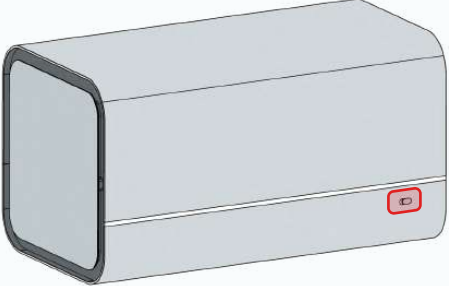
Problem: The light strips on Robot / Control Box / Motion Bar are not lit.

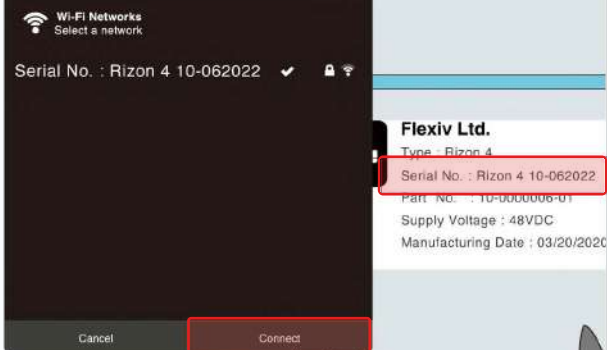
What actions you can take	What you will observe
 Ensure that: 1 Connect the robot to the Control Box with cable for robot-Control Box 2 Toggle the switch on the Control Box	 The light strips on the Control Box/robot light up in blue
 Ensure that: 1 Connect the Motion Bar to the Control Box 2 Toggle the switch on the Control Box	 Robot state indicator on the Motion Bar lights up in yellow

10.1 Troubleshooting

10.1.4 Network Issue

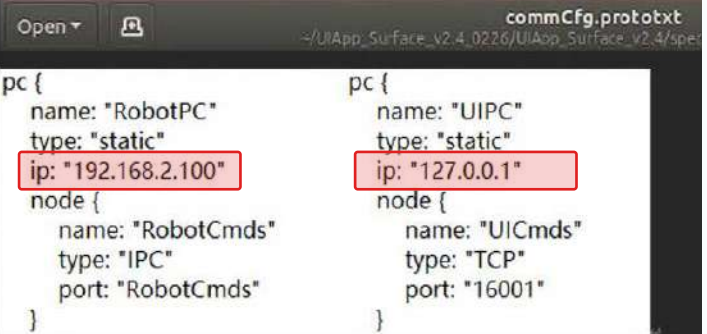
A. Wi-Fi is not connected







router configuration

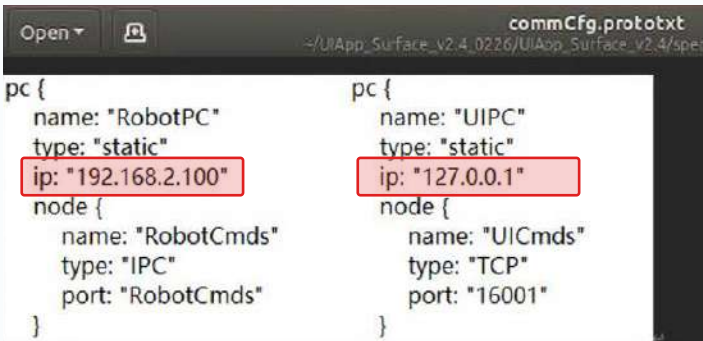


- Turn on the Wi-Fi switch on the Control Box
- On the Teach Pendant, connect to the Wi-Fi SSID that matches the robot's serial number
- Configure the router file provided by customer service
- On the Teach Pendant, tap  ConfigIP
Ensure that:
RobotPC IP is 192.168.2.100
UIPC IP is 127.0.0.1
If not, key in the correct value.

10.1 Troubleshooting

10.1.4 Network Issue

B. Network cable is not connected

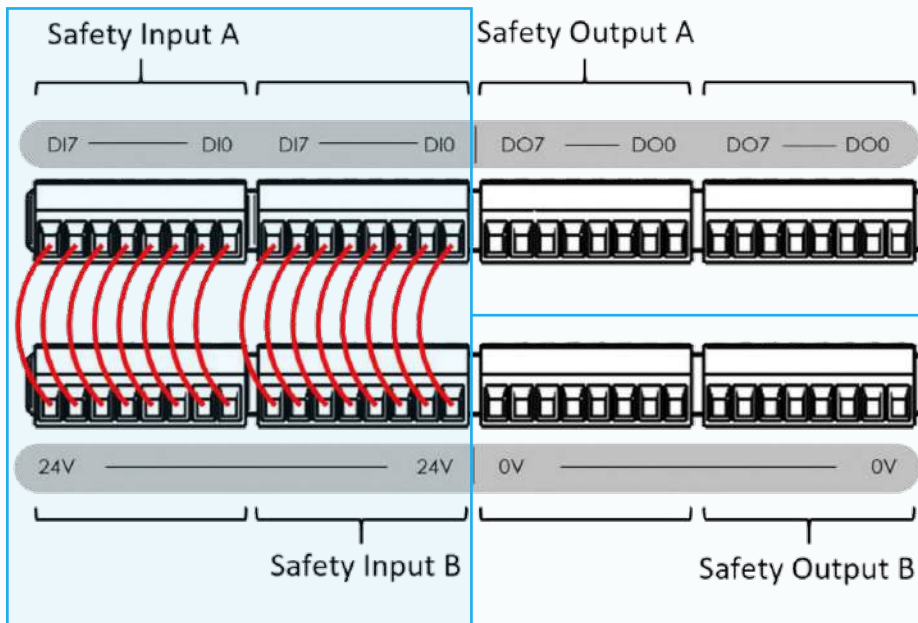
1 	1 Make a wired connection using the adapter
2  <pre> pc { name: "RobotPC" type: "static" ip: "192.168.2.100" node { name: "RobotCmds" type: "IPC" port: "RobotCmds" } } pc { name: "UIPC" type: "static" ip: "127.0.0.1" node { name: "UICmds" type: "TCP" port: "16001" } } </pre>	2 On the Teach Pendant, tap  ConfigIP Ensure that: RobotPC IP is 192.168.2.100 UIPC IP is 127.0.0.1 If not, key in the correct value.

***Note:** It is recommended to establish a wired connection in environments with high levels of interference.

10.1 Troubleshooting

10.1.5 Servo On Issue

Problem: The jumper wires of the Control Box are not correctly connected. As a result, the robot servo cannot be turned on even when the E-Stop button on the Motion Bar is released.



As shown in the illustration, ensure that:

Jumper wires are appropriately and correctly connected to the pins

10.1 Troubleshooting

10.1.6 Fault State Issue

Problem: Robot state is displayed as **Fault** on Flexiv Elements.

1. With fault clearing notification



If there is a fault clearing notification , you can tap the state below to clear the fault, and the state will return to normal.

Click to clear fault

2. Without fault clearing notification



If there is no fault clearing notification, you need to manually reboot the Control Box to clear the fault. The state will return to normal after the reboot is complete.